arXiv:submit/8345298 [math.DG] 8 Nov 2026

GLOBAL CLASSIFICATION OF THE FLAT SINGULAR ESCOBAR-YAMABE EQUATION

FELIPE G. BOGO

ABSTRACT. In every dimension $n \geq 3$, we classify positive solutions of the flat Escobar–Yamabe equation on the Euclidean half-space with a possible isolated boundary singularity at the origin. Solutions with a removable singularity are boundary bubbles. Up to dilation, the nonremovable solutions are the cylindrical solution and a family of periodic Delaunay solutions uniquely parametrized by the necksize $\varepsilon \in (0,1)$, the minimum of the normalized boundary trace. In each fundamental period T_ε, the trace has exactly two critical points, a nondegenerate minimum and a nondegenerate maximum. The period and the solution centered at the trace minimum and rescaled to period one depend smoothly on ε. As $\varepsilon \to 0$, $T_\varepsilon = \frac{4}{n-2}\log(1/\varepsilon) + O(1)$. As $\varepsilon \to 1$, the solutions converge to the cylindrical solution and their periods tend to that of its linearized oscillation. We use uniqueness in the odd and even classes for linear Robin equations to obtain reflection symmetry and continuation in the necksize.

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1. INTRODUCTION

Let $n \geq 3$. We study positive solutions of the singular Escobar–Yamabe problem

$$\begin{cases} \Delta U = 0 & \text{for } x_n > 0, \\ \partial_{x_n} U + \dfrac{n-2}{2} u^{\frac{n}{n-2}} = 0 & \text{on } \partial\mathbb{R}^n_+ \setminus \{0\}, \end{cases} \tag{1.1}$$

where $\mathbb{R}^n_+ = \{x = (\bar{x}, x_n) \in \mathbb{R}^n : x_n \geq 0\}$ and $u(\bar{x}) = U(\bar{x}, 0)$. We assume that U is smooth on $\mathbb{R}^n_+ \setminus \{0\}$. The conformal metric $g = U^{4/(n-2)}\delta$, where δ is the Euclidean metric, is scalar flat and has boundary mean curvature $h_g = 1$ away from 0. We take h_g to be the average of the principal curvatures with respect to the outward unit normal.

The relation with the fractional Yamabe problem is described by the conformal extension results of Chang and González [10] and Case and Chang [9]. At order $s = 1/2$, the Poisson extension of a solution of

$$(-\Delta_{\mathbb{R}^{n-1}})^{1/2} u = \frac{n-2}{2} u^{\frac{n}{n-2}} \qquad \text{in } \mathbb{R}^{n-1} \setminus \{0\}$$

solves (1.1) by the extension formula of Caffarelli and Silvestre [8].

We say that the singularity at 0 is removable if the trace u extends continuously across 0. In the nonremovable case, Caffarelli, Jin, Sire and Xiong [7, Theorem 1.1] prove that

$$c_1 |\bar{x}|^{-\frac{n-2}{2}} \leq u(\bar{x}) \leq c_2 |\bar{x}|^{-\frac{n-2}{2}}$$

for some $c_1, c_2 > 0$ and all nonzero $\bar{x}$ sufficiently close to 0. If the singularity is removable, the weak equation extends across 0 by the argument in Section 4.1. The half-space Liouville theorem of Jin, Li and Xiong [16, Theorem 1.8] then gives the boundary bubbles in (1.5). The classification for regular solutions is due to Li and Zhu [19, Theorem 1.2].

The case without boundary was considered by Caffarelli, Gidas and Spruck [6, Theorem 8.1(a) and Corollary 8.2(b)]. Positive solutions of

$$\Delta w + \frac{n(n-2)}{4} w^{\frac{n+2}{n-2}} = 0 \qquad \text{in } \mathbb{R}^n \setminus \{0\} \tag{1.2}$$

are standard bubbles if the singularity is removable and are radial if it is nonremovable. For a radial solution, the Emden–Fowler transformation is

$$w(r) = r^{-\frac{n-2}{2}} v(t), \qquad r = |x|, \qquad t = -\log r.$$

This gives the ordinary differential equation

$$v'' + \frac{n(n-2)}{4} v^{\frac{n+2}{n-2}} - \frac{(n-2)^2}{4} v = 0. \tag{1.3}$$

The Hamiltonian energy

$$H(v, v') = (v')^2 - \frac{(n-2)^2}{4} v^2 + \frac{(n-2)^2}{4} v^{\frac{2n}{n-2}} \tag{1.4}$$

is constant along solutions v of (1.3). Its periodic orbits give the Fowler solutions, also called Delaunay-type solutions after Delaunay's surfaces of revolution [14].

Proposition 1.1 (Mazzeo and Pacard [21, Proposition 1])**.** *For every energy H in the interval $(-(n-2)((n-2)/n)^{n/2}/2,\, 0)$ there is a unique positive bounded solution v of (1.3) with that energy and with $v'(0) = 0$ and $v''(0) > 0$. This solution is periodic, takes values in $(0, 1)$ and can be parametrized by its minimum*

$$\varepsilon = v(0) \in \left(0, \left(\frac{n-2}{n}\right)^{\frac{n-2}{4}}\right).$$

At the left endpoint of the energy interval the only positive bounded solution is the constant

$$v \equiv \left(\frac{n-2}{n}\right)^{\frac{n-2}{4}}.$$

At energy zero, a nonnegative bounded solution is either zero or

$$v(t) = (\cosh(t - t_0))^{-\frac{n-2}{2}} \qquad \text{for some } t_0 \in \mathbb{R}.$$

For the singular Escobar–Yamabe problem, Caffarelli, Jin, Sire and Xiong [7, Theorem 1.3] show that every solution with a nonremovable singularity is cylindrically symmetric, that is,

$$U(\bar{x}, x_n) = U(|\bar{x}|, x_n).$$

The Emden–Fowler transform can depend on both t and the angle with the x_n-axis, so the reduced equation is a partial differential equation. DelaTorre and González [13] studied the fractional equation, its Hamiltonian and the linearization at the cylindrical solution. DelaTorre, del Pino, González and Wei [12, Theorem 1.1] constructed periodic solutions by a variational argument. At order $1/2$, their theorem applies when the half-space has dimension $n \geq 4$. Cruz-Blázquez, DelaTorre and Ruiz [11, Section 1] asked whether every solution of the fractional equation whose Emden–Fowler transform is bounded above and away from zero is periodic in t. Andrade, DelaTorre, do Ó, Ratzkin and Wei [2, Theorem 1] classified the singular solutions for fractional orders sufficiently close to 1.

The normalized Emden–Fowler transform v in Section 2.1 solves (2.1). Its boundary coefficient c and the normalization factor b_n are

$$c = 2\left(\frac{\Gamma(n/4)}{\Gamma((n-2)/4)}\right)^2, \qquad b_n = \left(\frac{2c}{n-2}\right)^{\frac{n-2}{2}}.$$

For cylindrically symmetric solutions we write $\tilde{v}(t)$ for the trace on $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$. The cylindrical solution C_n has trace 1 in this normalization. We call the minimum of $\tilde{v}$ the *necksize*.

Theorem A. *Let $n \geq 3$. For every $\varepsilon \in (0,1)$, there is a unique positive smooth cylindrically symmetric solution v_ε of (2.1) with bounded trace and*

$$\tilde{v}_\varepsilon(0) = \min_{t\in\mathbb{R}} \tilde{v}_\varepsilon(t) = \varepsilon.$$

The solution is even in t and periodic. Its fundamental period T_ε is twice the first positive critical point of $\tilde{v}_\varepsilon$. The trace $\tilde{v}_\varepsilon$ has one minimum and one maximum per period, with $\tilde{v}_\varepsilon''(0) > 0$, $\tilde{v}_\varepsilon''(T_\varepsilon/2) < 0$ and $\tilde{v}_\varepsilon'(t) > 0$ for $0 < t < T_\varepsilon/2$.

Every positive smooth cylindrically symmetric solution v of (2.1) with trace bounded above and away from zero is either C_n or has the form $v(t,\theta) = v_\varepsilon(t+\tau,\theta)$ for a unique $\varepsilon \in (0,1)$ and a phase $\tau \in \mathbb{R}$ determined modulo T_ε. We set $v_1 = C_n$.

With $v_\varepsilon^{(1)}(s,\theta) = v_\varepsilon(T_\varepsilon s,\theta)$, the map $\varepsilon \mapsto (T_\varepsilon, v_\varepsilon^{(1)})$ is smooth from $(0,1)$ into

$$(0,\infty) \times C^{k,\gamma}_{1,\mathrm{sym},\mathrm{even}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$$

for every integer $k \geq 0$ and $0 < \gamma < 1$.

Idea of the proof of Theorem A. Proposition 3.1 gives reflection symmetry about each critical point of $\tilde{v}$ and determines an even solution by $\tilde{v}(0)$. The differences of solutions and the Jacobi fields used in that proposition satisfy linear equations with a coefficient $b(t)$ given by (3.10). For $n \geq 5$, we estimate the trace by decomposing the solution along the first eigenfunction and its orthogonal complement. For $n = 3, 4$, we obtain a contraction for the trace using the eigenfunctions for a fixed value of the boundary coefficient.

Testing (2.1) against C_n gives (3.16), which yields two distinct critical points of $\tilde{v}$ for every noncylindrical solution under consideration. Composing the reflections about these points gives a period. Odd uniqueness for the Jacobi field $\partial_t v$ gives nondegeneracy of the critical points. Even uniqueness gives uniqueness for a prescribed necksize after translating a minimum to $t = 0$.

Minimization on a compact cylinder of sufficiently large period gives one noncylindrical solution. After rescaling to period one, we treat the period as an unknown and prescribe $\tilde{v}(0)$. Even uniqueness and the Fredholm alternative give invertibility of the linearization of this augmented problem. The implicit function theorem and compactness then give, respectively, openness and relative closedness of the set of attained necksizes in $(0,1)$.

As $\varepsilon \to 1$, v_ε converges to C_n and T_ε tends to the period determined by the linearization there (Proposition 3.7). We prove that $\varepsilon \to 0$

implies,

$$T_\varepsilon = \frac{4}{n-2}\log(1/\varepsilon) + O(1)$$

in Corollary 3.9.

Theorem B. *Let $n \geq 3$ and let U be a positive solution of (1.1), smooth on $\mathbb{R}^n_+ \setminus \{0\}$. Exactly one of the following holds.*

(i) *The singularity at 0 is removable, and U is the bubble*

$$U(\bar{x}, x_n) = \left(\frac{2\lambda}{|\bar{x}-a|^2 + (x_n+\lambda)^2}\right)^{\frac{n-2}{2}} \tag{1.5}$$

for unique parameters $a \in \mathbb{R}^{n-1}$ and $\lambda > 0$.

(ii) *The singularity at 0 is nonremovable. There exist a unique necksize $\varepsilon \in (0,1]$ and a phase $\tau \in \mathbb{R}$ such that, for $r > 0$ and $\theta \in \mathbb{S}^{n-1}_+$,*

$$U(r\theta) = b_n r^{-(n-2)/2}\, v_\varepsilon\Big(-\log r + \tau, \theta\Big),$$

where v_ε is given by Theorem A. For $0 < \varepsilon < 1$, the phase is unique modulo T_ε. For $\varepsilon = 1$, all phases give the same cylindrical solution.

Conversely, both formulas give positive solutions of (1.1). The solutions in (i) *extend smoothly across 0, and those in* (ii) *have a nonremovable singularity there.*

Idea of the proof of Theorem B. If u extends continuously across 0, the weak equation for U extends across 0 and the regular half-space classification gives (1.5). In the nonremovable case, we apply the local two-sided bounds at 0 to U and its Kelvin transform $U_{0,1}$ from (2.2). The trace of $U_{0,1}$ is also singular at 0, since Kelvin inversion sends bubbles to bubbles. This gives two-sided bounds for $|x|^{(n-2)/2}U(x)$ on $\mathbb{R}^n_+ \setminus \{0\}$. Cylindrical symmetry and Theorem A then give the formula in (ii).

In the companion paper [4], we use these results to study the Jacobi fields of the Delaunay family and the asymptotic behavior of solutions near an isolated boundary singularity. For a smooth background metric, two-sided bounds on the boundary restriction give convergence to a fixed Delaunay solution. The corresponding problem without boundary was studied by Korevaar, Mazzeo, Pacard and Schoen [18] and by Marques [20].

The classification also supplies models for constructing singular solutions of the boundary Yamabe problem on manifolds with boundary. In the classical Yamabe problem, Silva Santos [25] used gluing to construct

complete conformal metrics with Delaunay-type ends on punctured closed manifolds under nondegeneracy and curvature hypotheses. For the fractional Yamabe equation, Ao, DelaTorre, González and Wei [3, Theorem 1.1] used Delaunay solutions in a gluing construction with prescribed isolated singularities.

Structure of the paper. In Section 2, Section 2.1 introduces the normalized Emden–Fowler transform and Kelvin inversion. Section 2.2 gives uniform regularity and compactness on the cylinder, and Section 2.3 specifies the linearized equations and periodic function spaces.

In Section 3, Section 3.1 proves rigidity and periodicity. Section 3.2 constructs the family and proves its smooth dependence on the necksize. Section 3.3 records the Hamiltonian and determines the limiting behavior of the period.

In Section 4, Section 4.1 treats removable singularities and obtains the global two-sided bounds in the nonremovable case. Section 4.2 uses these results and Theorem A to complete the proof of Theorem B.

In Appendix A, Section A.1 controls the extension on $\mathbb{R} \times \mathbb{S}^{n-1}_+$ by bounds on its trace. Section A.2 bounds the first derivative of the trace and proves second-order inequalities for powers of the trace, which are used to estimate the boundary coefficients. Section A.3 gives the eigenvalue bounds and the resolvent identity. In Appendix B, Section B.1 estimates a weighted integral of the trace, and Section B.2 uses this estimate to prove uniqueness in each parity for $n \geq 5$. Appendix C obtains the corresponding uniqueness statements for $n = 3, 4$ from integral equations for the trace and estimates for their kernels.

Relation to concurrent work. While this manuscript was in preparation, Jin, Liao and Xiong submitted their preprint [17] to arXiv on November 5, 2026. Their Theorems 1.1 and 2.1 give the same half-space classification. They parametrize the linear solution space by two projections of the Cauchy data in Proposition 2.10 and classify the nonlinear solutions by their Hamiltonian energy. Their Proposition A.2 also gives the necksize parametrization by showing that the necksize is a real-analytic function of the energy with a real-analytic inverse. Their Lemma A.1 proves uniqueness for even solutions from $Y(0, L)$ under the additional bound $|b'| \leq b$.

The scalar resolvent identity in the first line of (A.30) also appears in [17, (2.20)]. The weighted identity (B.5) is analogous to [17, (A.4)]. To obtain the weighted estimate, both papers decompose the solution along the first Robin eigenfunction and its orthogonal complement and integrate the corresponding identity against $e^{-(n-1)t}$. Appendices A and B give the derivations used in this paper.

We use uniqueness from $Y(0, L)$ in the even class and from $\partial_t Y(0, L)$ in the odd class. We deduce periodicity by composing reflections about trace critical points and construct the family by continuation directly in the necksize.

Corollary 3.9 gives the rate of divergence of the period at small necksize, $T_\varepsilon = \frac{4}{n-2}\log(1/\varepsilon) + O(1)$. Proposition A.3 gives second-order inequalities for powers of the trace in dimensions $n \geq 5$. For $n = 3, 4$, Theorem C.1 establishes uniqueness in each parity from the corresponding boundary datum, assuming only that b is smooth and even and $0 \leq b \leq n/2$. These estimates and the different proof of the classification are the reasons for presenting our results here.

Statement on the use of AI. The need for Proposition 3.1 was identified before we used AI. The model GPT-Sol 6 was instrumental in finding the method of proof of that proposition. We also used the model to revise the manuscript. All mathematical statements, proofs, and citations were independently checked by the author, who takes full responsibility for the contents of the article. The final manuscript was written and revised by the author.

Acknowledgments. This work forms part of my Ph.D. thesis at Universidade Federal Fluminense. I am grateful to my advisor, Sérgio Almaraz, for his guidance and advice throughout this work. I also thank Xuezhang Chen for his warm welcome at Nanjing University and his generous support during my stay.

This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) - Finance Code 001. I received support under process No. 88887.815246/2023-00 and a scholarship through the Programa Institucional de Doutorado Sanduíche no Exterior (PDSE), process No. 88881.127484/2025-01.

2. Preliminaries

Section 2.1 gives the cylindrical formulation of (1.1) and describes the action of dilations and Kelvin inversion. Section 2.2 establishes uniform regularity and compactness for solutions bounded above and away from zero. In Section 2.3, we introduce the linearized equations and the estimates for Jacobi fields with polynomial growth used in the rigidity argument. We also define the periodic Hölder spaces for the local parametrization by necksize.

2.1. **Notation.** We use the half-space and trace conventions of Section 1. We introduce the normalized Emden–Fowler variables, write the interior equation and the boundary condition on the cylinder, and express dilations as translations in t. Cylindrical symmetry means invariance under rotations of $\bar{x}$ with x_n fixed. For cylindrically symmetric functions, we give the angular form of the Laplacian, the conormal derivative at the equator, and the regularity condition at the pole. We then record the Kelvin transform and its action in t when the center of inversion is the origin.

Write $x = r\theta$ with $r > 0$ and $\theta \in \mathbb{S}^{n-1}_+ = \{\theta \in \mathbb{S}^{n-1} : \theta_n \geq 0\}$. In the normalization of the introduction, the Emden–Fowler variables are

$$t = -\log r, \qquad v(t,\theta) = b_n^{-1} r^{\frac{n-2}{2}} U(r\theta).$$

We write $\tilde{v}$ for the restriction of v to $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$. Thus $\tilde{v}(t,\vartheta) = v(t,(\vartheta,0))$ for $\vartheta \in \mathbb{S}^{n-2}$. The dilation $U_\lambda(x) = \lambda^{(n-2)/2}U(\lambda x)$ corresponds to $v_\lambda(t,\theta) = v(t-\log\lambda,\theta)$.

In these variables, (1.1) becomes

$$\begin{cases} v_{tt} + \Delta_{\mathbb{S}^{n-1}_+} v - \dfrac{(n-2)^2}{4} v = 0 & \text{in } \mathbb{R}\times\mathbb{S}^{n-1}_+, \\ \partial_{\nu_{\mathbb{S}^{n-1}_+}} v = c\,\tilde{v}^{\frac{n}{n-2}} & \text{on } \mathbb{R}\times\partial\mathbb{S}^{n-1}_+. \end{cases} \tag{2.1}$$

Here $\Delta_{\mathbb{S}^{n-1}_+}$ is the Laplace–Beltrami operator of the round metric, and $\nu_{\mathbb{S}^{n-1}_+}$ is the outward unit conormal along $\partial\mathbb{S}^{n-1}_+$. The cylinder carries the product metric with volume element $\mathrm{d}\mu = \mathrm{d}t\,\mathrm{d}\theta$.

For $\theta \in \mathbb{S}^{n-1}_+$, let $\beta = \arccos\theta_n$ and $L = \pi/2$. Then

$$\theta = (\sin\beta\,\vartheta, \cos\beta), \qquad 0 \leq \beta \leq L, \qquad \vartheta \in \mathbb{S}^{n-2}.$$

For cylindrically symmetric functions, we write $v = v(t,\beta)$ and $\tilde{v}(t) = v(t,L)$. For $0 < \beta < L$, the Laplace–Beltrami operator on such functions is

$$\Delta_{\mathbb{S}^{n-1}_+} v = \omega^{-1}(\omega v_\beta)_\beta = v_{\beta\beta} + (n-2)\cot\beta\, v_\beta, \qquad \omega(\beta) = \sin^{n-2}\beta.$$

On $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$ we have $\partial_{\nu_{\mathbb{S}^{n-1}_+}} v = v_\beta(t,L)$. For each fixed t and ϑ, the restriction of v to $s\mapsto(\sin s\,\vartheta,\cos s)$ is smooth and even near $s=0$ by cylindrical symmetry. Hence $v_\beta(t,0)=0$.

For $a=(\bar a,0)$ and $\rho>0$, the Kelvin transform is

$$U_{a,\rho}(x)=\left(\frac{\rho}{|x-a|}\right)^{n-2}U\left(a+\frac{\rho^2(x-a)}{|x-a|^2}\right). \tag{2.2}$$

It preserves (1.1) on the inverted domain. For $a=0$, the map $r\mapsto\rho^2/r$ gives the cylindrical action $t\mapsto -t-2\log\rho$.

2.2. **Uniform regularity and compactness of solutions.** We define uniformly local Hölder norms on the cylinder and its boundary. For solutions with common positive lower and upper bounds, we first obtain a Hölder estimate in half-space coordinates and then use Schauder estimates on the cylinder to control higher derivatives. The estimates are uniform under translations in t and give convergence of a subsequence on every finite cylinder. They also apply when the boundary coefficient is 1.

For an integer $k\ge 0$ and $0<\gamma<1$, the space $C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ consists of locally $C^{k,\gamma}$ functions for which

$$\|f\|_{C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)}=\sup_{\tau\in\mathbb{R}}\|f\|_{C^{k,\gamma}([\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+)}<\infty.$$

The same definition applies on $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$.

Proposition 2.1. *Let v be a positive smooth solution of (2.1) with $m\le v\le M$ on $\mathbb{R}\times\mathbb{S}^{n-1}_+$ for some $m,M>0$. For every integer $k\ge 0$ and $0<\gamma<1$, there is a constant $C=C(n,m,M,k,\gamma)$ such that*

$$\|v\|_{C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)}\le C. \tag{2.3}$$

Every sequence of such solutions with the same bounds has a subsequence converging in $C^{k,\gamma}$ on each finite cylinder, for every $k\ge 0$ and $0<\gamma<1$, to a solution with the same bounds. The same conclusions hold when the boundary coefficient is 1.

Proof. Let U be the solution of (1.1) corresponding to v. For $\tau\in\mathbb{R}$, the dilation in Section 2.1 gives

$$U_{e^{-\tau}}(r\theta)=b_n r^{-\frac{n-2}{2}}v(\tau-\log r,\theta).$$

Consider the half-annulus

$$A^+=\{x\in\mathbb{R}^n_+ : e^{-3}<|x|<e^3\}.$$

The bound $v\le M$ controls $U_{e^{-\tau}}$ on A^+, including its trace on $A^+\cap\partial\mathbb{R}^n_+$, uniformly in τ. Choose finitely many cylinders $B^{n-1}_\rho(\bar y)\times[0,\rho)$ contained

in A^+ whose cylinders of half the radius cover $\{e^{-1} \le |x| \le e\}$ near $\partial\mathbb{R}^n_+$. By (1.1), $-\partial_{x_n} U_{e^{-\tau}}$ is uniformly bounded in L^n on each base. Since n exceeds the boundary dimension $n-1$, the Hölder estimate of Jin, Li and Xiong [16, Proposition 2.6(iii)] at order $1/2$ applies. Combining it with the interior estimates for harmonic functions and returning to cylindrical variables gives

$$\|v\|_{C^{0,\alpha}([\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+)} \le C(n, M)$$

for some $0 < \alpha < 1$ independent of τ.

For $j \ge 0$, applying the local Schauder estimates for the Neumann problem [1, Theorem 9.3] to (2.1) with conormal derivative $c\tilde{v}^{n/(n-2)}$ on $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$ gives

$$\|v\|_{C^{j+1,\alpha}([\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+)} \le C\left(M + \|c\tilde{v}^{\frac{n}{n-2}}\|_{C^{j,\alpha}([\tau-2,\tau+2]\times\partial\mathbb{S}^{n-1}_+)}\right).$$

The constant C is independent of τ by translation invariance. To bound the Hölder quotients on $[\tau-2, \tau+2]$, we use the interval of radius 1 centered at the midpoint of the two time coordinates when they differ by at most 2. For the remaining pairs, the supremum norms control the Hölder quotients. Since $s \mapsto cs^{n/(n-2)}$ is smooth on $[m, M]$, the chain rule bounds the right-hand side uniformly in τ whenever v has a uniform $C^{j,\alpha}_{\mathrm{uloc}}$ bound. Induction gives these bounds for all $j \ge 0$. The bound in $C^{k+1,\alpha}_{\mathrm{uloc}}$ implies (2.3) for every $0 < \gamma < 1$. When the boundary coefficient is 1, we apply the estimates to $c^{-(n-2)/2}v$.

For a sequence with the same bounds, diagonal extraction on $[-j, j] \times \mathbb{S}^{n-1}_+$ gives convergence in C^j there for every integer $j \ge 1$. Taking $j \ge k+1$ large enough to cover a given finite cylinder gives convergence in $C^{k,\gamma}$ on that cylinder. The limit satisfies (2.1) and the same pointwise bounds. □

2.3. **The linearized equations and periodic function spaces.** We linearize the interior equation and the boundary condition and define Jacobi fields with polynomial growth. Under uniform bounds on the linearized boundary coefficient and its derivatives, local estimates give Hölder bounds with the same order of growth as the field. We then define periodic Hölder spaces on the cylinder and its boundary and specify the subspaces with cylindrical symmetry and prescribed parity in t. Along an even, cylindrically symmetric periodic solution, the linearized operators preserve the period and the symmetry conditions.

Let v be a solution of (2.1). For a smooth function z on $\mathbb{R} \times \mathbb{S}^{n-1}_+$, write $\tilde{z}$ for its restriction to $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$. The linearized interior operator

$\mathcal{A}$ and the boundary operator $\mathcal{B}$ are

$$\mathcal{A}z = -\partial_t^2 z - \Delta_{\mathbb{S}^{n-1}_+} z + \frac{(n-2)^2}{4} z,$$
$$\mathcal{B}z = \partial_{\nu_{\mathbb{S}^{n-1}_+}} z - \frac{n}{n-2} c\, \tilde{v}^{\frac{2}{n-2}} \tilde{z}.$$

When the choice of v needs to be explicit, we write $\mathcal{B}_v$ for $\mathcal{B}$. A smooth function Φ is a Jacobi field along v if

$$\begin{cases} \mathcal{A}\Phi = 0 & \text{in } \mathbb{R} \times \mathbb{S}^{n-1}_+, \\ \mathcal{B}\Phi = 0 & \text{on } \mathbb{R} \times \partial\mathbb{S}^{n-1}_+. \end{cases} \tag{2.4}$$

A Jacobi field Φ has polynomial growth of order at most $q \geq 0$ if there is a constant $C > 0$ such that

$$|\Phi(t,\theta)| \leq C(1+t^2)^{q/2} \qquad \text{for all } (t,\theta) \in \mathbb{R} \times \mathbb{S}^{n-1}_+. \tag{2.5}$$

Assume that the coefficient $\frac{n}{n-2} c\, \tilde{v}^{2/(n-2)}$ in $\mathcal{B}$ and all its derivatives are uniformly bounded on $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$. These bounds follow from Proposition 2.1 when $m \leq v \leq M$ for positive constants m and M. For every integer $k \geq 0$ and $0 < \gamma < 1$, the local Schauder estimates for (2.4) give

$$\|\Phi\|_{C^{k,\gamma}([\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+)} \leq C_{k,\gamma} \|\Phi\|_{C^0([\tau-2,\tau+2]\times\mathbb{S}^{n-1}_+)},$$

where $C_{k,\gamma}$ is independent of τ and depends only on n, k, γ and the bounds for the coefficient of $\tilde{z}$ in $\mathcal{B}$ and its derivatives. Since $1 + t^2 \leq 9(1+\tau^2)$ for $|t-\tau| \leq 2$, (2.5) implies

$$\|\Phi\|_{C^{k,\gamma}([\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+)} \leq C'(1+\tau^2)^{q/2},$$

where C' is independent of τ.

For $T > 0$, we denote by $C^{k,\gamma}_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ the subspace of $C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ consisting of T-periodic functions,

$$f(t+T,\theta) = f(t,\theta) \qquad \text{for every } (t,\theta) \in \mathbb{R} \times \mathbb{S}^{n-1}_+,$$

equipped with the norm $\|f\|_{C^{k,\gamma}([0,T]\times\mathbb{S}^{n-1}_+)}$. The same convention applies on $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$.

The subscripts sym, even and odd indicate cylindrical symmetry, evenness in t and oddness in t, respectively. On $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$, the subscript sym means independence of $\vartheta \in \mathbb{S}^{n-2}$. We combine these subscripts with T when imposing the corresponding conditions on T-periodic functions. Each condition defines a closed subspace. When v is T-periodic, cylindrically symmetric and even in t, the operators $\mathcal{A}$ and $\mathcal{B}$ preserve the period and the indicated symmetries.

3. The Delaunay family

We prove Theorem A for cylindrically symmetric solutions with positive two-sided trace bounds. Reflection symmetry, periodicity and uniqueness for solutions with a prescribed trace minimum at $t = 0$ are established in Section 3.1. The local parametrization and compactness give the family for every necksize in $(0, 1)$. We conclude with the Hamiltonian and the behavior of the fundamental period at both ends of the family.

3.1. **Rigidity and periodicity.**

The two-sided trace bounds control the solution on the whole cylinder. They allow us to apply linear uniqueness to differences of solutions and to Jacobi fields.

Let C_n be the solution of the Dirichlet problem for $-\Delta_{\mathbb{S}^{n-1}_+} + (n-2)^2/4$ with boundary value 1. The Dirichlet form is coercive, and the maximum principle gives $C_n > 0$. Uniqueness gives cylindrical symmetry. The constant-mode Dirichlet-to-Neumann symbol at zero frequency in [13, Theorem 1.1], with boundary dimension $n-1$ and order $1/2$, equals c (see (3.49)). Thus, with β, L and ω as in Section 2.1, C_n satisfies

$$\begin{cases} (\omega C_n')' = \dfrac{(n-2)^2}{4}\omega C_n, \\ C_n(L) = 1, \qquad C_n'(L) = c. \end{cases} \tag{3.1}$$

Smoothness and cylindrical symmetry give $C_n'(0) = 0$, as in Section 2.1.

Let $\mathscr{B}_n$ be the class of positive cylindrically symmetric solutions of (2.1) that are smooth up to $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$ and satisfy, for some $m, M > 0$,

$$m \le \tilde{v}(t) \le M \qquad \text{for every } t \in \mathbb{R}. \tag{3.2}$$

For $v \in \mathscr{B}_n$, set

$$\widehat{v} = c^{\frac{n-2}{2}} v, \qquad \widehat{v}(t, L) = c^{\frac{n-2}{2}} \tilde{v}(t). \tag{3.3}$$

Then (2.1) gives

$$\begin{cases} \mathcal{A}\widehat{v} = 0 & \text{in } \mathbb{R} \times \mathbb{S}^{n-1}_+, \\ \partial_\beta \widehat{v}(t, L) = \widehat{v}(t, L)^{\frac{n}{n-2}} & \text{for } t \in \mathbb{R}. \end{cases} \tag{3.4}$$

Applying Proposition A.1 to $\widehat{v}$ and dividing by the factor in (3.3) gives

$$mC_n(\beta) \le v(t, \beta) \le MC_n(\beta) \qquad \text{for every } (t, \beta) \in \mathbb{R} \times [0, L]. \tag{3.5}$$

Let v_1 and v_2 be bounded solutions of $\mathcal{A}v_i = 0$ for $i = 1, 2$ in $\mathbb{R} \times \mathbb{S}^{n-1}_+$ whose traces $\tilde{v}_1$ and $\tilde{v}_2$ depend only on t. If $\tilde{v}_1 \le \tilde{v}_2$ on $\mathbb{R}$, the maximum principle argument in the proof of Proposition A.1, applied to $v_1 - v_2$,

gives $v_1 \le v_2$ in $\mathbb{R}\times\mathbb{S}^{n-1}_+$. Set $\eta = \|\tilde v_1 - \tilde v_2\|_{L^\infty(\mathbb{R})}$. The traces of $v_2 \pm \eta C_n$ are $\tilde v_2 \pm \eta$, so $v_2 - \eta C_n \le v_1 \le v_2 + \eta C_n$. Equivalently,

$$|v_1(t,\theta) - v_2(t,\theta)| \le \|\tilde v_1 - \tilde v_2\|_{L^\infty(\mathbb{R})} C_n(\theta). \tag{3.6}$$

If h is a period of $\tilde v$, then $v(t+h,\theta)$ and $v(t,\theta)$ are bounded solutions with the same trace, so they coincide by (3.6). The converse follows by restriction to $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$. Thus

$$h \text{ is a period of } v \quad \text{if and only if} \quad h \text{ is a period of } \tilde v. \tag{3.7}$$

If $\tilde v$ is constant, say $\tilde v \equiv a > 0$, then (3.6) gives $v = aC_n$. The boundary condition in (2.1) and the value of $C_n'(L)$ in (3.1) give

$$ac = aC_n'(L) = \partial_\beta v(t,L) = c\, a^{\frac{n}{n-2}}.$$

Dividing by ac gives $a^{2/(n-2)} = 1$, so $a = 1$. In particular, C_n is the unique cylindrical solution in $\mathscr{B}_n$.

Proposition A.2, applied to $\widehat v = c^{\frac{n-2}{2}} v$, bounds the trace uniformly over $\mathscr{B}_n$. Indeed, the nonnegative derivative term in (A.2) gives $c^2 \tilde v(t)^{\frac{4}{n-2}} \le (n-2)^2/4$. Thus

$$\tilde v(t) \le M_n := \left(\frac{n-2}{2c}\right)^{\frac{n-2}{2}} \qquad \text{for every } t\in\mathbb{R}. \tag{3.8}$$

Fix $\delta > 0$. For every $v \in \mathscr{B}_n$ with $\inf_{\mathbb{R}} \tilde v \ge \delta$, the bounds in (3.5) hold with $m = \delta$ and $M = M_n$. Since C_n is positive on $\mathbb{S}^{n-1}_+$, these bounds control v above and away from zero in terms of n and δ. By Proposition 2.1, for every integer $k \ge 0$ and $0 < \gamma < 1$ there is a constant $C = C(n,\delta,k,\gamma)$ such that

$$\sup_{\tau\in\mathbb{R}} \|v\|_{C^{k,\gamma}([\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+)} \le C. \tag{3.9}$$

Let $v_1, v_2 \in \mathscr{B}_n$ and set $w = \widehat v_1 - \widehat v_2$. By the fundamental theorem of calculus, the difference of the boundary nonlinearities is $b(t)w(t,L)$, where

$$b(t) = \frac{n}{n-2}\int_0^1 \Big((1-s)\widehat v_1(t,L) + s\widehat v_2(t,L)\Big)^{\frac{2}{n-2}}\, ds. \tag{3.10}$$

This formula also defines $b(t)$ where the traces agree. Subtracting the equations for $\widehat v_1$ and $\widehat v_2$ gives

$$\begin{cases} \mathcal{A} w = 0 & \text{in } \mathbb{R}\times\mathbb{S}^{n-1}_+, \\ \partial_\beta w(t,L) = b(t) w(t,L) & \text{on } \mathbb{R}\times\partial\mathbb{S}^{n-1}_+. \end{cases} \tag{3.11}$$

When $v_1 = v_2 = v$, the coefficient in (3.10) is $\frac{n}{n-2}\widehat v(t,L)^{2/(n-2)} = \frac{n}{n-2} c\tilde v(t)^{2/(n-2)}$. Thus a cylindrically symmetric Jacobi field Φ along v

satisfies

(3.12)
$$\begin{cases} \mathcal{A}\Phi = 0 & \text{in } \mathbb{R}\times\mathbb{S}^{n-1}_+, \\ \mathcal{B}\Phi = \partial_\beta\Phi(t,L) - \dfrac{n}{n-2}\widehat{v}(t,L)^{\frac{2}{n-2}}\Phi(t,L) = 0 & \text{on } \mathbb{R}\times\partial\mathbb{S}^{n-1}_+. \end{cases}$$

Proposition 3.1. *Let $n \geq 3$.*

(i) *Let $v \in \mathscr{B}_n$. If $\tilde{v}'(t_0) = 0$ at some $t_0 \in \mathbb{R}$, then*
$$v(t_0 + t, \theta) = v(t_0 - t, \theta) \qquad \text{for every } (t,\theta) \in \mathbb{R}\times\mathbb{S}^{n-1}_+.$$

(ii) *Let $v_1, v_2 \in \mathscr{B}_n$ be even in t and satisfy $\tilde{v}_1(0) = \tilde{v}_2(0)$. Then $v_1 = v_2$ on $\mathbb{R}\times\mathbb{S}^{n-1}_+$.*

(iii) *Let $v \in \mathscr{B}_n$ be even in t, and let Φ be a cylindrically symmetric Jacobi field along v with polynomial growth in the sense of (2.5). If Φ is odd in t and $\partial_t\Phi(0,L) = 0$, then $\Phi \equiv 0$. If Φ is even in t and $\Phi(0,L) = 0$, then $\Phi \equiv 0$.*

Proof. The trace bound in (3.8) gives $\widehat{v}(t,L) \leq ((n-2)/2)^{(n-2)/2}$. Thus the boundary coefficients in (3.10) and (3.12) satisfy

$$0 < b(t) \leq \frac{n}{2} \qquad (t \in \mathbb{R}). \tag{3.13}$$

For $n \geq 5$, Corollary A.4 also gives $|b'| \leq b$ for the coefficients in the three cases below. For the reflected difference and for the Jacobi equation along an even solution, (A.16) gives $|b'(t)|/b(t) \leq (n-2)t/2$ for $t \geq 0$, since $n \geq 5$ and $\tanh s \leq s$ for $s \geq 0$.

For (i), set
$$w_{t_0}(t,\theta) = \widehat{v}(t_0 + t, \theta) - \widehat{v}(t_0 - t, \theta).$$
The two functions solve (3.4) and have the same trace bounds as $\widehat{v}$. The change $t \mapsto -t$ interchanges their traces, and the change $s \mapsto 1 - s$ in (3.10) shows that b is even. Since $\partial_t\widehat{v}(t_0, L) = 0$, we have

$$w_{t_0}(-t,\theta) = -w_{t_0}(t,\theta), \qquad \partial_t w_{t_0}(0,L) = 2\partial_t\widehat{v}(t_0,L) = 0. \tag{3.14}$$

By (3.9), w_{t_0} and its first derivatives are bounded. The equation for w_{t_0} and (3.14) give $w_{t_0} = 0$ by Lemma B.1(i) if $n \geq 5$, and by Theorem C.1(i) if $n = 3, 4$. Hence v is symmetric about $t = t_0$.

For (ii), $w = \widehat{v}_1 - \widehat{v}_2$ and b are even in t. The two traces agree at $t = 0$, so $w(0,L) = 0$. The function w and its first derivatives are bounded by (3.9). Applying Lemma B.1(ii) if $n \geq 5$ and Theorem C.1(ii) if $n = 3, 4$ gives $w = 0$, so $v_1 = v_2$.

For (iii), b is even because v is even. The positive lower trace bound and (3.9) give uniform bounds for b and its derivatives. The estimates in Section 2.3 therefore give the same order of polynomial growth for

the local Hölder norms of Φ. We apply Lemma B.1 if $n \geq 5$ and Theorem C.1 if $n = 3, 4$, using the corresponding part for the parity of Φ. □

To locate critical points of the trace, we integrate the interior equation against C_n. For $v \in \mathscr{B}_n$, set

$$I(t) = \int_0^L \omega(\beta) C_n(\beta) v(t, \beta) \, \mathrm{d}\beta. \tag{3.15}$$

If M is an upper bound for $\tilde{v}$, then (3.5) gives

$$0 < I(t) \leq M \int_0^L \omega(\beta) C_n(\beta)^2 \, \mathrm{d}\beta \qquad \text{for all } t \in \mathbb{R}.$$

Differentiating under the integral and using the interior equation in (2.1), we obtain

$$\begin{aligned} I''(t) &= \int_0^L C_n \left(\frac{(n-2)^2}{4} \omega v - \partial_\beta(\omega \partial_\beta v) \right) \, \mathrm{d}\beta \\ &= \int_0^L \left[v(\omega C_n')' - C_n \partial_\beta(\omega \partial_\beta v) \right] \, \mathrm{d}\beta \\ &= \Big[\omega(C_n' v - C_n \partial_\beta v) \Big]_{\beta=0}^{\beta=L}, \end{aligned}$$

where the second equality follows from (3.1) and the last from integration by parts. The endpoint term at $\beta = 0$ vanishes since $\omega(0) = 0$. At $\beta = L$, we have $\omega(L) = C_n(L) = 1$ and $C_n'(L) = c$. The boundary condition in (2.1) thus gives

$$I''(t) = c\Big(\tilde{v}(t) - \tilde{v}(t)^{\frac{n}{n-2}}\Big). \tag{3.16}$$

Lemma 3.2. *Let $v \in \mathscr{B}_n \setminus \{C_n\}$. Then $\tilde{v}$ takes values both below and above 1 and has at least two distinct critical points.*

Proof. If $\tilde{v} \geq 1$ on $\mathbb{R}$, then I is concave by (3.16). If $\tilde{v} \leq 1$ on $\mathbb{R}$, then I is convex. In either case, boundedness on $\mathbb{R}$ forces I to be constant. Since $c\tilde{v} > 0$, (3.16) gives $\tilde{v}^{2/(n-2)} \equiv 1$. The trace is therefore identically 1, and (3.6) gives $v = C_n$. For $v \neq C_n$, there must be $t_-, t_+ \in \mathbb{R}$ such that $\tilde{v}(t_-) < 1 < \tilde{v}(t_+)$. The corresponding conclusion for the fractional Yamabe equation under positive two-sided bounds is proved in [11, Proposition 2.5].

Suppose that $\tilde{v}$ is monotone on $[R, \infty)$. The bounds in (3.2) give a limit $\ell \in (0, \infty)$ as $t \to +\infty$. By (3.16),

$$\lim_{t \to +\infty} I''(t) = c\Big(\ell - \ell^{\frac{n}{n-2}}\Big).$$

If this limit were nonzero, there would be $\delta > 0$ and $\sigma \in \{-1, 1\}$ such that $\sigma I''(t) \geq \delta$ for all $t \geq R$, after increasing R. The integral formula for $I(t)$ based at R then gives

$$\begin{aligned}\sigma I(t) &= \sigma I(R) + (t-R)\sigma I'(R) + \int_R^t (t-s)\sigma I''(s)\, \mathrm{d}s \\ &\geq \sigma I(R) + (t-R)\sigma I'(R) + \frac{\delta}{2}(t-R)^2.\end{aligned}$$

The right-hand side tends to $+\infty$, whereas I is bounded. Thus $\ell = \ell^{\frac{n}{n-2}}$, and $\ell > 0$ gives $\ell = 1$. If $\tilde{v}$ is monotone near $-\infty$, the same argument applied to $I(-t)$ and $\tilde{v}(-t)$ gives $\tilde{v}(t) \to 1$ as $t \to -\infty$.

If $\tilde{v}$ had fewer than two critical points, its derivative would have a fixed strict sign on each tail, and $\tilde{v}$ would tend to 1 at both ends. Choose R large enough that t_- and t_+ lie in $(-R, R)$ and the values $\tilde{v}(\pm R)$ lie strictly between $\tilde{v}(t_-)$ and $\tilde{v}(t_+)$. The minimum and maximum on $[-R, R]$ are then attained at distinct interior points. Both points are critical, a contradiction. □

The following proposition proves the periodicity conjectured in [11, Section 1] at order 1/2 for solutions in $\mathscr{B}_n$ and describes the critical points of their traces.

Proposition 3.3. *Let $n \geq 3$ and $v \in \mathscr{B}_n$. If $v \neq C_n$, then v is periodic in t and has the following properties.*

(i) *At each critical point t_0 of $\tilde{v}$, we have $\tilde{v}''(t_0) \neq 0$, and v is symmetric about $t = t_0$.*
(ii) *Let t_0 be a minimum point of $\tilde{v}$, and let t_1 be the first critical point to its right. Then $T = 2(t_1 - t_0)$ is a period of v. The derivative $\tilde{v}'$ is positive between t_0 and t_1 and negative between t_1 and $t_0 + T$.*
(iii) *The next minimum after t_0 occurs at $t_0 + T$, and t_1 is the unique maximum point of the trace between these minima. The periods of v and $\tilde{v}$ are exactly the integer multiples of T.*

Proof. For $v \neq C_n$, Lemma 3.2 gives two critical points $\tau_0 < \tau_1$ of $\tilde{v}$. By Proposition 3.1(i), reflection about either point preserves v. Reflecting first about τ_0 and then about τ_1 gives

$$v\Big(t + 2(\tau_1 - \tau_0), \theta\Big) = v(t, \theta) \qquad \text{for every } (t, \theta) \in \mathbb{R} \times \mathbb{S}^{n-1}_+. \tag{3.17}$$

At any critical point t_0, the function $v_{t_0}(t,\theta) = v(t_0+t,\theta)$ is even in t. Its derivative $\Phi = \partial_t v_{t_0}$ is odd and satisfies

$$\begin{cases} \mathcal{A}\Phi = 0 & \text{in } \mathbb{R}\times\mathbb{S}^{n-1}_+, \\ \mathcal{B}_{v_{t_0}}\Phi = 0 & \text{on } \mathbb{R}\times\partial\mathbb{S}^{n-1}_+, \end{cases}$$

by differentiating (2.1). It is bounded by (3.9) and has polynomial growth with $q = 0$. Since $\Phi(t,L) = \tilde{v}'(t_0+t)$, a vanishing second derivative $\tilde{v}''(t_0)$ would give $\partial_t\Phi(0,L) = 0$. Proposition 3.1(iii) would then give $\Phi \equiv 0$, making v independent of t and hence equal to C_n. Thus $\tilde{v}''(t_0) \neq 0$. In particular, every zero of $\tilde{v}'$ is simple and isolated.

By periodicity, $\tilde{v}$ attains its minimum at some t_0, and adding a positive period to t_0 gives another minimum. The set of critical points is closed, and t_0 is isolated in it. Hence there is a first critical point t_1 to the right of t_0.

Since $\tilde{v}''(t_0) > 0$, the derivative is positive to the right of t_0 for t sufficiently close to t_0. It stays positive between t_0 and t_1 because there is no critical point between them. Differentiating the reflection identity about t_1 gives

$$\tilde{v}'(2t_1 - t) = -\tilde{v}'(t).$$

Consequently the derivative is negative between t_1 and $2t_1 - t_0$. The reflections about t_0 and t_1 give the period $T = 2(t_1 - t_0)$ as in (3.17), and $2t_1 - t_0 = t_0 + T$.

The trace thus increases from its minimum at t_0 to its maximum at t_1 and returns to its minimum at $t_0 + T$. By periodicity, the minimum points are $t_0 + jT$ for integers j. A period of $\tilde{v}$ sends t_0 to one of these points, so it is an integer multiple of T. Conversely, every integer multiple of T is a period of v and of $\tilde{v}$. □

If the trace of a noncylindrical solution has a minimum at t_0, we use $v(t_0+t,\theta)$ to place this minimum at $t = 0$. By Proposition 3.3(iii), the difference between any two minimum points is a period of v, so all choices give the same centered solution. The lower bound in (3.2) and the trace value below 1 from Lemma 3.2 place the minimum in $(0,1)$. We call it the necksize and write

$$\varepsilon = \tilde{v}(0) = \min_{t\in\mathbb{R}} \tilde{v}(t). \tag{3.18}$$

By Proposition 3.3(i)–(ii), the centered solution is even in t, and its trace increases strictly from its minimum at 0 to its maximum at $T/2$.

3.2. **Parametrization by necksize.** We use the minimum of the trace to parametrize centered solutions, first locally and then on all of $(0,1)$.

Fix a centered noncylindrical solution $v_0 \in \mathscr{B}_n$ with necksize $\varepsilon_0 \in (0,1)$ and fundamental period T as in Proposition 3.3. Let M_T be the compact cylinder obtained from $[0,T] \times \mathbb{S}^{n-1}_+$ by identifying $(0,\theta)$ with (T,θ) for each $\theta \in \mathbb{S}^{n-1}_+$. We equip M_T with the product metric and use t for the cylindrical coordinate on its periodic lift. For $0 < \gamma < 1$, set

$$\begin{aligned} \mathscr{X}_\gamma &= C^{2,\gamma}_{T,\mathrm{sym},\mathrm{even}}(\mathbb{R} \times \mathbb{S}^{n-1}_+), \\ \mathscr{X}^+_\gamma &= \{v \in \mathscr{X}_\gamma : \min_{[0,T]} \tilde{v} > 0\}, \\ \mathscr{Y}^0_\gamma &= C^{0,\gamma}_{T,\mathrm{sym},\mathrm{even}}(\mathbb{R} \times \mathbb{S}^{n-1}_+) \times C^{1,\gamma}_{T,\mathrm{sym},\mathrm{even}}(\mathbb{R} \times \partial\mathbb{S}^{n-1}_+), \\ \mathscr{Y}_\gamma &= \mathscr{Y}^0_\gamma \times \mathbb{R}. \end{aligned} \tag{3.19}$$

The set $\mathscr{X}^+_\gamma$ is open in $\mathscr{X}_\gamma$ because restriction to the compact boundary ∂M_T is continuous in the uniform norm.

The factor $\lambda > 0$ allows the period to vary from T to λT. On M_T, we define $\mathfrak{F} : \mathscr{X}^+_\gamma \times (0,\infty) \times \mathbb{R} \to \mathscr{Y}_\gamma$ by

$$\mathfrak{F}(v,\lambda,\varepsilon) = \begin{pmatrix} -\lambda^{-2}\partial_t^2 v - \Delta_{\mathbb{S}^{n-1}_+} v + \dfrac{(n-2)^2}{4} v \\ \partial_\beta v(\cdot, L) - c\tilde{v}^{\frac{n}{n-2}} \\ \tilde{v}(0) - \varepsilon \end{pmatrix}. \tag{3.20}$$

In particular, $\mathfrak{F}(v_0, 1, \varepsilon_0) = 0$.

Proposition 3.4. *Let $n \geq 3$ and $0 < \gamma < 1$, and let v_0, ε_0 and T be as above. There exist an open interval $I \Subset (0,1)$ containing ε_0, an open neighborhood $\mathcal{N}$ of $(v_0, 1)$ in $\mathscr{X}^+_\gamma \times (0,\infty)$, and a smooth map $\varepsilon \mapsto (v^{(T)}_\varepsilon, \lambda_\varepsilon)$ from I into $\mathcal{N}$. At $\varepsilon = \varepsilon_0$ it takes the value $(v_0, 1)$, and its graph is given by*

$$\mathfrak{F}^{-1}(0) \cap (\mathcal{N} \times I) = \{(v^{(T)}_\varepsilon, \lambda_\varepsilon, \varepsilon) : \varepsilon \in I\}. \tag{3.21}$$

For each $\varepsilon \in I$, define

$$\begin{aligned} v_\varepsilon(t,\theta) &= v^{(T)}_\varepsilon(t/\lambda_\varepsilon, \theta), \\ T_\varepsilon &= \lambda_\varepsilon T. \end{aligned} \tag{3.22}$$

Then $v_\varepsilon \in \mathscr{B}_n$ is centered and noncylindrical, with necksize ε. Its trace first returns to its minimum at $t = T_\varepsilon$, which is the fundamental period of v_ε.

For every integer $k \geq 0$, the map $\varepsilon \mapsto (v^{(T)}_\varepsilon, \lambda_\varepsilon)$ is smooth with values in $C^{k,\gamma}_{T,\mathrm{sym},\mathrm{even}}(\mathbb{R} \times \mathbb{S}^{n-1}_+) \times (0,\infty)$.

Proof. Restriction to $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$ defines a bounded linear map

$$\mathscr{X}_\gamma \ni v \longmapsto \tilde{v} \in C^{1,\gamma}_{T,\mathrm{sym},\mathrm{even}}(\mathbb{R} \times \partial\mathbb{S}^{n-1}_+).$$

For v in a sufficiently small neighborhood of any point of $\mathscr{X}_\gamma^+$, the trace values lie in a compact subinterval of $(0,\infty)$. Taylor's formula and the multiplication estimate in the boundary Hölder space therefore give smoothness of $v \mapsto c\,\tilde{v}^{n/(n-2)}$. The remaining terms of $\mathfrak{F}$ are linear in (v,ε) with coefficients smooth in $\lambda > 0$, so $\mathfrak{F}$ is smooth.

Let $\mathcal{A}$ be the interior operator and $\mathcal{B}_{v_0}$ the linearized boundary operator at v_0, as in Section 2.3. The derivative of $\mathfrak{F}$ with respect to (v,λ) at $(v_0,1,\varepsilon_0)$ is the bounded linear operator $\mathcal{D} : \mathscr{X}_\gamma \times \mathbb{R} \to \mathscr{Y}_\gamma$ given by

$$\mathcal{D}(z,\mu) = \Big(\mathcal{A}z + 2\mu\partial_t^2 v_0,\ \mathcal{B}_{v_0} z,\ z(0,L)\Big). \tag{3.23}$$

The kernel of $\mathcal{D}$ consists of the pairs (z,μ) satisfying

$$\begin{aligned} \mathcal{A}z &= -2\mu\partial_t^2 v_0, \\ \mathcal{B}_{v_0} z &= 0, \\ z(0,L) &= 0. \end{aligned} \tag{3.24}$$

Differentiating (2.1) in t shows that $\partial_t v_0$ is a Jacobi field along v_0. The product rule gives

$$\begin{aligned} \mathcal{A}(t\partial_t v_0) &= -2\partial_t^2 v_0, \\ \mathcal{B}_{v_0}(t\partial_t v_0) &= 0. \end{aligned} \tag{3.25}$$

Thus, for $(z,\mu) \in \ker \mathcal{D}$, the function $\Phi = z - \mu t\partial_t v_0$ satisfies the Jacobi equation along v_0. The positivity and smoothness of $\tilde{v}_0$ give smooth coefficients in the boundary condition. Starting from $\Phi \in C^{2,\gamma}_{\mathrm{loc}}$, successive Schauder estimates give smoothness up to $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$.

Both terms in Φ are cylindrically symmetric. The function z is even, and $t\partial_t v_0$ is the product of two odd functions, so Φ is even. The correction vanishes at $t = 0$, giving $\Phi(0,L) = z(0,L) = 0$. Since z and $\partial_t v_0$ are continuous and T-periodic, they are bounded. Consequently,

$$|\Phi(t,\theta)| \le \|z\|_{C^0} + |\mu|\,|t|\,\|\partial_t v_0\|_{C^0} \le C(1+t^2)^{1/2}.$$

Thus Φ has polynomial growth of order at most 1 in the sense of (2.5), and Proposition 3.1(iii) gives $\Phi = 0$. Using $z = \mu t\partial_t v_0$ and periodicity, we obtain

$$0 = z(t+T,\theta) - z(t,\theta) = \mu T\partial_t v_0(t,\theta).$$

Since v_0 is noncylindrical, $\partial_t v_0$ is nonzero somewhere. Hence $\mu = 0$ and $z = 0$.

Consider the Neumann operator

$$\mathcal{P} : \mathscr{X}_\gamma \to \mathscr{Y}^0_\gamma, \qquad \mathcal{P}z = (\mathcal{A}z, \partial_\beta z(\cdot, L)).$$

For $(f,g) \in \mathscr{Y}_\gamma^0$, periodicity allows us to regard f and g as functions on M_T and ∂M_T, respectively. On M_T, the associated bilinear form is

$$(z,\varphi) \longmapsto \int_{M_T} \left(\langle \nabla z, \nabla \varphi \rangle + \frac{(n-2)^2}{4} z\varphi \right) \mathrm{d}V.$$

It is bounded and coercive on $H^1(M_T)$ because its diagonal terms control the L^2 norms of ∇z and z. The interior and boundary data define a bounded linear functional on $H^1(M_T)$ by the trace theorem. Lax–Milgram gives a unique weak solution.

Schauder regularity for the Neumann problem gives $z \in C^{2,\gamma}(M_T)$, since $f \in C^{0,\gamma}(M_T)$ and $g \in C^{1,\gamma}(\partial M_T)$. The reflection $t \mapsto -t$ and the rotations defining cylindrical symmetry preserve $\mathcal{A}$, the normal derivative and both data. By uniqueness, the solution is invariant under these isometries. Its periodic lift therefore belongs to $\mathscr{X}_\gamma$. Thus $\mathcal{P}$ is bijective, and the bounded inverse theorem makes it an isomorphism.

The operator $\mathcal{P} \oplus \mathrm{Id}_{\mathbb{R}}$ is an isomorphism from $\mathscr{X}_\gamma \times \mathbb{R}$ to $\mathscr{Y}_\gamma^0 \times \mathbb{R} = \mathscr{Y}_\gamma$. By (3.23), the difference between $\mathcal{D}$ and this isomorphism is

$$(z,\mu) \longmapsto \left(2\mu \partial_t^2 v_0,\ -\frac{n}{n-2} c\, \tilde{v}_0^{\frac{2}{n-2}} \tilde{z},\ z(0,L) - \mu \right).$$

The first and last components have finite rank. The boundary component is compact because restriction and multiplication by the smooth coefficient are bounded into $C^{2,\gamma}(\partial M_T)$, which embeds compactly into $C^{1,\gamma}(\partial M_T)$. Hence the difference is compact. By stability of the Fredholm index under compact perturbations [5, § 6.4, item 1(c)], $\mathcal{D}$ is Fredholm of index zero. Since its kernel is trivial, its range has codimension zero in $\mathscr{Y}_\gamma$, so $\mathcal{D}$ is surjective. The bounded inverse theorem makes $\mathcal{D}$ an isomorphism. The implicit function theorem gives (3.21).

Since $\min_{M_T} v_0 > 0$, we may shrink $\mathcal{N}$ and I so that each $v_\varepsilon^{(T)}$ is positive on M_T and close to v_0 in $C^2(M_T)$.

By Proposition 3.3(i)–(ii), $\tilde{v}_0''(0) > 0$, $\tilde{v}_0''(T/2) < 0$, and $\tilde{v}_0'$ is positive on $(0, T/2)$. For $v \in \mathscr{X}_\gamma$, evenness and periodicity give $\tilde{v}'(0) = \tilde{v}'(T/2) = 0$. When v is sufficiently close to v_0 in $C^2(M_T)$, the signs of $\tilde{v}_0''$ persist near 0 and $T/2$. Integrating from these two points gives $\tilde{v}' > 0$ near both ends of $(0, T/2)$. On the compact interval between those neighborhoods, $\tilde{v}_0'$ has a positive lower bound, so $\tilde{v}' > 0$ there as well. Reflection about $T/2$ gives strict decrease on $(T/2, T)$. Applied to $v_\varepsilon^{(T)}$, this shows that the trace has its minimum at 0, where it equals ε by (3.20).

The function v_ε defined in (3.22) solves (2.1) because $\mathfrak{F}(v_\varepsilon^{(T)}, \lambda_\varepsilon, \varepsilon) = 0$. Since $\tilde{v}_\varepsilon \geq \varepsilon > 0$, the boundary term $c\, \tilde{v}_\varepsilon^{n/(n-2)}$ has the Hölder regularity of $\tilde{v}_\varepsilon$. Successive applications of the boundary Schauder estimates give

smoothness up to $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$. The function v_ε is positive and cylindrically symmetric, and its periodic trace is bounded with minimum $\varepsilon>0$. Thus $v_\varepsilon\in\mathscr{B}_n$.

The trace attains its minimum at 0 and is strictly greater than ε on $(0,T_\varepsilon)$. Since v_ε is T_ε-periodic, this shows that it is centered and noncylindrical and that T_ε is its fundamental period.

It remains to prove smooth dependence in the higher Hölder norms on M_T. Let $\mathcal{D}_\varepsilon$ be the derivative of $\mathfrak{F}$ with respect to (v,λ) at $(v_\varepsilon^{(T)},\lambda_\varepsilon,\varepsilon)$. Smoothness of $\mathfrak{F}$ and continuity of the family in $\mathscr{X}_\gamma\times(0,\infty)$ give $\mathcal{D}_\varepsilon\to\mathcal{D}$ in operator norm as $\varepsilon\to\varepsilon_0$. We may therefore restrict I so that

$$\left\|\mathcal{D}^{-1}(\mathcal{D}_\varepsilon-\mathcal{D})\right\|<1\qquad\text{for every }\varepsilon\in I.$$

The operator $\mathcal{D}^{-1}\mathcal{D}_\varepsilon$ is the identity plus $\mathcal{D}^{-1}(\mathcal{D}_\varepsilon-\mathcal{D})$, so it is invertible by the Neumann series. It follows that $\mathcal{D}_\varepsilon:\mathscr{X}_\gamma\times\mathbb{R}\to\mathscr{Y}_\gamma$ is an isomorphism throughout I.

Fix $\varepsilon_*\in I$ and $k\geq 2$. For $v\in C^{k,\gamma}_{T,\mathrm{sym},\mathrm{even}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ with positive trace, the formula (3.20) defines a smooth map with values in

$$C^{k-2,\gamma}_{T,\mathrm{sym},\mathrm{even}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)\times C^{k-1,\gamma}_{T,\mathrm{sym},\mathrm{even}}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)\times\mathbb{R}.$$

The smoothness follows from the same composition argument. Let $f\in C^{k-2,\gamma}(M_T)$ and $g\in C^{k-1,\gamma}(\partial M_T)$ be even and cylindrically symmetric, and let $a\in\mathbb{R}$. Since $(f,g,a)\in\mathscr{Y}_\gamma$, the isomorphism $\mathcal{D}_{\varepsilon_*}$ gives a unique pair $(z,\mu)\in\mathscr{X}_\gamma\times\mathbb{R}$ satisfying $\mathcal{D}_{\varepsilon_*}(z,\mu)=(f,g,a)$. Surjectivity in the higher Hölder spaces follows once $z\in C^{k,\gamma}(M_T)$. The interior equation for z is uniformly elliptic, and its right-hand side differs from f by a multiple of the smooth function $\partial_t^2 v_{\varepsilon_*}^{(T)}$. The boundary condition expresses the normal derivative as g plus the trace of z multiplied by a smooth function. Thus, if $z\in C^{j,\gamma}(M_T)$ for $2\leq j<k$, the interior right-hand side lies in $C^{j-1,\gamma}(M_T)$ and the normal derivative lies in $C^{j,\gamma}(\partial M_T)$. Schauder regularity then gives $z\in C^{j+1,\gamma}(M_T)$. Starting from $z\in C^{2,\gamma}(M_T)$ gives $z\in C^{k,\gamma}(M_T)$ by iteration. Injectivity follows from uniqueness in $\mathscr{X}_\gamma\times\mathbb{R}$, and the bounded inverse theorem gives an isomorphism in the higher Hölder spaces.

The implicit function theorem now gives a smooth family in $C^{k,\gamma}_{T,\mathrm{sym},\mathrm{even}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)\times(0,\infty)$ near ε_*. It passes through $(v_{\varepsilon_*}^{(T)},\lambda_{\varepsilon_*})$ and, by continuous embedding into $\mathscr{X}_\gamma\times(0,\infty)$, lies in $\mathcal{N}$ for ε close to ε_*. The graph identity (3.21) identifies the two families. Since ε_* was arbitrary, this proves smooth dependence on I for $k\geq 2$. The continuous inclusions from $C^{2,\gamma}$ into $C^{k,\gamma}$ give the cases $k=0,1$. □

The local parametrization gives openness of the set of realized neck-sizes, and the uniform estimates give relative closedness in $(0,1)$.

Theorem A. *Let $n \geq 3$. For every $\varepsilon \in (0,1)$ there is a unique solution $v_\varepsilon \in \mathscr{B}_n$ satisfying*

$$\tilde{v}_\varepsilon(0) = \min_{t\in\mathbb{R}} \tilde{v}_\varepsilon(t) = \varepsilon. \tag{3.26}$$

The solution v_ε is noncylindrical, even in t, and periodic. Denote its fundamental period by T_ε.

(i) *Let M_n be the upper bound in (3.8). The trace attains its maximum at $T_\varepsilon/2$, and this maximum is strictly greater than 1. For every $t \in \mathbb{R}$,*

$$\varepsilon \leq \tilde{v}_\varepsilon(t) \leq \tilde{v}_\varepsilon(T_\varepsilon/2) \leq M_n. \tag{3.27}$$

The corresponding bounds on the cylinder are

$$\varepsilon C_n(\theta) \leq v_\varepsilon(t,\theta) \leq M_n C_n(\theta) \qquad \text{for all } (t,\theta) \in \mathbb{R}\times\mathbb{S}^{n-1}_+.$$

(ii) *If t_1 is the first positive critical point of $\tilde{v}_\varepsilon$, then*

$$T_\varepsilon = 2t_1. \tag{3.28}$$

The periods of v_ε and $\tilde{v}_\varepsilon$ are precisely the integer multiples of T_ε. The critical points of $\tilde{v}_\varepsilon$ consist of the minimum points $t_j^{\min}$ and the maximum points $t_j^{\max}$, given by

$$\begin{aligned} t_j^{\min} &= jT_\varepsilon, \\ t_j^{\max} &= \left(j+\tfrac{1}{2}\right)T_\varepsilon, \end{aligned} \qquad j \in \mathbb{Z}. \tag{3.29}$$

We have $\tilde{v}_\varepsilon''(0) > 0$ and $\tilde{v}_\varepsilon''(T_\varepsilon/2) < 0$. Moreover,

$$\tilde{v}_\varepsilon'(t) = -\tilde{v}_\varepsilon'(T_\varepsilon - t) > 0 \qquad \text{for } 0 < t < T_\varepsilon/2. \tag{3.30}$$

(iii) *The only cylindrical solution in $\mathscr{B}_n$ is C_n, and we set $v_1 = C_n$. Every noncylindrical solution $v \in \mathscr{B}_n$ has the form*

$$v(t,\theta) = v_\varepsilon(t+\tau,\theta) \tag{3.31}$$

for a unique $\varepsilon \in (0,1)$ and some $\tau \in \mathbb{R}$. Two choices of τ give the same solution exactly when their difference is an integer multiple of T_ε.

(iv) *Define the 1-periodic function $v_\varepsilon^{(1)}$ by*

$$v_\varepsilon^{(1)}(s,\theta) = v_\varepsilon(T_\varepsilon s,\theta).$$

For every integer $k \geq 0$ and $0 < \gamma < 1$, the map

$$\begin{aligned} (0,1) &\longrightarrow (0,\infty)\times C^{k,\gamma}_{1,\mathrm{sym},\mathrm{even}}(\mathbb{R}\times\mathbb{S}^{n-1}_+), \\ \varepsilon &\longmapsto (T_\varepsilon, v_\varepsilon^{(1)}) \end{aligned} \tag{3.32}$$

is smooth.

Proof. Let

$$E = \Big\{\varepsilon \in (0,1) : \tilde{v}(0) = \min_{t\in\mathbb{R}} \tilde{v}(t) = \varepsilon \text{ for some } v \in \mathscr{B}_n\Big\}. \tag{3.33}$$

To show that E is nonempty, we construct a periodic solution on M_T, with $T > 0$ to be chosen. Set $q = 2(n-1)/(n-2)$ and minimize

$$Q_T(V) = \frac{\displaystyle\int_{M_T} \left(|\nabla V|^2 + \frac{(n-2)^2}{4} V^2\right) \mathrm{d}\mu}{\displaystyle\left(\int_{\partial M_T} |V|^q \,\mathrm{d}S\right)^{2/q}}$$

over cylindrically symmetric functions in $H^1(M_T)$ with nonzero trace. Here $\mathrm{d}\mu$ is the product volume element and $\mathrm{d}S$ is the induced boundary measure. In this class the trace depends only on t. The trace theorem gives a trace in $H^{1/2}$ of the circle of length T, which embeds compactly in L^q for every finite q. Normalizing the denominator to 1 gives a bounded minimizing sequence in $H^1(M_T)$. Its weak limit retains the trace norm by compactness and minimizes Q_T by lower semicontinuity.

Taking the absolute value gives a nonnegative minimizer. Its Euler–Lagrange equation is the interior equation in (2.1) with $\partial_\beta V = \lambda \tilde{V}^{q-1}$ on ∂M_T, where $\lambda > 0$ is the minimum of Q_T under the trace normalization. Averaging any test function over the rotations defining cylindrical symmetry gives the weak equation for all test functions. The trace belongs to $L^r(\partial M_T)$ for every finite r, and so does $\lambda \tilde{V}^{q-1}$. For $r > n$, duality gives $L^r(\partial M_T) \hookrightarrow W^{-1/r,r}(\partial M_T)$, and Neumann regularity gives $V \in W^{1,r}(M_T)$. By Sobolev embedding, $\tilde{V}$ is Hölder continuous. Since $q > 2$, the normal derivative $\lambda \tilde{V}^{q-1}$ has the same Hölder exponent. The Schauder estimates then give $V \in C^{1,\gamma}(M_T)$ for some $0 < \gamma < 1$. The strong maximum principle and the boundary point lemma give positivity on M_T, since a zero on the boundary would have zero normal derivative. The boundary condition then gives smoothness up to ∂M_T. Multiplication by $(\lambda/c)^{1/(q-2)}$ makes its boundary condition the one in (2.1).

Among functions independent of t, the least value of Q_T is $Q_T(C_n)$. Indeed, for a fixed constant trace, its multiple of C_n minimizes the numerator by the interior equation and integration by parts. For $V_s = C_n(1 + s\cos(2\pi t/T))$, expansion of the two integrals gives

$$\frac{Q_T(V_s)}{Q_T(C_n)} = 1 + \frac{s^2}{2}\left[\frac{(2\pi/T)^2}{c}\int_0^L \omega C_n^2 \,\mathrm{d}\beta - (q-2)\right] + O(s^3).$$

Here we used (3.1) to compute $\int_0^L \omega\Big((C_n')^2 + \frac{(n-2)^2}{4} C_n^2\Big)\,\mathrm{d}\beta = c$. The coefficient of s^2 is negative for sufficiently large T. For such T the

minimizing solution depends on t and its periodic lift belongs to $\mathscr{B}_n$. Its trace is nonconstant, by uniqueness of the bounded extension in (3.6).

Integrating (3.16) over a period gives $\int_0^T (\tilde{v} - \tilde{v}^{n/(n-2)})\,\mathrm{d}t = 0$ for this normalized solution v. Since the trace is positive and nonconstant, it takes values below and above 1. Choosing the origin of t at a minimum thus gives a necksize $\varepsilon_0 \in E$.

Now fix $\varepsilon_0 \in E$ and choose $v_0 \in \mathscr{B}_n$ whose trace attains its minimum ε_0 at 0. Since $\varepsilon_0 < 1$ and the only cylindrical solution has trace 1, v_0 is noncylindrical. Proposition 3.4, applied at v_0, gives an open interval $I \subset (0,1)$ containing ε_0. For each $\varepsilon \in I$, it gives a solution in $\mathscr{B}_n$ whose trace attains its minimum ε at 0. Thus $I \subset E$, proving that E is open in $(0,1)$.

Let (ε_j) be a sequence in E converging to $\varepsilon_\infty \in (0,1)$. Choose $v_j \in \mathscr{B}_n$ whose trace attains its minimum ε_j at 0. For all sufficiently large j, we have $\varepsilon_j \geq \varepsilon_\infty/2$, so $\tilde{v}_j(t) \geq \varepsilon_\infty/2$ for every $t \in \mathbb{R}$. The uniform upper bound (3.8) gives $\tilde{v}_j(t) \leq M_n$. Applying the extension bounds in (3.5) with these two constants gives

$$\tfrac{1}{2}\varepsilon_\infty C_n(\theta) \leq v_j(t,\theta) \leq M_n C_n(\theta) \qquad \text{for all } (t,\theta) \in \mathbb{R} \times \mathbb{S}^{n-1}_+. \tag{3.34}$$

The uniform estimates in (3.9) and a diagonal argument give convergence, along a subsequence, in $C^k([-R,R] \times \mathbb{S}^{n-1}_+)$ for every $k \geq 0$ and $R > 0$ to a solution v_∞ of (2.1). The bounds and cylindrical symmetry pass to the limit, so $v_\infty \in \mathscr{B}_n$. Since $\tilde{v}_j(t) \geq \tilde{v}_j(0) = \varepsilon_j$ for all t, passing to the limit gives $\tilde{v}_\infty(t) \geq \tilde{v}_\infty(0) = \varepsilon_\infty$ for all t. Hence $\varepsilon_\infty \in E$, and E is relatively closed in $(0,1)$. By connectedness,

$$E = (0,1). \tag{3.35}$$

For any solution satisfying (3.26), the trace has derivative zero at 0, so Proposition 3.1(i) gives evenness in t. Two such solutions for the same ε have the same trace value at 0 and hence agree by Proposition 3.1(ii). Denote this unique solution by v_ε. It is noncylindrical because $\varepsilon < 1$.

Proposition 3.3 gives the periods and the critical points in (ii), including the signs of the second derivatives at 0 and $T_\varepsilon/2$. It also gives $\tilde{v}_\varepsilon'(t) > 0$ for $0 < t < T_\varepsilon/2$. Evenness and periodicity imply $\tilde{v}_\varepsilon(T_\varepsilon - t) = \tilde{v}_\varepsilon(t)$. Differentiating this identity gives (3.30).

For (i), the maximum at $T_\varepsilon/2$ is greater than 1 by Lemma 3.2 and at most M_n by (3.8). Applying (3.5) with trace bounds ε and M_n gives the bounds for v_ε on the cylinder.

Section 3.1 identifies C_n as the only cylindrical solution. For a noncylindrical solution $v \in \mathscr{B}_n$, choose t_0 at which its trace attains its minimum and set $\varepsilon = \tilde{v}(t_0)$. The function $(t,\theta) \mapsto v(t+t_0,\theta)$ has trace minimum ε at 0, so by uniqueness it equals v_ε. It follows that

$v(t,\theta) = v_\varepsilon(t - t_0, \theta)$, which gives (3.31) with $\tau = -t_0$. The parameter ε is determined by the minimum of $\tilde{v}$. If two values τ_1 and τ_2 give the same solution, the substitution $s = t + \tau_2$ shows that $\tau_1 - \tau_2$ is a period of v_ε. Conversely, any such period gives equality. By (ii), these periods are precisely the integer multiples of T_ε. This proves (iii).

For (iv), fix $\varepsilon_0 \in (0,1)$ and set $T_0 = T_{\varepsilon_0}$. Proposition 3.4, applied at v_{ε_0}, gives the smooth map $\varepsilon \mapsto (v_\varepsilon^{(T_0)}, \lambda_\varepsilon)$ near ε_0. The function $v_\varepsilon^{(T_0)}(t/\lambda_\varepsilon, \theta)$ has trace minimum ε at 0. By uniqueness it equals v_ε, and (3.22) gives $T_\varepsilon = \lambda_\varepsilon T_0$. Consequently,

$$v_\varepsilon^{(1)}(s,\theta) = v_\varepsilon(T_\varepsilon s, \theta) = v_\varepsilon^{(T_0)}(T_0 s, \theta). \tag{3.36}$$

Since T_0 is fixed, the substitution $t = T_0 s$ gives a bounded linear map from the periodic Hölder space of period T_0 to the corresponding space of period 1. The local map therefore gives smooth dependence of $(T_\varepsilon, v_\varepsilon^{(1)})$ near ε_0 in (3.32). Since ε_0 was arbitrary, this proves (iv). □

Smooth dependence in Theorem A(iv) gives compactness on the cylinder of period 1 for necksizes in a compact subinterval of $(0,1)$.

Lemma 3.5. *Let $0 < \varepsilon_- \le \varepsilon_+ < 1$ and let (ε_j) be a sequence in $[\varepsilon_-, \varepsilon_+]$. Write $v_j = v_{\varepsilon_j}$ and $T_j = T_{\varepsilon_j}$, as in Theorem A. After passing to a subsequence, $\varepsilon_j \to \varepsilon_\infty$ for some $\varepsilon_\infty \in [\varepsilon_-, \varepsilon_+]$. With $v_\infty = v_{\varepsilon_\infty}$ and $T_\infty = T_{\varepsilon_\infty}$, we have*

$$\begin{gathered} T_j \longrightarrow T_\infty, \\ v_j \longrightarrow v_\infty \quad \text{in } C^{k,\gamma}_{\mathrm{loc}}(\mathbb{R} \times \mathbb{S}^{n-1}_+). \end{gathered} \tag{3.37}$$

On the cylinder of period T_∞, we also have

$$\begin{gathered} v_{\varepsilon_j}^{(1)}(t/T_\infty, \theta) = v_j\Big((T_j/T_\infty)t, \theta\Big) \longrightarrow v_\infty(t,\theta) \\ \text{in } C^{k,\gamma}_{T_\infty,\mathrm{sym,even}}(\mathbb{R} \times \mathbb{S}^{n-1}_+). \end{gathered} \tag{3.38}$$

Both function convergences hold along the same subsequence for every integer $k \ge 0$ and every $0 < \gamma < 1$.

Proof. By compactness of $[\varepsilon_-, \varepsilon_+]$, we may pass to a subsequence for which $\varepsilon_j \to \varepsilon_\infty$ with $\varepsilon_\infty \in [\varepsilon_-, \varepsilon_+]$. The continuity of the map in (3.32) gives

$$T_j \longrightarrow T_\infty \qquad \text{and} \qquad v_{\varepsilon_j}^{(1)} \longrightarrow v_{\varepsilon_\infty}^{(1)} \quad \text{in } C^{k,\gamma}_{1,\mathrm{sym,even}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$$

for every integer $k \ge 0$ and every $0 < \gamma < 1$, along the subsequence chosen using only (ε_j). Composition with the fixed dilation $s = t/T_\infty$ preserves the symmetries and is bounded between the periodic Hölder spaces of periods 1 and T_∞. Each t-derivative contributes a factor of T_∞^{-1}, and the Hölder quotients are controlled by the Lipschitz constant

of the dilation in product coordinates. Since $v^{(1)}_{\varepsilon_\infty}(t/T_\infty,\theta) = v_\infty(t,\theta)$, this gives (3.38).

For the local convergence, fix $R > 0$ and write $K_R = [-R,R] \times \mathbb{S}^{n-1}_+$. For $|t| \le R$ and large j, both t/T_j and t/T_∞ belong to $[-2R/T_\infty, 2R/T_\infty]$. The derivatives through order $k+1$ of $v^{(1)}_{\varepsilon_j}$ converge uniformly on the cylinder over this interval, and $t/T_j \to t/T_\infty$ uniformly. In $v_j(t,\theta) = v^{(1)}_{\varepsilon_j}(t/T_j,\theta)$, each t-derivative contributes a factor T_j^{-1}. The chain rule and uniform continuity of the limiting derivatives give $v_j \to v_\infty$ in $C^{k+1}(K_R)$. On K_R, the mean value estimate in product charts bounds the Hölder seminorms of the derivatives of order k by the C^{k+1} norm. This proves the local convergence in (3.37). □

3.3. **The Hamiltonian and the limiting periods.** We record the conserved Hamiltonian of the Delaunay family and evaluate it at C_n and at a trace minimum. The even Jacobi field at C_n determines the limit of the fundamental period as $\varepsilon \to 1$. As $\varepsilon \to 0$, the period grows logarithmically.

For $v \in \mathscr{B}_n$, define the Hamiltonian divided by the area of $\mathbb{S}^{n-2}$ by

(3.39)
$$H_v(t) = \frac{1}{2}\int_0^L \omega(\beta)\left(v_t(t,\beta)^2 - v_\beta(t,\beta)^2 - \frac{(n-2)^2}{4}v(t,\beta)^2\right)\mathrm{d}\beta + \frac{c(n-2)}{2(n-1)}\tilde{v}(t)^{\frac{2(n-1)}{n-2}}.$$

This is the cylindrical form of the Hamiltonian of DelaTorre and González [13, Theorem 1.2] at order 1/2. Differentiating in t and integrating by parts in β gives

(3.40)
$$H_v'(t) = \int_0^L \omega v_t\left(v_{tt} + \omega^{-1}(\omega v_\beta)_\beta - \frac{(n-2)^2}{4}v\right)\mathrm{d}\beta - [\omega v_\beta v_t]_0^L + c\,\tilde{v}(t)^{\frac{n}{n-2}}\tilde{v}'(t).$$

The integral vanishes by the interior equation in (2.1), and the term at $\beta = 0$ vanishes since $\omega(0) = 0$. At $\beta = L$, we have $\omega(L) = 1$ and $v_t(t,L) = \tilde{v}'(t)$, so the boundary condition in (2.1) gives

(3.41)
$$H_v'(t) = \Big(c\,\tilde{v}(t)^{\frac{n}{n-2}} - v_\beta(t,L)\Big)\tilde{v}'(t) = 0.$$

We denote this constant by $H(v)$. For the extension $b_n v$, the Hamiltonian with integrals over $\mathbb{S}^{n-1}_+$ and its boundary is $|\mathbb{S}^{n-2}|b_n^2 H(v)$. The factor b_n^2 in the boundary term follows from $\frac{n-2}{2}b_n^{2/(n-2)} = c$.

Multiplying the equation for C_n in (3.1) by C_n and integrating by parts gives

$$\int_0^L \omega \left((C_n')^2 + \frac{(n-2)^2}{4} C_n^2 \right) \mathrm{d}\beta = [\omega C_n C_n']_0^L = c, \tag{3.42}$$

where we used $C_n(L) = 1$ and $C_n'(L) = c$. Hence (3.39) gives

$$H(C_n) = -\frac{c}{2(n-1)}. \tag{3.43}$$

For $0 < \varepsilon < 1$, write

$$v_\varepsilon(0, \beta) = \varepsilon C_n(\beta) + R_\varepsilon(\beta), \qquad R_\varepsilon(L) = 0. \tag{3.44}$$

By (3.1) and the product rule,

$$\begin{aligned} \int_0^L \omega \left(C_n' R_\varepsilon' + \frac{(n-2)^2}{4} C_n R_\varepsilon \right) \mathrm{d}\beta &= \int_0^L (\omega C_n' R_\varepsilon)' \, \mathrm{d}\beta \\ &= [\omega C_n' R_\varepsilon]_0^L = 0. \end{aligned} \tag{3.45}$$

The boundary terms vanish because $\omega(0) = 0$ and $R_\varepsilon(L) = 0$. By evenness in t, $\partial_t v_\varepsilon(0, \theta) = 0$ for $\theta \in \mathbb{S}_+^{n-1}$. Thus (3.39) at $t = 0$ and identities (3.42) and (3.45) give

$$\begin{aligned} H(v_\varepsilon) = & -\frac{c\varepsilon^2}{2} \left(1 - \frac{n-2}{n-1} \varepsilon^{\frac{2}{n-2}} \right) \\ & - \frac{1}{2} \int_0^L \omega \left((R_\varepsilon')^2 + \frac{(n-2)^2}{4} R_\varepsilon^2 \right) \mathrm{d}\beta < 0. \end{aligned} \tag{3.46}$$

The factor in parentheses is positive for $0 < \varepsilon < 1$.

The limiting period at C_n is determined by the Dirichlet-to-Neumann operator associated with the interior equation in (2.1).

For $\xi \in \mathbb{R}$, let A_ξ be the smooth cylindrically symmetric solution of

$$(\omega A_\xi')' = \left(\frac{(n-2)^2}{4} + \xi^2 \right) \omega A_\xi, \qquad A_\xi(L) = 1. \tag{3.47}$$

Existence, uniqueness and smooth dependence on ξ follow from the Dirichlet problem for $-\Delta_{\mathbb{S}_+^{n-1}} + (n-2)^2/4 + \xi^2$ on $\mathbb{S}_+^{n-1}$ with boundary value 1 on $\partial \mathbb{S}_+^{n-1}$.

The bounded extension of $e^{i\xi t}$ that solves the interior equation in (2.1) is $e^{i\xi t} A_\xi(\beta)$. By the conormal convention of Section 2.1,

$$\partial_\beta \left(e^{i\xi t} A_\xi(\beta) \right) \Big|_{\beta = L} = A_\xi'(L) e^{i\xi t}. \tag{3.48}$$

The Dirichlet-to-Neumann operator therefore sends $e^{i\xi t}$ to $\Theta_n(\xi) e^{i\xi t}$, where $\Theta_n(\xi) = A_\xi'(L)$. In boundary dimension $n-1$ and at order $1/2$,

the formula of DelaTorre and González [13, Theorem 1.1] for functions independent of the spherical variable gives

$$\Theta_n(\xi) = 2\left|\frac{\Gamma(\frac{n}{4}+\frac{i\xi}{2})}{\Gamma(\frac{n-2}{4}+\frac{i\xi}{2})}\right|^2, \qquad \Theta_n(0) = c. \tag{3.49}$$

Here $A_0 = C_n$, which also gives the stated value at $\xi = 0$.

The equation in (3.47) depends on ξ through ξ^2, so uniqueness gives $A_{-\xi} = A_\xi$ and $\Theta_n(-\xi) = \Theta_n(\xi)$. Green's identity for A_ξ and A_η gives

$$\Theta_n(\xi) - \Theta_n(\eta) = (\xi^2 - \eta^2)\int_0^L \omega A_\xi A_\eta \, \mathrm{d}\beta.$$

Here $A_\xi(L) = A_\eta(L) = \omega(L) = 1$ and the term at $\beta = 0$ vanishes since $\omega(0) = 0$. Dividing by $\xi - \eta$ and letting $\eta \to \xi$, we obtain

$$\Theta_n'(\xi) = 2\xi\int_0^L \omega A_\xi^2 \, \mathrm{d}\beta > 0 \qquad \text{for } \xi > 0. \tag{3.50}$$

Applying Stirling's formula to the quotient in (3.49) gives $\Theta_n(\xi)/|\xi| \to 1$ as $|\xi| \to \infty$. Since Θ_n is strictly increasing on $(0,\infty)$ and $\Theta_n(0) = c < \frac{n}{n-2}c$, there is a unique $\xi_* > 0$ such that

$$\Theta_n(\xi_*) = \frac{n}{n-2}c, \qquad T_* = \frac{2\pi}{\xi_*}. \tag{3.51}$$

For $\xi > 0$, set $Z_\xi(t,\beta) = A_\xi(\beta)\cos(\xi t)$. By (3.47), Z_ξ satisfies the interior equation $\mathcal{A}Z_\xi = 0$. Using $A_\xi(L) = 1$ and taking the real part in (3.48) gives its trace and conormal derivative on $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$,

$$Z_\xi(t,L) = \cos(\xi t), \qquad \partial_\beta Z_\xi(t,L) = \Theta_n(\xi)\cos(\xi t). \tag{3.52}$$

The boundary condition linearized at C_n, whose trace is 1, is $\partial_\beta Z(t,L) = \frac{n}{n-2}c\,Z(t,L)$. Thus Z_ξ satisfies this condition precisely when $\Theta_n(\xi) = \frac{n}{n-2}c$. By (3.51), this occurs at $\xi = \xi_*$. The Jacobi field Z_{ξ_*} has fundamental period $T_* = 2\pi/\xi_*$, as is seen from its trace. This is the period obtained in [13, Theorem 1.3] for the linearized equation at order $1/2$ in boundary dimension $n-1$ when the trace is independent of the spherical variable.

As $\varepsilon \to 1$, we compare the trace of Z_{ξ_*} with that of the difference $v_\varepsilon - C_n$ after normalization at an extremum where $|\tilde{v}_\varepsilon - 1|$ is largest. The bound for $|\partial_t \widehat{v}(t,L)|/\widehat{v}(t,L)$ in (A.2) gives the divergence of the period as $\varepsilon \to 0$.

Proposition 3.7. *Let $n \geq 3$ and let $(v_\varepsilon, T_\varepsilon)_{0<\varepsilon<1}$ be the family in Theorem A.*

(i) *For every integer* $k \geq 0$ *and every* $0 < \gamma < 1$,

$$\lim_{\varepsilon \to 1} \|v_\varepsilon - C_n\|_{C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)} = 0. \tag{3.53}$$

(ii) *Let* $\xi_* > 0$ *be the unique solution of* $\Theta_n(\xi_*) = \frac{n}{n-2}c$ *and set* $T_* = \frac{2\pi}{\xi_*}$. *Then*

$$\lim_{\varepsilon \to 1} T_\varepsilon = T_*, \qquad \lim_{\varepsilon \to 0} T_\varepsilon = +\infty. \tag{3.54}$$

Proof. Let (ε_j) be a sequence in $(0,1)$ converging to 1, and write $v_j = v_{\varepsilon_j}$ and $T_j = T_{\varepsilon_j}$. For all sufficiently large j, Theorem A gives

$$\begin{aligned} &\tfrac{1}{2} \leq \varepsilon_j \leq \tilde{v}_j(t) \leq M_n, \\ &\tfrac{1}{2} C_n(\theta) \leq v_j(t,\theta) \leq M_n C_n(\theta) \end{aligned} \qquad \text{for } (t,\theta) \in \mathbb{R}\times\mathbb{S}^{n-1}_+. \tag{3.55}$$

Set $a_j = \|\tilde{v}_j - 1\|_{L^\infty(\mathbb{R})}$. The minimum of $\tilde{v}_j$ is ε_j at $t = 0$, and its maximum is attained at $t = T_j/2$. Thus

$$a_j = \max\{1 - \varepsilon_j,\ \tilde{v}_j(T_j/2) - 1\}.$$

Since $1 - \varepsilon_j \to 0$, if a_j did not tend to zero, there would be a subsequence with $\tilde{v}_j(T_j/2) \geq 1 + \delta$ for some $\delta > 0$. The functions $v_j(t + T_j/2, \theta)$ satisfy (3.55). By Proposition 2.1, a further subsequence converges in $C^k([-R,R]\times\mathbb{S}^{n-1}_+)$ for every $k \geq 0$ and $R > 0$ to a solution $v_\infty \in \mathscr{B}_n$. The lower bound $\tilde{v}_j \geq \varepsilon_j$ passes to the limit as $\tilde{v}_\infty \geq 1$, while the bound at the maxima gives $\tilde{v}_\infty(0) \geq 1 + \delta$. Since the trace of C_n is 1, Lemma 3.2 requires $\tilde{v}_\infty$ to take a value below 1. This contradiction shows that $a_j \to 0$.

For large j, the trace bounds $0 < 1 - a_j \leq \tilde{v}_j \leq 1 + a_j$ give, by (3.5),

$$|v_j(t,\theta) - C_n(\theta)| \leq a_j C_n(\theta) \qquad \text{on } \mathbb{R}\times\mathbb{S}^{n-1}_+. \tag{3.56}$$

The difference $D_j = v_j - C_n$ satisfies $\mathcal{A}D_j = 0$, where $\mathcal{A}$ is defined in (2.4). Subtracting the boundary conditions for v_j and C_n gives

$$\partial_\beta D_j(t,L) = q_j(t) D_j(t,L), \qquad q_j(t) = \frac{n}{n-2} c \int_0^1 \left(1 - s + s\tilde{v}_j(t)\right)^{\frac{2}{n-2}} \mathrm{d}s.$$

For $0 \leq s \leq 1$, the convex combination $1 - s + s\tilde{v}_j(t)$ lies in $[1/2, M_n]$ by (3.55). The function $x \mapsto x^{2/(n-2)}$ is smooth on this interval. The uniform regularity bounds for $\tilde{v}_j$ in Proposition 2.1 therefore bound q_j in $C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R})$ for every k and γ.

For each $\tau \in \mathbb{R}$, the local Schauder estimate of Section 2.3 bounds the $C^{k,\gamma}$ norm of D_j on $[\tau - 1, \tau + 1]\times\mathbb{S}^{n-1}_+$ by its L^∞ norm on $[\tau - 2, \tau + 2]\times\mathbb{S}^{n-1}_+$. The constant is independent of j and τ by the

uniform bounds on q_j and the invariance of $\mathcal{A}$ under translation in t. Taking the supremum over τ and using (3.56), we obtain

$$\begin{aligned} \|D_j\|_{C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)} &\leq C_{k,\gamma}\|D_j\|_{L^\infty(\mathbb{R}\times\mathbb{S}^{n-1}_+)} \\ &\leq C'_{k,\gamma}a_j \longrightarrow 0. \end{aligned} \tag{3.57}$$

This proves (i).

To determine the limit of T_j, set $s_j = 0$ if $1-\varepsilon_j \geq \tilde{v}_j(T_j/2) - 1$ and $s_j = T_j/2$ otherwise. Then $|\tilde{v}_j(s_j) - 1| = a_j > 0$. Define

$$Z_j(t,\theta) = \frac{v_j(t+s_j,\theta) - C_n(\theta)}{\tilde{v}_j(s_j) - 1}, \qquad h_j(t) = Z_j(t,L). \tag{3.58}$$

Evenness gives $v_j(s_j - t,\theta) = v_j(t - s_j,\theta)$. Since $2s_j$ is either 0 or T_j, periodicity gives $v_j(t-s_j,\theta) = v_j(t+s_j,\theta)$. Hence Z_j is even in t. By (3.56), $|Z_j| \leq C_n$, and the normalization gives $h_j(0) = 1$. In particular, $h_j'(0) = 0$. The equations for D_j yield

$$\begin{cases} \mathcal{A}Z_j = 0 & \text{in } \mathbb{R}\times\mathbb{S}^{n-1}_+, \\ \partial_\beta Z_j(t,L) = q_j(t+s_j)h_j(t) & \text{for } t\in\mathbb{R}. \end{cases} \tag{3.59}$$

Dividing (3.57) by a_j gives uniform bounds for Z_j in $C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ for every k and γ. By (i) and the formula for q_j, we also have

$$q_j(\cdot + s_j) \longrightarrow \frac{n}{n-2}c \qquad \text{in } C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R})$$

for every k and γ, since the norm is invariant under translation. Every subsequence of (Z_j) therefore has a further subsequence converging in $C^k([-R,R]\times\mathbb{S}^{n-1}_+)$ for every $k\geq 0$ and $R>0$. The limit Z is bounded, even and cylindrically symmetric, and $Z(0,L) = 1$. Passing to the limit in (3.59), we obtain

$$\begin{cases} \mathcal{A}Z = 0 & \text{in } \mathbb{R}\times\mathbb{S}^{n-1}_+, \\ \partial_\beta Z(t,L) = \dfrac{n}{n-2}c\,Z(t,L) & \text{for } t\in\mathbb{R}. \end{cases} \tag{3.60}$$

The Jacobi field $Z_{\xi_*}(t,\beta) = A_{\xi_*}(\beta)\cos(\xi_* t)$ constructed above is bounded, even and cylindrically symmetric, and $Z_{\xi_*}(0,L) = 1$. The difference $Z - Z_{\xi_*}$ is therefore a bounded even Jacobi field along C_n with trace zero at $t=0$. Proposition 3.1(iii) gives $Z = Z_{\xi_*}$. This identifies every subsequential limit and yields

$$h_j \longrightarrow h, \qquad h(t) = \cos(\xi_* t), \tag{3.61}$$

in $C^k([-R,R])$ for every $k\geq 0$ and $R>0$.

Since s_j is a trace extremum, the first positive critical point of h_j is $T_j/2$ by Theorem A(ii). Fix $0<\eta<T_*/4$ and choose $r\in(0,T_*/2-\eta)$

so small that $h'' < 0$ on $[0, r]$. For large j, convergence in C^2 gives $h_j'' < 0$ on $[0, r]$. As $h_j'(0) = 0$, integration gives $h_j' < 0$ on $(0, r]$. On $[r, T_*/2 - \eta]$, the derivative h' is bounded above by a strictly negative constant, so uniform convergence gives $h_j' < 0$ there as well. Finally, $h'(T_*/2 + \eta) > 0$, and therefore $h_j'(T_*/2 + \eta) > 0$ for large j. The first positive zero of h_j' consequently satisfies

$$T_*/2 - \eta < T_j/2 < T_*/2 + \eta.$$

Since η and the sequence (ε_j) converging to 1 were arbitrary, this proves $\lim_{\varepsilon \to 1} T_\varepsilon = T_*$.

Finally, since $\tilde{v}_\varepsilon(t) = c^{-\frac{n-2}{2}} \widehat{v}_\varepsilon(t, L)$ by (3.3), the gradient bound in (A.2) gives $|\tilde{v}_\varepsilon'|/\tilde{v}_\varepsilon \leq (n-2)/2$. Integrating on $[0, T_\varepsilon/2]$ and using (3.27), we obtain

$$\log \frac{1}{\varepsilon} < \log \frac{\tilde{v}_\varepsilon(T_\varepsilon/2)}{\tilde{v}_\varepsilon(0)} \leq \frac{n-2}{4} T_\varepsilon.$$

Thus $T_\varepsilon \to +\infty$ as $\varepsilon \to 0$. □

A prescribed period T may contain several repetitions of a solution with fundamental period T_ε.

Corollary 3.8. *Let $\varepsilon \in (0, 1)$ and $T > 0$. There exists a T-periodic solution in $\mathscr{B}_n$ with necksize ε if and only if $T = mT_\varepsilon$ for some positive integer m. In this case, the solutions are precisely $v_\varepsilon(t + \tau, \theta)$ with $\tau \in \mathbb{R}$.*

Proof. By Theorem A(iii), the solutions with necksize ε are $v_\varepsilon(t + \tau, \theta)$ with $\tau \in \mathbb{R}$. They have the same periods as v_ε, so Theorem A(ii) gives the criterion for T. □

For small ε, split the increasing half-period at the first point where $\tilde{v}_\varepsilon = \delta$, with $\delta > 0$ fixed and small. The first interval gives the logarithmic term, and the second has uniformly bounded length.

Corollary 3.9. *Let $n \geq 3$ and let T_ε be the fundamental period of v_ε in Theorem A. There exist $\varepsilon_0 \in (0, 1)$ and $C > 0$, depending only on n, such that*

$$\left| T_\varepsilon - \frac{4}{n-2} \log \frac{1}{\varepsilon} \right| \leq C \qquad \text{for every } \varepsilon \in (0, \varepsilon_0).$$

Proof. By (3.27) and the boundary condition in (2.1),

$$\begin{aligned} 0 < v_\varepsilon(t, \theta) &\leq M_n C_n(\theta), \\ 0 < \tilde{v}_\varepsilon(t) &\leq M_n, \\ 0 < \partial_\beta v_\varepsilon(t, L) = c\, \tilde{v}_\varepsilon(t)^{\frac{n}{n-2}} &\leq c M_n^{\frac{n}{n-2}}. \end{aligned} \tag{3.62}$$

The Harnack inequality in the proof of Proposition A.1, applied to $c^{\frac{n-2}{2}} v_\varepsilon$ on $[t-1, t+1] \times \mathbb{S}^{n-1}_+$, compares $v_\varepsilon(t, \beta)$ with $\tilde{v}_\varepsilon(t)$, with a constant independent of t, β and ε because $c\,\tilde{v}_\varepsilon(t)^{2/(n-2)} \leq cM_n^{2/(n-2)}$. Put $a = (n-2)/2$ and set

$$J(t) = \int_0^L \omega(\beta) v_\varepsilon(t, \beta)\, \mathrm{d}\beta. \tag{3.63}$$

Integrating against ω gives

$$K^{-1}\tilde{v}_\varepsilon(t) \leq J(t) \leq K\tilde{v}_\varepsilon(t) \qquad \text{for every } t \in \mathbb{R},$$

where $K \geq 1$ depends only on n.

Integrating the interior equation in (2.1) against ω gives

$$J''(t) = a^2 J(t) - [\omega(\beta)\partial_\beta v_\varepsilon(t, \beta)]_0^L.$$

The term at $\beta = 0$ vanishes since $\omega(0) = 0$. As $\omega(L) = 1$, the boundary condition in (2.1) gives

$$J'' = (a^2 - q)J, \qquad q(t) = \frac{c\,\tilde{v}_\varepsilon(t)^{n/(n-2)}}{J(t)}.$$

Since v_ε is even in t, we have $J'(0) = 0$.

Fix $\delta \in (0, 1)$ so small that $cK\delta^{2/(n-2)} \leq a^2/2$. For $0 < \varepsilon < \delta$, the trace increases strictly from ε at $t = 0$ to a value greater than 1 at $t = T_\varepsilon/2$. Hence there is a unique $s_\varepsilon \in (0, T_\varepsilon/2)$ with $\tilde{v}_\varepsilon(s_\varepsilon) = \delta$. On $[0, s_\varepsilon]$, the lower bound for J gives

$$0 \leq q(t) \leq cK\tilde{v}_\varepsilon(t)^{2/(n-2)} \leq cK\delta^{2/(n-2)} \leq \frac{a^2}{2}.$$

Thus $J'' \geq a^2 J/2$ on this interval, and $J'(0) = 0$ gives $J' \geq 0$. Multiplying $J'' \leq a^2 J$ by $2J'$ and integrating from 0 to t also gives

$$J'(t)^2 \leq a^2\Big(J(t)^2 - J(0)^2\Big).$$

In particular, $0 \leq J' \leq aJ$ on $[0, s_\varepsilon]$.

Set $F = J' + aJ$. On $[0, s_\varepsilon]$ we have

$$aJ \leq F \leq 2aJ, \qquad F' = aF - qJ, \qquad (\log F)' = a - \frac{qJ}{F}.$$

Since $J/F \leq 1/a$ and $q \leq a^2/2$, we have $(\log F)' \geq a/2$. Integrating from t to s_ε and using $J(s_\varepsilon) \leq K\delta$ yields

$$J(t) \leq \frac{F(t)}{a} \leq \frac{F(s_\varepsilon)}{a} e^{-\frac{a}{2}(s_\varepsilon - t)} \leq 2K\delta\, e^{-\frac{a}{2}(s_\varepsilon - t)} \qquad (0 \leq t \leq s_\varepsilon).$$

The inequality $\tilde{v}_\varepsilon \le KJ$ also gives $q \le CJ^{2/(n-2)}$. Consequently,

$$\int_0^{s_\varepsilon} q(t)\,\mathrm{d}t \le C\delta^{2/(n-2)} \int_0^\infty e^{-\frac{a}{n-2}r}\,\mathrm{d}r \le C,$$

with a bound independent of ε. Integrating the identity for $(\log F)'$ now gives

$$0 \le as_\varepsilon - \log\frac{F(s_\varepsilon)}{F(0)} = \int_0^{s_\varepsilon} \frac{q(t)J(t)}{F(t)}\,\mathrm{d}t \le C.$$

Since $F(0) = aJ(0)$ and $aJ(s_\varepsilon) \le F(s_\varepsilon) \le 2aJ(s_\varepsilon)$, the trace values at 0 and s_ε give

$$\frac{\delta}{K^2\varepsilon} \le \frac{F(s_\varepsilon)}{F(0)} \le \frac{2K^2\delta}{\varepsilon}.$$

As δ is fixed in terms of n, it follows that

$$\left|s_\varepsilon - \frac{1}{a}\log\frac{1}{\varepsilon}\right| \le C \qquad \text{for } 0 < \varepsilon < \delta.$$

To bound the length of $[s_\varepsilon, T_\varepsilon/2]$, let σ_ε be the unique point in $(s_\varepsilon, T_\varepsilon/2)$ such that $\tilde{v}_\varepsilon(\sigma_\varepsilon) = 1$. Let I be as in (3.15) for $v = v_\varepsilon$. The bound $v_\varepsilon \le M_n C_n$ gives $0 < I \le C$, and symmetry at 0 and $T_\varepsilon/2$ gives $I'(0) = I'(T_\varepsilon/2) = 0$. By (A.2) and (3.62), $|\tilde{v}_\varepsilon'| \le (n-2)M_n/2$. Choose $\ell = (1-\delta)/((n-2)M_n)$. Then $\sigma_\varepsilon - s_\varepsilon \ge 2\ell$ and

$$\delta \le \tilde{v}_\varepsilon(t) \le \frac{1+\delta}{2} \qquad \text{for } s_\varepsilon \le t \le s_\varepsilon + \ell.$$

By (3.16), $I'' \ge 0$ on $[0, \sigma_\varepsilon]$ and $I'' \ge d > 0$ on $[s_\varepsilon, s_\varepsilon + \ell]$, with d depending only on n and δ. Since $I'(0) = 0$, this gives $I' \ge d\ell$ on $[s_\varepsilon + \ell, \sigma_\varepsilon]$. Integration and boundedness of I give

$$\sigma_\varepsilon - s_\varepsilon \le \ell + \frac{C}{d\ell}.$$

On $[\sigma_\varepsilon, T_\varepsilon/2]$, the trace is increasing and at least 1. Since $x \mapsto x - x^{n/(n-2)}$ is decreasing on $[1, \infty)$, I' is concave there by (3.16). Its graph lies above the chord joining $I'(\sigma_\varepsilon) \ge d\ell$ and $I'(T_\varepsilon/2) = 0$. Integrating the chord gives

$$\frac{d\ell}{2}\big(T_\varepsilon/2 - \sigma_\varepsilon\big) \le I(T_\varepsilon/2) - I(\sigma_\varepsilon) \le C.$$

Thus $T_\varepsilon/2 - s_\varepsilon$ is bounded independently of ε. Together with the estimate for s_ε, this gives the asserted inequality. □

The coefficient $4/(n-2)$ also occurs in the period estimate for the classical Yamabe family [21, Proposition 2].

4. Global classification

We use the half-space Liouville theorem to classify the solutions with a removable singularity at 0. For solutions with a nonremovable singularity, we use the classification of $\mathscr{B}_n$ in Theorem A.

4.1. **Global solutions and the Kelvin reduction.** We first classify the solutions of (1.1) whose traces $u = U(\cdot, 0)$ extend continuously across 0. We then use this classification and Kelvin inversion to obtain global two-sided bounds for $|x|^{(n-2)/2}U(x)$ when the singularity at 0 is nonremovable. These bounds and cylindrical symmetry imply that the normalized Emden–Fowler transform belongs to $\mathscr{B}_n$.

We normalize the coefficient in the boundary condition to 1 by setting

$$\widehat{U} = \left(\frac{n-2}{2}\right)^{\frac{n-2}{2}} U, \qquad \widehat{u} = \widehat{U}(\cdot, 0).$$

The boundary condition becomes $-\partial_{x_n}\widehat{U}(\bar{x}, 0) = \widehat{u}(\bar{x})^{n/(n-2)}$ on $\mathbb{R}^{n-1} \setminus \{0\}$. Thus $\widehat{U}$ satisfies the extension problem studied in [7, 16] at order $1/2$ and in boundary dimension $n-1$.

We first show that U is a bubble if $|\bar{x}|^{(n-2)/2}\widehat{u}(\bar{x}) \to 0$ as $\bar{x} \to 0$. By the annular Harnack inequality [7, Lemma 3.2, (21)], $\widehat{U}(r\theta) = o(r^{-(n-2)/2})$ uniformly for $\theta \in \mathbb{S}^{n-1}_+$.

We use the barrier argument in the proof of [7, Proposition 3.4]. With $x = r\theta$ as in Section 2.1, set

$$w_\varepsilon(r\theta) = \left(Cr^{-\frac{n-2}{4}} + \varepsilon r^{-\frac{n-2}{2}}\right)(1 - \delta\theta_n), \qquad \varepsilon > 0.$$

At $\delta = 0$, both radial powers have negative Laplacian because their exponents lie in $(2-n, 0)$. By continuity on $\mathbb{S}^{n-1}_+$ and homogeneity of each term, w_ε is positive and $\Delta w_\varepsilon < 0$ for sufficiently small $0 < \delta < 1$. On $\partial\mathbb{R}^n_+ \setminus \{0\}$,

$$-\partial_{x_n} w_\varepsilon(\bar{x}, 0) = \frac{\delta}{|\bar{x}|} w_\varepsilon(\bar{x}, 0).$$

Choose $\rho \in (0, 1)$ so that $\widehat{u}(\bar{x})^{2/(n-2)} < \delta/|\bar{x}|$ for $0 < |\bar{x}| < \rho$. Take C large enough that $w_\varepsilon \geq \widehat{U}$ on $\{x \in \mathbb{R}^n_+ : |x| = \rho\}$ for every $\varepsilon > 0$. For fixed ε, the Harnack estimate gives $\widehat{U}/w_\varepsilon \to 0$ as $x \to 0$.

If $\widehat{U}/w_\varepsilon$ exceeds 1 for $0 < |x| < \rho$, its maximum is attained either at an interior point or on the flat boundary. At an interior maximum, the product rule gives $0 = \Delta\widehat{U} \leq (\widehat{U}/w_\varepsilon)\Delta w_\varepsilon < 0$. At a maximum on the flat boundary, $-\partial_{x_n}$ of the quotient is nonnegative, whereas

$$-\partial_{x_n}\left(\frac{\widehat{U}}{w_\varepsilon}\right) = \left(\widehat{u}^{2/(n-2)} - \frac{\delta}{|\bar{x}|}\right)\frac{\widehat{U}}{w_\varepsilon} < 0.$$

Thus $\widehat{U} \le w_\varepsilon$, and letting $\varepsilon \to 0^+$ gives

$$\widehat{U}(r\theta) \le Cr^{-\frac{n-2}{4}} \qquad \text{for } 0 < r < \rho, \quad \theta \in \mathbb{S}^{n-1}_+.$$

For $0 < r < \rho/2$, the functions $r^{(n-2)/4}\widehat{U}(r\,\cdot)$ are uniformly bounded on $\{x \in \mathbb{R}^n_+ : 1/2 < |x| < 2\}$. Their boundary condition has coefficient $r^{1/2}$ in front of the nonlinearity. The local Hölder estimate used in the proof of Proposition 2.1 therefore gives uniform Hölder bounds in a neighborhood of $\mathbb{S}^{n-1}_+$. Since $s \mapsto s^{n/(n-2)}$ is Lipschitz on bounded intervals in $[0,\infty)$, the boundary condition gives uniform Hölder bounds with the same exponent for the normal derivatives of the rescaled functions. The first Neumann Schauder estimate in that proof, together with the interior gradient estimate, gives a uniform gradient bound on $\mathbb{S}^{n-1}_+$. Returning to the original variables gives

$$|\nabla \widehat{U}(r\theta)| \le Cr^{-1-\frac{n-2}{4}}.$$

The two bounds imply $\widehat{U} \in H^1(\{x \in \mathbb{R}^n : x_n > 0,\ |x| < \rho/2\})$ since $2 + (n-2)/2 < n$.

For a smooth test function supported in $|x| < \rho/2$, integrate by parts on $\{x \in \mathbb{R}^n_+ : x_n > 0,\ r < |x| < \rho/2\}$. The integral on $r\mathbb{S}^{n-1}_+$ is $O(r^{3(n-2)/4})$ by the gradient bound. The flat boundary term converges because $\widehat{u}^{n/(n-2)} = O(|\bar{x}|^{-n/4})$ is locally integrable. Letting $r \to 0$ gives the weak equation across 0. Together with smoothness away from 0, this shows that $\widehat{U}$ is a nonzero nonnegative weak solution on the whole half-space and belongs to H^1_{loc} up to the boundary. We can therefore apply the half-space Liouville theorem of Jin, Li and Xiong [16, Theorem 1.8] with boundary dimension $n-1$ and fractional order $1/2$. Returning to the original normalization, we obtain

$$u(\bar{x}) = A\Big(|\bar{x} - a|^2 + \lambda^2\Big)^{-\frac{n-2}{2}} \qquad \Big(A > 0,\ a \in \mathbb{R}^{n-1},\ \lambda > 0\Big),$$

and U is the Poisson extension of u. For $z = (a, -\lambda)$, the function $A|x - z|^{2-n}$ is bounded and harmonic on the half-space and has trace u. It therefore equals U. In particular, the singularity is removable. Conversely, if u extends continuously across 0, then $|\bar{x}|^{(n-2)/2}\widehat{u}(\bar{x}) \to 0$, and the same argument gives this formula for U.

For $U(x) = A|x - z|^{2-n}$, the boundary condition in (1.1) gives

$$(n-2)\lambda A|(\bar{x}, 0) - z|^{-n} = \frac{n-2}{2} A^{\frac{n}{n-2}}|(\bar{x}, 0) - z|^{-n}.$$

It follows that $A^{2/(n-2)} = 2\lambda$, so U has the form (1.5).

Proposition 4.1. *Let $n \geq 3$ and let U be a positive solution of (1.1), smooth on $\mathbb{R}^n_+ \setminus \{0\}$. Suppose that $u = U(\cdot, 0)$ has a nonremovable singularity at 0. Then there exist constants $0 < m \leq M < \infty$ such that*

$$m|x|^{-\frac{n-2}{2}} \leq U(x) \leq M|x|^{-\frac{n-2}{2}} \qquad \text{for every } x \in \mathbb{R}^n_+ \setminus \{0\}. \tag{4.1}$$

In cylindrical variables, the normalized Emden–Fowler transform of U, given by

$$v(t, \theta) = b_n^{-1} e^{-\frac{n-2}{2}t} U(e^{-t}\theta), \qquad (t, \theta) \in \mathbb{R} \times \mathbb{S}^{n-1}_+, \tag{4.2}$$

belongs to $\mathscr{B}_n$.

Proof. By [7, Proposition 3.1], $|\bar{x}|^{(n-2)/2}\widehat{u}(\bar{x})$ is bounded above near 0. If its lower limit at 0 were zero, [7, Proposition 3.3] would give convergence to zero, and the preceding removability argument would give a continuous extension of u. Thus it is bounded away from zero near 0. For small $r > 0$, the Harnack inequality [7, Lemma 3.2, (21)] applies on

$$\{x \in \mathbb{R}^n_+ : x_n > 0,\ r/2 < |x| < 2r\},$$

with a constant independent of r. By continuity up to $\partial\mathbb{R}^n_+$, it bounds $\widehat{U}(r\theta)$ above and below by positive multiples of $\widehat{u}(r\vartheta)$, uniformly for $\theta \in \mathbb{S}^{n-1}_+$ and $\vartheta \in \mathbb{S}^{n-2}$. Since $\widehat{U}$ is a fixed positive multiple of U, the bounds for $|\bar{x}|^{(n-2)/2}\widehat{u}(\bar{x})$ give (4.1) near 0.

Take $a = 0$ and $\rho = 1$ in (2.2). The function

$$U_{0,1}(x) = |x|^{2-n} U\left(\frac{x}{|x|^2}\right)$$

satisfies (1.1). If $U_{0,1}(\cdot, 0)$ extended continuously across 0, then $U_{0,1}$ would be a bubble of the form (1.5). For a bubble with parameters (a, λ), put $z = (a, -\lambda)$. The unit Kelvin transform gives the bubble with parameters $(a/|z|^2, \lambda/|z|^2)$. Since $(U_{0,1})_{0,1} = U$, it would follow that U is a bubble, contrary to the nonremovability of the singularity of u at 0.

The estimates near 0 therefore apply to both U and $U_{0,1}$. By Section 2.1, $U_{0,1}$ corresponds to $t \mapsto -t$. In terms of $r = e^{-t}$, this gives

$$r^{\frac{n-2}{2}} U(r\theta) = (r^{-1})^{\frac{n-2}{2}} U_{0,1}(r^{-1}\theta).$$

Choose $r_0 \in (0, 1)$ so that the local estimates hold for both functions when $0 < |x| < r_0$. The identity gives the bounds for $|x| > r_0^{-1}$. On the compact set $\{x \in \mathbb{R}^n_+ : r_0 \leq |x| \leq r_0^{-1}\}$, the function $|x|^{(n-2)/2}U(x)$ is continuous and positive. Its minimum and maximum on this set complete the bounds in (4.1).

By [7, Theorem 1.3], $\widehat{U}$ is invariant under rotations of $\bar{x}$, and so is U. Thus the function v in (4.2) is cylindrically symmetric and satisfies (2.1) by Section 2.1. Moreover, (4.1) gives

$$mb_n^{-1} \le \tilde{v}(t) \le Mb_n^{-1} \qquad \text{for every } t \in \mathbb{R}.$$

Therefore $v \in \mathscr{B}_n$. □

4.2. **Proof of Theorem B.** We recover the parameters of a bubble from $u = U(\cdot, 0)$. When the singularity at 0 is nonremovable, we use Theorem A to identify the necksize and phase. As in that theorem, we set $v_1 = C_n$.

Theorem B. *Let $n \ge 3$ and let U be a positive solution of (1.1), smooth on $\mathbb{R}^n_+ \setminus \{0\}$. Exactly one of the following holds.*

(i) *The singularity at 0 is removable, and U is the bubble*

$$U(\bar{x}, x_n) = \left(\frac{2\lambda}{|\bar{x} - a|^2 + (x_n + \lambda)^2} \right)^{\frac{n-2}{2}} \tag{4.3}$$

for unique parameters $a \in \mathbb{R}^{n-1}$ and $\lambda > 0$.

(ii) *The singularity at 0 is nonremovable. There exist a unique necksize $\varepsilon \in (0, 1]$ and a phase $\tau \in \mathbb{R}$ such that, writing $x = r\theta$ with $r > 0$ and $\theta \in \mathbb{S}^{n-1}_+$, we have*

$$U(r\theta) = b_n r^{-(n-2)/2}\, v_\varepsilon\Big(-\log r + \tau, \theta\Big). \tag{4.4}$$

For $0 < \varepsilon < 1$, the phase is unique modulo the fundamental period T_ε. For $\varepsilon = 1$, all choices of τ give the same cylindrical solution.

Conversely, both formulas give positive solutions of (1.1). The bubbles in (i) *extend smoothly across 0, and the solutions in* (ii) *have a nonremovable singularity there.*

Proof. If the singularity at 0 is removable, Section 4.1 gives the bubble in (4.3). Its trace is

$$u(\bar{x}) = U(\bar{x}, 0) = \left(\frac{2\lambda}{|\bar{x} - a|^2 + \lambda^2} \right)^{\frac{n-2}{2}}.$$

The unique maximum of u is attained at $\bar{x} = a$ and is equal to $(2/\lambda)^{(n-2)/2}$. This determines a and λ.

If the singularity at 0 is nonremovable, Proposition 4.1 shows that the normalized transform v defined in (4.2) belongs to $\mathscr{B}_n$. By Theorem A(iii), either $v = C_n$ or $v(t, \theta) = v_\varepsilon(t + \tau, \theta)$ for some $0 < \varepsilon < 1$ and $\tau \in \mathbb{R}$. The inverse transformation gives (4.4), with $\varepsilon = 1$ in the

cylindrical case. Since $v_1 = C_n$ is independent of t, every choice of τ gives the same solution in this case.

For any $\vartheta \in \mathbb{S}^{n-2}$, the normalized trace is

$$\tilde{v}(t) = b_n^{-1} e^{-\frac{n-2}{2}t} u(e^{-t}\vartheta) = \tilde{v}_\varepsilon(t+\tau).$$

Thus $\varepsilon = \min_{t\in\mathbb{R}} \tilde{v}(t)$, including the cylindrical case, where $\tilde{v}_1 \equiv 1$. Since the Emden–Fowler transformation is invertible, Theorem A(iii) gives uniqueness of τ modulo T_ε when $0 < \varepsilon < 1$.

Conversely, the bubbles in (4.3) solve (1.1) by the computation in Section 4.1 and are smooth across 0. By Section 2.1, the functions in (4.4) solve (1.1) because v_ε satisfies (2.1). For $r > 0$ and $\vartheta \in \mathbb{S}^{n-2}$, their traces are

$$u(r\vartheta) = b_n r^{-(n-2)/2}\, \tilde{v}_\varepsilon(-\log r + \tau) \geq b_n \varepsilon r^{-(n-2)/2}.$$

The lower bound tends to $+\infty$ as $r \to 0$, so the singularity at 0 is nonremovable. □

Appendix A. Estimates for the extension and its boundary trace

We establish the estimates used in Proposition 3.1. For $n \geq 3$, we bound $\widehat{v}$ on $\mathbb{R} \times \mathbb{S}^{n-1}_+$ in terms of two-sided bounds for $\widehat{v}(t, L)$ and estimate $\partial_t \widehat{v}(t, L)$. For $n \geq 5$, we also prove second-order differential inequalities for $\widehat{v}(t, L)$ and use them to estimate $b(t)$ in (A.13), the coefficient in the boundary condition for differences of solutions and for Jacobi fields.

We then consider the restriction of $\mathcal{A}$ from Section 2.3 to functions $f = f(\beta)$ with boundary condition $f'(L) = bf(L)$, where $b \geq 0$. For $n \geq 5$, we estimate the first eigenvalue and its positive eigenfunction and bound the gap between the first two eigenvalues from below. We conclude with a resolvent identity valid for all $n \geq 3$.

A.1. **Bounds for the extension.** We use bounds for $\widehat{v}(t, L)$ to obtain a comparison with C_n on $\mathbb{R} \times \mathbb{S}^{n-1}_+$. We also prove uniqueness for the bounded extension with trace $\widehat{v}(t, L)$ and obtain uniform regularity for its translates in t.

By cylindrical symmetry, the normalization $\widehat{v}$ in (3.3) depends only on t and β. With β, L and ω as in Section 2.1, (3.4) takes the form

$$\begin{cases} \widehat{v}_{tt} + \omega^{-1}(\omega \widehat{v}_\beta)_\beta - \dfrac{(n-2)^2}{4}\widehat{v} = 0 & \text{for } (t,\beta) \in \mathbb{R} \times (0, L), \\ \widehat{v}_\beta(t, L) = \widehat{v}(t, L)^{\frac{n}{n-2}} & \text{for } t \in \mathbb{R}. \end{cases} \tag{A.1}$$

Proposition A.1. *Let $n \geq 3$ and let $\widehat{v}$ be a positive smooth cylindrically symmetric solution of (A.1) on $\mathbb{R} \times \mathbb{S}^{n-1}_+$. Suppose that there are constants $m, M > 0$ such that*

$$m \leq \widehat{v}(t, L) \leq M \qquad \text{for all } t \in \mathbb{R}.$$

Then

$$mC_n(\theta) \leq \widehat{v}(t, \theta) \leq MC_n(\theta) \qquad \text{on } \mathbb{R} \times \mathbb{S}^{n-1}_+.$$

Here C_n is given by (3.1). The function $\widehat{v}$ is the unique bounded solution of $\mathcal{A}w = 0$ on $\mathbb{R} \times \mathbb{S}^{n-1}_+$ with $w(t, (\vartheta, 0)) = \widehat{v}(t, L)$ for every $(t, \vartheta) \in \mathbb{R} \times \mathbb{S}^{n-2}$, where $\mathcal{A}$ is as in Section 2.3. For every integer $k \geq 0$ and $0 < \gamma < 1$, $\widehat{v}$ belongs to $C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$, and the family $\{\widehat{v}(t + \tau, \theta) : \tau \in \mathbb{R}\}$ is precompact in $C^{k,\gamma}_{\mathrm{loc}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$.

Proof. Write the boundary condition in (A.1) as $\widehat{v}_\beta(t, L) = b(t)\widehat{v}(t, L)$, where

$$b(t) = \widehat{v}(t, L)^{\frac{2}{n-2}} \leq M^{\frac{2}{n-2}}.$$

Set $\rho = L/4$. In the coordinates (t, β, ϑ) of Section 2.1, the set of points in $\mathbb{R} \times \mathbb{S}^{n-1}_+$ at distance less than ρ from $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$ is

$$\{(t, \beta, \vartheta) : t \in \mathbb{R},\ L - \rho < \beta \leq L,\ \vartheta \in \mathbb{S}^{n-2}\},$$

since the distance to $\partial\mathbb{S}^{n-1}_+$ is $L - \beta$. Choose a fixed smooth function $\chi(\beta)$ that is zero for $\beta \leq L - \rho$ and equal to 1 for $\beta \geq L - \rho/2$. Set $X_b = b(t)\chi(\beta)\partial_\beta$, with $X_b = 0$ where $\beta \leq L - \rho$. Both X_b and

$$\operatorname{div} X_b = b(t)\Big(\chi'(\beta) + (n-2)\cot\beta\,\chi(\beta)\Big)$$

are bounded in terms of n and M. The divergence only involves differentiation in β, since X_b has no component in the t direction. At $\beta = L$, $\langle X_b, \partial_\beta\rangle = b(t)$, so $\langle \nabla\widehat{v} - \widehat{v}X_b, \partial_\beta\rangle = 0$ by (A.1). Applying the divergence theorem to $\widehat{v}\phi X_b$ in the weak formulation of (A.1) gives

$$\int_{\mathbb{R}\times\mathbb{S}^{n-1}_+} \begin{bmatrix} \langle \nabla\widehat{v} - \widehat{v}X_b, \nabla\phi\rangle \\ -\phi\langle X_b, \nabla\widehat{v}\rangle + \left(\dfrac{(n-2)^2}{4} - \operatorname{div} X_b\right)\widehat{v}\phi \end{bmatrix} \mathrm{d}\mu = 0,$$

for every $\phi \in C^\infty_c(\mathbb{R} \times \mathbb{S}^{n-1}_+)$, with support allowed to meet $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$. Here the gradient and divergence are taken with respect to the product metric of Section 2.1, with volume element $\mathrm{d}\mu$.

Let $\iota(t, \theta) = (t, (\theta_1, \ldots, \theta_{n-1}, -\theta_n))$. In the coordinate $s = \beta - L$ on $|s| < \rho$, the product metric is $\mathrm{d}t^2 + \mathrm{d}s^2 + \cos^2 s\, g_{\mathbb{S}^{n-2}}$ and $\iota(t, s, \vartheta) = (t, -s, \vartheta)$. Extend $\widehat{v}$ and the bounded scalar $\operatorname{div} X_b$ to $\iota(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ by composition with ι, and extend X_b to this set by the differential of ι. Write $\widehat{v}^{\,e}$, $(\operatorname{div} X_b)^e$ and X_b^e for the extensions. For $0 < s < \rho$,

$$X_b^e(t, s, \vartheta) = -b(t)\chi(L - s)\partial_s.$$

The two pieces defining $\widehat{v}^{\,e}$ have the same trace on $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$, so $\widehat{v}^{\,e}\in H^1_{\mathrm{loc}}(\mathbb{R}\times\mathbb{S}^{n-1})$.

Let $\psi\in C_c^\infty(\mathbb{R}\times\mathbb{S}^{n-1})$ and set

$$\phi=(\psi+\psi\circ\iota)|_{\mathbb{R}\times\mathbb{S}^{n-1}_+}.$$

Split the integral over $\mathbb{R}\times\mathbb{S}^{n-1}$ into the integrals over $\mathbb{R}\times\mathbb{S}^{n-1}_+$ and $\iota(\mathbb{R}\times\mathbb{S}^{n-1}_+)$. On the second set, change variables by ι. Since ι preserves the metric and volume element, we obtain

$$\begin{aligned}&\int_{\mathbb{R}\times\mathbb{S}^{n-1}}\left[\begin{array}{l}\langle\nabla\widehat{v}^{\,e}-\widehat{v}^{\,e}X_b^e,\nabla\psi\rangle\\ -\psi\langle X_b^e,\nabla\widehat{v}^{\,e}\rangle+\left(\dfrac{(n-2)^2}{4}-(\operatorname{div}X_b)^e\right)\widehat{v}^{\,e}\psi\end{array}\right]\mathrm{d}\mu\\ &\quad=\int_{\mathbb{R}\times\mathbb{S}^{n-1}_+}\left[\begin{array}{l}\langle\nabla\widehat{v}-\widehat{v}X_b,\nabla\phi\rangle\\ -\phi\langle X_b,\nabla\widehat{v}\rangle+\left(\dfrac{(n-2)^2}{4}-\operatorname{div}X_b\right)\widehat{v}\phi\end{array}\right]\mathrm{d}\mu=0.\end{aligned}$$

Thus $\widehat{v}^{\,e}$ satisfies a homogeneous weak equation on $\mathbb{R}\times\mathbb{S}^{n-1}$. In local coordinates, the product metric gives a uniformly elliptic principal part in divergence form and the lower-order coefficients are bounded in terms of n and M. Applying the interior Harnack inequality [24, Theorem 7] on a fixed chain of overlapping charts in $(-2,2)\times\mathbb{S}^{n-1}$, translated by τ in t, gives

$$\begin{aligned}\sup_{[\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+}\widehat{v}&\le C\inf_{[\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+}\widehat{v}\\ &\le C\widehat{v}(\tau,L)\le CM\qquad(\tau\in\mathbb{R}).\end{aligned}$$

Here $C=C(n,M)$ is independent of τ and m.

We now use the maximum principle for the interior operator $\mathcal{A}$ of Section 2.3. Let w be a bounded solution of $\mathcal{A}w=0$ on $\mathbb{R}\times\mathbb{S}^{n-1}_+$ with $w|_{\mathbb{R}\times\partial\mathbb{S}^{n-1}_+}\le 0$. Fix $0<\delta<(n-2)/2$ and $\varepsilon>0$, and set $w_\varepsilon=w-\varepsilon\cosh(\delta t)$. Then

$$\mathcal{A}w_\varepsilon=-\varepsilon\left(\frac{(n-2)^2}{4}-\delta^2\right)\cosh(\delta t)<0.$$

We have $w_\varepsilon\le 0$ on $[-R,R]\times\partial\mathbb{S}^{n-1}_+$ and on $\{-R,R\}\times\mathbb{S}^{n-1}_+$ whenever $\varepsilon\cosh(\delta R)\ge\|w\|_{L^\infty(\mathbb{R}\times\mathbb{S}^{n-1}_+)}$. The maximum principle on $[-R,R]\times\mathbb{S}^{n-1}_+$ gives $w_\varepsilon\le 0$ there for every sufficiently large R. Hence $w_\varepsilon\le 0$ on the whole cylinder. Letting $\varepsilon\to 0^+$ gives $w\le 0$.

By (3.1), $\mathcal{A}C_n=0$ and $C_n(L)=1$. The Harnack bound gives boundedness of $\widehat{v}-MC_n$ and $mC_n-\widehat{v}$, whose values at $\beta=L$ are nonpositive. The comparison gives $mC_n\le\widehat{v}\le MC_n$. If w_1,w_2 are

bounded solutions with $w_1|_{\mathbb{R}\times\partial\mathbb{S}^{n-1}_+} = w_2|_{\mathbb{R}\times\partial\mathbb{S}^{n-1}_+}$, the same argument applied to $w_1 - w_2$ and $w_2 - w_1$ gives $w_1 = w_2$.

Since $C_n > 0$ on $\mathbb{S}^{n-1}_+$, which is compact, the comparison gives

$$0 < m \min_{\mathbb{S}^{n-1}_+} C_n \le \widehat{v}(t,\theta) \le M \max_{\mathbb{S}^{n-1}_+} C_n \qquad \text{for all } (t,\theta) \in \mathbb{R}\times\mathbb{S}^{n-1}_+.$$

For $\tau \in \mathbb{R}$, set $\widehat{v}_\tau(t,\theta) = \widehat{v}(t+\tau,\theta)$. By translation invariance, $\widehat{v}_\tau$ solves (A.1) and satisfies the same bounds. Proposition 2.1 for the boundary coefficient 1 gives, for every integer $k \ge 0$ and $0 < \gamma < 1$,

$$\sup_{\tau\in\mathbb{R}} \|\widehat{v}_\tau\|_{C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)} \le C(n,m,M,k,\gamma).$$

By the compactness assertion of the same proposition, every sequence $(\widehat{v}_{\tau_j})_{j\ge1}$ has a subsequence converging in $C^{k,\gamma}_{\mathrm{loc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$. □

A.2. **Bounds for the trace.** We prove a gradient bound on $\mathbb{R}\times(0,L]$ which controls $\partial_t\widehat{v}(t,L)$ for $n \ge 3$. For $n \ge 5$, we use the gradient bound to derive differential inequalities for $\widehat{v}(t,L)$ and then estimate the boundary coefficient in (A.13). After changing the normalization, the boundary estimate below and the first-order bound for the secant coefficient also appear in [17, Proposition 2.2 and Lemma 2.3].

Proposition A.2. *Let $n \ge 3$ and let $\widehat{v}$ satisfy the hypotheses of Proposition A.1. On $\mathbb{R}\times(0,L]$, consider the metric*

$$g_* = \sin^{-2}\beta\,(\,\mathrm{d}t^2 + \mathrm{d}\beta^2).$$

Then

$$\left|\nabla_{g_*} \log\left(\sin^{\frac{n-2}{2}}\beta\,\widehat{v}\right)\right|^2_{g_*} \le \frac{(n-2)^2}{4}.$$

In particular,

$$\left(\frac{\partial_t\widehat{v}(t,L)}{\widehat{v}(t,L)}\right)^2 + \widehat{v}(t,L)^{\frac{4}{n-2}} \le \frac{(n-2)^2}{4} \qquad \text{for all } t\in\mathbb{R}. \tag{A.2}$$

Proof. The interior equation in (A.1) gives

$$(\partial_t^2 + \partial_\beta^2)\left(\sin^{\frac{n-2}{2}}\beta\,\widehat{v}\right) = \frac{(n-2)(n-4)}{4}\csc^2\beta\,\sin^{\frac{n-2}{2}}\beta\,\widehat{v}.$$

Since $\Delta_{g_*} = \sin^2\beta(\partial_t^2 + \partial_\beta^2)$, we obtain

$$\Delta_{g_*}\log\left(\sin^{\frac{n-2}{2}}\beta\,\widehat{v}\right) = \frac{(n-2)(n-4)}{4} - \left|\nabla_{g_*}\log\left(\sin^{\frac{n-2}{2}}\beta\,\widehat{v}\right)\right|^2_{g_*}. \tag{A.3}$$

Define

$$\Psi(t,\beta) = \frac{\left|\nabla_{g_*}\log\left(\sin^{\frac{n-2}{2}}\beta\,\widehat{v}\right)\right|^2_{g_*} - \frac{(n-2)^2}{4}}{\sin\beta\,\widehat{v}(t,\beta)^{\frac{2}{n-2}}}.$$

In the coordinates of Section 2.1, cylindrical symmetry and smoothness give an even smooth extension of $\widehat{v}(t,\beta)$ across $\beta = 0$. In particular, $\partial_\beta \widehat{v}(t,0) = 0$. Using the even extension and the uniform estimates in Proposition A.1 in the definition of Ψ gives

$$\Psi(t,\beta) = O(\beta) \qquad \text{as } \beta \to 0,$$

uniformly in t. The same estimates bound Ψ on $\mathbb{R} \times [\delta, L]$ for each $\delta > 0$, so $\sup_{\mathbb{R}\times(0,L]} \Psi$ is finite.

Suppose that $\sup_{\mathbb{R}\times(0,L]} \Psi > 0$ and choose (t_j, β_j) such that

$$\Psi(t_j,\beta_j) \longrightarrow \sup_{\mathbb{R}\times(0,L]} \Psi.$$

Since $\Psi(t,\beta) = O(\beta)$ uniformly in t, there is $\delta_0 > 0$ such that $\beta_j \geq \delta_0$ for all sufficiently large j. After passing to a subsequence, $\beta_j \to \beta_\infty \in [\delta_0, L]$. By Proposition A.1, a further subsequence of $\widehat{v}(t + t_j, \theta)$ converges in $C^k([-R,R] \times \mathbb{S}^{n-1}_+)$ for every $R > 0$ and every integer $k \geq 0$. We denote the limit again by $\widehat{v}$. For this limit, Ψ attains a positive maximum at $(0, \beta_\infty)$.

Suppose first that the maximum occurs with $0 < \beta < L$. Since $\nabla_{g_*} \Psi = 0$ there, differentiation of the definition of Ψ gives

$$\begin{aligned} \nabla_{g_*} \Big| \nabla_{g_*} \log\Big(\sin^{\tfrac{n-2}{2}} \beta\, \widehat{v}\Big) \Big|^2_{g_*} \\ = \frac{2\Psi \sin\beta\, \widehat{v}^{\frac{2}{n-2}}}{n-2} \nabla_{g_*} \log\Big(\sin^{\tfrac{n-2}{2}} \beta\, \widehat{v}\Big). \end{aligned} \tag{A.4}$$

For every smooth function f and tangent vector Y,

$$2\nabla^2_{g_*} f(\nabla_{g_*} f, Y) = Y\Big(|\nabla_{g_*} f|^2_{g_*}\Big).$$

The gradient on the right of (A.4) is nonzero since $\Psi > 0$. By (A.4) and the Hessian identity above, it is an eigenvector of $\nabla^2_{g_*} \log(\sin^{\frac{n-2}{2}} \beta\, \widehat{v})$, viewed as an endomorphism with respect to g_*. Its eigenvalue and the other eigenvalue are, respectively,

$$\frac{\Psi \sin\beta\, \widehat{v}^{\frac{2}{n-2}}}{n-2}, \qquad -\frac{n-2}{2} - \frac{n-1}{n-2} \Psi \sin\beta\, \widehat{v}^{\frac{2}{n-2}}.$$

Their sum is the Laplacian in (A.3).

The curvature of g_* is $-\sin^2\beta\, \partial^2_\beta(-\log\sin\beta) = -1$. Using the two Hessian eigenvalues in Bochner's formula and applying the product rule to $\Psi \sin\beta\, \widehat{v}^{2/(n-2)}$ at $\nabla_{g_*}\Psi = 0$, we obtain

$$\Delta_{g_*} \Psi = \Psi \left(n - 2 + \frac{2(n-1)}{n-2} \Psi \sin\beta\, \widehat{v}^{\frac{2}{n-2}} \right) > 0.$$

Here we used (A.3) and (A.4). This contradicts $\Delta_{g_*}\Psi \leq 0$ at the maximum.

It remains to consider $\beta_\infty = L$. The boundary condition in (A.1) gives

$$\partial_\beta \log\left(\sin^{\frac{n-2}{2}} \beta\, \widehat{v}\right)\Big|_{\beta=L} = \widehat{v}(t,L)^{\frac{2}{n-2}}.$$

Consequently,

$$\Psi(t,L) = \widehat{v}(t,L)^{-\frac{2}{n-2}}\left[(\partial_t \log \widehat{v}(t,L))^2 + \widehat{v}(t,L)^{\frac{4}{n-2}} - \frac{(n-2)^2}{4}\right].$$

Differentiating this identity gives

$$\begin{aligned}\partial_t \Psi(t,L) &= 2\widehat{v}(t,L)^{-\frac{2}{n-2}}\partial_t \log \widehat{v}(t,L)\\ &\quad\times\left[\partial_t^2 \log \widehat{v}(t,L) + \frac{2}{n-2}\widehat{v}(t,L)^{\frac{4}{n-2}} - \frac{\Psi(t,L)}{n-2}\widehat{v}(t,L)^{\frac{2}{n-2}}\right].\end{aligned}$$

At the maximum, $\partial_t \Psi = 0$ and $\partial_t^2 \Psi \leq 0$. If $\partial_t \log \widehat{v}(t,L) \neq 0$, the bracket vanishes. If $\partial_t \log \widehat{v}(t,L) = 0$, differentiating once more gives

$$\partial_t^2 \log \widehat{v}(t,L)\left[\partial_t^2 \log \widehat{v}(t,L) + \frac{\widehat{v}(t,L)^{\frac{4}{n-2}} + \frac{(n-2)^2}{4}}{n-2}\right] \leq 0.$$

Hence $\partial_t^2 \log \widehat{v}(t,L)$ lies between the two roots of this quadratic. In both cases,

$$\partial_t^2 \log \widehat{v}(t,L) + \frac{2}{n-2}\widehat{v}(t,L)^{\frac{4}{n-2}} \geq \frac{\Psi(t,L)}{n-2}\widehat{v}(t,L)^{\frac{2}{n-2}}.$$

Differentiating the boundary condition also gives

$$\partial_t \partial_\beta \log\left(\sin^{\frac{n-2}{2}} \beta\, \widehat{v}\right)\Big|_{\beta=L} = \frac{2}{n-2}\widehat{v}(t,L)^{\frac{2}{n-2}}\partial_t \log \widehat{v}(t,L).$$

Using (A.3) and $\partial_\beta(\sin^2\beta)|_{\beta=L} = 0$ in the definition of Ψ, we obtain

$$\begin{aligned}\partial_\beta \Psi(t,L) = -2\Bigg[&\left(1 - \frac{1}{n-2}\right)\Psi(t,L)\widehat{v}(t,L)^{\frac{2}{n-2}} + \partial_t^2 \log \widehat{v}(t,L)\\ &+ \frac{2}{n-2}\widehat{v}(t,L)^{\frac{4}{n-2}}\Bigg] \leq -2\Psi(t,L)\widehat{v}(t,L)^{\frac{2}{n-2}} < 0.\end{aligned}$$

This contradicts $\partial_\beta \Psi(t,L) \geq 0$ at a maximum on $\mathbb{R}\times(0,L]$. Therefore $\Psi \leq 0$ everywhere. At $\beta = L$, the inequality $\Psi(t,L) \leq 0$ gives (A.2). □

We use Proposition A.2 to compare $\partial_\beta \widehat{v}/\sin\beta$ with $\widehat{v}^{n/(n-2)}$ on $\mathbb{R}\times(0,L]$. Their β derivatives at L give the upper bound for $\partial_t^2 \widehat{v}(t,L)$. For the other inequality, we use the quotient of two positive solutions with the same linear boundary condition.

Proposition A.3. *Let $n \geq 5$ and let $\widehat{v}$ satisfy the hypotheses of Proposition A.1. For every $t \in \mathbb{R}$,*

$$\partial_t^2 \widehat{v}(t,L) \leq \frac{(n-2)^2}{4}\widehat{v}(t,L) - \frac{n}{n-2}\widehat{v}(t,L)^{\frac{n+2}{n-2}}, \tag{A.5}$$

$$\partial_t^2\left(\widehat{v}(t,L)^{-\frac{2}{n-2}}\right) < \widehat{v}(t,L)^{-\frac{2}{n-2}}. \tag{A.6}$$

Proof. Expanding the gradient bound in Proposition A.2 gives

$$\widehat{v}_t^2 + \widehat{v}_\beta^2 + (n-2)\cot\beta\,\widehat{v}\widehat{v}_\beta \leq \frac{(n-2)^2}{4}\widehat{v}^2.$$

Indeed, the bound is

$$\sin^2\beta\left[\left(\frac{\widehat{v}_t}{\widehat{v}}\right)^2 + \left(\frac{\widehat{v}_\beta}{\widehat{v}} + \frac{n-2}{2}\cot\beta\right)^2\right] \leq \frac{(n-2)^2}{4},$$

and $\csc^2\beta - \cot^2\beta = 1$. Differentiating the interior equation in (A.1) in β and applying the chain rule to $\widehat{v}^{n/(n-2)}$ gives

$$\begin{aligned}&\left(-\partial_t^2 - \partial_\beta^2 - n\cot\beta\,\partial_\beta + \frac{n^2}{4}\right)\left(\frac{\widehat{v}_\beta}{\sin\beta} - \widehat{v}^{\frac{n}{n-2}}\right)\\ &\quad= \frac{2n}{(n-2)^2}\widehat{v}^{\frac{4-n}{n-2}}\left[\widehat{v}_t^2 + \widehat{v}_\beta^2 + (n-2)\cot\beta\,\widehat{v}\widehat{v}_\beta - \frac{(n-2)^2}{4}\widehat{v}^2\right] \leq 0.\end{aligned}$$

The identity

$$\frac{\widehat{v}_\beta(t,\beta)}{\sin\beta} = \frac{\beta}{\sin\beta}\int_0^1 \partial_\beta^2\widehat{v}(t,s\beta)\,\mathrm{d}s$$

and evenness at $\beta = 0$ give a smooth even extension of the quotient. The function $\widehat{v}_\beta/\sin\beta - \widehat{v}^{n/(n-2)}$ can consequently be regarded as a smooth cylindrically symmetric function on $\mathbb{R}\times\mathbb{S}^{n+1}_+$ with product metric $\mathrm{d}t^2 + g_{\mathbb{S}^{n+1}}$, where $g_{\mathbb{S}^{n+1}}$ is the round metric of radius 1. It is bounded by Proposition A.1. Since $\Delta_{\mathbb{S}^{n+1}_+} = \partial_\beta^2 + n\cot\beta\,\partial_\beta$ on functions of β, we obtain

$$\left(-\partial_t^2 - \Delta_{\mathbb{S}^{n+1}_+} + \frac{n^2}{4}\right)\left(\frac{\widehat{v}_\beta}{\sin\beta} - \widehat{v}^{\frac{n}{n-2}}\right) \leq 0. \tag{A.7}$$

By the boundary condition in (A.1) and $\sin L = 1$,

$$\left(\frac{\partial_\beta\widehat{v}}{\sin\beta} - \widehat{v}^{\frac{n}{n-2}}\right)\Bigg|_{\beta=L} = \partial_\beta\widehat{v}(t,L) - \widehat{v}(t,L)^{\frac{n}{n-2}} = 0.$$

For $\varepsilon > 0$, the operator in (A.7) applied to $\widehat{v}_\beta/\sin\beta - \widehat{v}^{n/(n-2)} - \varepsilon\cosh t$ is at most $-\varepsilon(n^2/4-1)\cosh t < 0$. The function is nonpositive on $[-R,R]\times\partial\mathbb{S}^{n+1}_+$ by its value at $\beta = L$ and on $\{-R,R\}\times\mathbb{S}^{n+1}_+$ for sufficiently large R by boundedness of $\widehat{v}_\beta/\sin\beta - \widehat{v}^{n/(n-2)}$. As in

Proposition A.1, the maximum principle on $[-R,R]\times\mathbb{S}^{n+1}_+$ and the limit $\varepsilon\to 0^+$ give

$$\frac{\widehat{v}_\beta(t,\beta)}{\sin\beta}-\widehat{v}(t,\beta)^{\frac{n}{n-2}}\le 0\qquad\text{for }(t,\beta)\in\mathbb{R}\times[0,L].$$

Since equality holds at $\beta=L$,

$$\begin{aligned}0&\le\partial_\beta\left(\frac{\widehat{v}_\beta}{\sin\beta}-\widehat{v}^{\frac{n}{n-2}}\right)\Big|_{\beta=L}\\&=\frac{(n-2)^2}{4}\widehat{v}(t,L)-\partial_t^2\widehat{v}(t,L)-\frac{n}{n-2}\widehat{v}(t,L)^{\frac{n+2}{n-2}}.\end{aligned}$$

Here we used $\cot L=\cos L=0$, $\sin L=1$ and both equations in (A.1). This proves (A.5).

To prove (A.6), define

$$\begin{aligned}h_\pm(t,\beta)&=\frac{n-2}{2}\widehat{v}(t,\beta)-\cot\beta\,\widehat{v}_\beta(t,\beta)\pm\widehat{v}_t(t,\beta),\\f_\pm(t,\beta)&=e^{\pm t}\sin\beta\,h_\pm(t,\beta).\end{aligned}$$

We derive their equations directly from (A.1). Differentiating this equation in β and using $(\cot\beta)'=-\csc^2\beta$ and $(\cot\beta)''=2\cot\beta\csc^2\beta$ gives

$$\begin{aligned}&\left(\partial_t^2+\partial_\beta^2+n\cot\beta\,\partial_\beta-\frac{n^2-4}{4}\right)\left(\frac{n-2}{2}\widehat{v}-\cot\beta\,\widehat{v}_\beta\right)\\&\quad=2\widehat{v}_{\beta\beta}+2(n-2)\cot\beta\,\widehat{v}_\beta-\frac{(n-2)^2}{2}\widehat{v}=-2\widehat{v}_{tt}.\end{aligned}$$

Differentiating (A.1) in t also gives

$$\begin{aligned}&\left(\partial_t^2+\partial_\beta^2+n\cot\beta\,\partial_\beta-\frac{n^2-4}{4}\right)\widehat{v}_t\\&\quad=2\cot\beta\,\widehat{v}_{t\beta}-(n-2)\widehat{v}_t\\&\quad=-2\partial_t\left(\frac{n-2}{2}\widehat{v}-\cot\beta\,\widehat{v}_\beta\right).\end{aligned}$$

Adding and subtracting these identities yields, on $\mathbb{R}\times(0,L)$,

$$h_{\pm,tt}\pm 2h_{\pm,t}+h_{\pm,\beta\beta}+n\cot\beta\,h_{\pm,\beta}-\frac{n^2-4}{4}h_\pm=0.\tag{A.8}$$

For $f_\pm=e^{\pm t}\sin\beta\,h_\pm$, the product rule gives

$$\begin{aligned}f_{\pm,tt}&=e^{\pm t}\sin\beta\,(h_{\pm,tt}\pm 2h_{\pm,t}+h_\pm),\\f_{\pm,\beta\beta}+(n-2)\cot\beta\,f_{\pm,\beta}&=e^{\pm t}\sin\beta\Big[h_{\pm,\beta\beta}+n\cot\beta\,h_{\pm,\beta}\\&\qquad+\Big((n-2)\cot^2\beta-1\Big)h_\pm\Big].\end{aligned}$$

Since

$$(n-2)\cot^2\beta - \left(\frac{(n-2)^2}{4} + (n-2)\csc^2\beta\right) = -\frac{n^2-4}{4},$$

equation (A.8) gives, on $\mathbb{R}\times(0,L)$,

$$\begin{aligned} &f_{\pm,tt} + f_{\pm,\beta\beta} + (n-2)\cot\beta\, f_{\pm,\beta} \\ &\qquad - \left(\frac{(n-2)^2}{4} + (n-2)\csc^2\beta\right) f_\pm = 0. \end{aligned} \tag{A.9}$$

At $\beta = L$,

$$h_\pm(t,L) = \frac{n-2}{2}\widehat{v}(t,L) \pm \partial_t\widehat{v}(t,L).$$

Differentiating the definition of $h_\pm$ in β and using (A.1) gives

$$\begin{aligned} h_{\pm,\beta}(t,L) &= \frac{n}{2}\widehat{v}(t,L)^{\frac{n}{n-2}} \pm \frac{n}{n-2}\widehat{v}(t,L)^{\frac{2}{n-2}}\partial_t\widehat{v}(t,L) \\ &= \frac{n}{n-2}\widehat{v}(t,L)^{\frac{2}{n-2}} h_\pm(t,L). \end{aligned}$$

Since $\cos L = 0$,

$$f_{\pm,\beta}(t,L) = \frac{n}{n-2}\widehat{v}(t,L)^{\frac{2}{n-2}} f_\pm(t,L) \qquad (t\in\mathbb{R}). \tag{A.10}$$

Since $\cot\beta\,\widehat{v}_\beta = \cos\beta\,(\widehat{v}_\beta/\sin\beta)$, this term extends smoothly and evenly across $\beta = 0$. Thus $h_\pm$ are smooth on $\mathbb{R}\times\mathbb{S}^{n+1}_+$ and (A.8) holds there with $\Delta_{\mathbb{S}^{n+1}_+}$ in place of $\partial_\beta^2 + n\cot\beta\,\partial_\beta$. The integral formula for $\widehat{v}_\beta/\sin\beta$ and Proposition A.1 give uniform bounds for $h_\pm$ and their derivatives on $\mathbb{R}\times\mathbb{S}^{n+1}_+$.

Let $m > 0$ be a lower bound for $\widehat{v}(t,L)$ on $\mathbb{R}$. By (A.2),

$$\begin{aligned} h_\pm(t,L) &\geq \widehat{v}(t,L)\left(\frac{n-2}{2} - \left|\frac{\partial_t\widehat{v}(t,L)}{\widehat{v}(t,L)}\right|\right) \\ &\geq m\left(\frac{n-2}{2} - \sqrt{\frac{(n-2)^2}{4} - m^{\frac{4}{n-2}}}\right) > 0. \end{aligned}$$

This bound is independent of t and of the choice of sign. Suppose that one of these functions has $\inf_{\mathbb{R}\times[0,L]} h_\pm \leq 0$. Choose (t_j,β_j) such that

$$h_\pm(t_j,\beta_j) \longrightarrow \inf_{\mathbb{R}\times[0,L]} h_\pm \leq 0.$$

By Proposition A.1 and the definition of $h_\pm$, we may pass to a subsequence for which $\beta_j \to \beta_\infty \in [0,L]$ and $h_\pm(t+t_j,\beta)$ converges in $C^2([-R,R]\times\mathbb{S}^{n+1}_+)$ for every $R > 0$. Here convergence through $\beta = 0$ follows from the integral formula for $\widehat{v}_\beta/\sin\beta$. We denote the limit

again by $h_\pm$. It attains its minimum at $(0,\beta_\infty)$, where $h_\pm(0,\beta_\infty)\le 0$. The positive lower bound for $h_\pm(t,L)$ gives $\beta_\infty < L$. At this minimum,

$$\partial_t h_\pm = 0, \qquad (\partial_t^2 + \Delta_{\mathbb{S}^{n+1}})h_\pm \ge 0.$$

If $h_\pm(0,\beta_\infty) < 0$, then

$$(\partial_t^2 + \Delta_{\mathbb{S}^{n+1}})h_\pm \pm 2\partial_t h_\pm - \frac{n^2-4}{4}h_\pm > 0$$

there, contradicting (A.8). If $h_\pm(0,\beta_\infty) = 0$, the strong minimum principle applied to (A.8) gives $h_\pm \equiv 0$, contradicting the positive bound on $\mathbb{R}\times\partial\mathbb{S}^{n+1}_+$. Consequently

$$0 < \inf_{\mathbb{R}\times[0,L]} h_\pm \le \sup_{\mathbb{R}\times[0,L]} h_\pm < \infty.$$

Multiplying (A.9) for $f_\pm$ by $f_\mp$ and subtracting cancels the zeroth-order terms. Since $\omega = \sin^{n-2}\beta$, we obtain

$$\partial_t[\omega(f_- f_{+,t} - f_+ f_{-,t})] + \partial_\beta[\omega(f_- f_{+,\beta} - f_+ f_{-,\beta})] = 0.$$

Using

$$f_+ f_- = \sin^2\beta\, h_+ h_-, \qquad \log(f_+/f_-) = 2t + \log(h_+/h_-),$$

and dividing by 2, we obtain

$$\begin{aligned}\partial_t\left[\sin^n\beta\, h_+h_-\left(1+\frac{1}{2}\partial_t\log\frac{h_+}{h_-}\right)\right]\\ +\frac{1}{2}\partial_\beta\left[\sin^n\beta\, h_+h_-\,\partial_\beta\log\frac{h_+}{h_-}\right]=0.\end{aligned} \tag{A.11}$$

The boundary condition (A.10) gives

$$\partial_\beta\log\frac{h_+}{h_-}(t,L) = 0.$$

The uniform positive bounds and derivative bounds for $h_\pm$ give constants $c_1, c_2 > 0$ such that

$$c_1\beta^n \le \sin^n\beta\, h_+(t,\beta)h_-(t,\beta) \le c_2\beta^n \qquad (t\in\mathbb{R},\ 0<\beta\le L).$$

They also give

$$\partial_t\log(h_+/h_-) = O(1), \qquad \partial_\beta\log(h_+/h_-) = O(\beta) \qquad \text{as } \beta\to 0,$$

uniformly in t. The second estimate follows from evenness at $\beta = 0$.

For $(t,\beta)\in\mathbb{R}\times[0,L]$, define

$$k(t,\beta) = \int_0^\beta \sin^n s\, h_+(t,s)h_-(t,s)\left(1+\frac{1}{2}\partial_t\log\frac{h_+(t,s)}{h_-(t,s)}\right)\,\mathrm{d}s.$$

Then $k = O(\beta^{n+1})$ uniformly in t, and

$$k_\beta(t,\beta) = \sin^n \beta\, h_+ h_- \left(1 + \frac{1}{2}\partial_t \log \frac{h_+}{h_-}\right),$$
$$k_t(t,\beta) = -\frac{1}{2}\sin^n \beta\, h_+ h_-\, \partial_\beta \log \frac{h_+}{h_-}.$$

The second identity follows by differentiating the integral and using (A.11). The term at $s = 0$ vanishes because it is $O(s^{n+1})$. Dividing the two identities by $\sin^n \beta\, h_+ h_-$ and differentiating gives

$$\partial_t \left(\frac{k_t}{\sin^n \beta\, h_+ h_-}\right) + \partial_\beta \left(\frac{k_\beta}{\sin^n \beta\, h_+ h_-}\right) = 0 \quad \text{on } \mathbb{R} \times (0, L). \tag{A.12}$$

We have $k(t,0) = 0$. Since (A.10) gives $\partial_\beta \log(h_+/h_-)(t,L) = 0$, the formula for k_t gives $k_t(t,L) = 0$. Hence $k(t,L) = J$ for a real constant J.

We claim that $\min(0,J) \le k \le \max(0,J)$. The formula for k_β and the uniform bounds for $h_\pm$ and their derivatives give $|k_\beta| \le C$ on $\mathbb{R} \times [0,L]$. Together with $k(t,0) = 0$ and $k(t,L) = J$, this gives

$$|k(t,\beta)| \le C\beta, \qquad |k(t,\beta) - J| \le C(L - \beta).$$

Suppose that $\sup_{\mathbb{R}\times[0,L]} k > \max(0,J)$ and choose (t_j, β_j) such that $k(t_j,\beta_j) \to \sup_{\mathbb{R}\times[0,L]} k$. The two bounds give $\beta_j \in [\delta, L - \delta]$ for some $\delta > 0$ and all sufficiently large j. After passing to a subsequence, $\beta_j \to \beta_\infty \in [\delta, L-\delta]$. By Proposition A.1 and the formulas for $h_\pm$ and k, the translates $h_\pm(t+t_j,\beta)$ and $k(t+t_j,\beta)$ converge in $C^2([-R,R]\times[0,L])$ for every $R > 0$ along a further subsequence. At $\beta = 0$, this convergence follows from the integral formula for $\widehat{v}_\beta/\sin\beta$. The limiting $h_\pm$ are bounded below by positive constants. The limit of $k(t+t_j,\beta)$ therefore satisfies (A.12) with the limiting coefficients and has the same values 0 and J at $\beta = 0$ and $\beta = L$. Its maximum is attained at $(0,\beta_\infty)$. The strong maximum principle on $\mathbb{R} \times (0,L)$ makes the limit constant, which contradicts its value 0 at $\beta = 0$. The same argument applied to $-k$, with boundary value $-J$ at $\beta = L$, proves the lower bound.

If $J < 0$, the function k is nonconstant because its boundary values are 0 and J. The strong minimum principle applied to (A.12) gives $k > J$ for $0 < \beta < L$. The coefficient in this equation is smooth and positive near each point (t,L). The boundary point lemma at the minimum $k(t,L) = J$ therefore gives

$$k_\beta(t,L) < 0.$$

If $J = 0$, the bound $\min(0, J) \le k \le \max(0, J)$ gives $k \equiv 0$ and $k_\beta(t, L) = 0$. In either case,

$$\begin{aligned} 0 &\ge \frac{k_\beta(t,L)}{h_+(t,L)h_-(t,L)} \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\left[t + \frac{1}{2}\log\frac{\frac{n-2}{2}\widehat{v}(t,L) + \partial_t\widehat{v}(t,L)}{\frac{n-2}{2}\widehat{v}(t,L) - \partial_t\widehat{v}(t,L)}\right]. \end{aligned}$$

The uniform bounds for $h_\pm(t, L)$ bound the quotient above and below by positive constants. Hence $t + \frac{1}{2}\log(h_+(t, L)/h_-(t, L)) = t + O(1)$ tends to $+\infty$ as $t \to +\infty$, contradicting its nonpositive derivative. Hence $J > 0$.

For $J > 0$, the strong maximum principle for (A.12) gives $k < J$ on $\mathbb{R}\times(0, L)$, since $k(t, 0) = 0$. The boundary point lemma at the maximum $k(t, L) = J$ gives $k_\beta(t, L) > 0$. Dividing by $h_+(t, L)h_-(t, L) > 0$ and differentiating the logarithm gives

$$1 + \frac{\frac{n-2}{2}\left[\widehat{v}(t,L)\,\partial_t^2\widehat{v}(t,L) - (\partial_t\widehat{v}(t,L))^2\right]}{\frac{(n-2)^2}{4}\widehat{v}(t,L)^2 - (\partial_t\widehat{v}(t,L))^2} > 0.$$

The denominator equals $h_+(t, L)h_-(t, L) > 0$. Multiplying by this product and dividing by $\frac{n-2}{2}\widehat{v}(t, L)^2$ yields

$$\frac{\partial_t^2\widehat{v}(t,L)}{\widehat{v}(t,L)} > \frac{n}{n-2}\left(\frac{\partial_t\widehat{v}(t,L)}{\widehat{v}(t,L)}\right)^2 - \frac{n-2}{2}.$$

Finally,

$$\begin{aligned} \partial_t^2\left(\widehat{v}(t,L)^{-\frac{2}{n-2}}\right) &= \frac{2}{n-2}\widehat{v}(t,L)^{-\frac{2}{n-2}}\left[\frac{n}{n-2}\left(\frac{\partial_t\widehat{v}(t,L)}{\widehat{v}(t,L)}\right)^2 - \frac{\partial_t^2\widehat{v}(t,L)}{\widehat{v}(t,L)}\right] \\ &< \widehat{v}(t,L)^{-\frac{2}{n-2}}, \end{aligned}$$

which proves (A.6). □

Corollary A.4. *Let $n \ge 5$, and let $\widehat{v}_1, \widehat{v}_2$ satisfy the hypotheses of Proposition A.1, with constants $m, M > 0$ such that*

$$m \le \widehat{v}_i(t, L) \le M \qquad (t \in \mathbb{R},\ i = 1, 2).$$

Suppose either that both traces are even in t, or that, for some $t_0 \in \mathbb{R}$ and some solution $\widehat{v}$ satisfying the same hypotheses,

$$\widehat{v}_1(t, L) = \widehat{v}(t_0 + t, L), \qquad \widehat{v}_2(t, L) = \widehat{v}(t_0 - t, L).$$

The coefficient

$$b(t) = \frac{n}{n-2}\int_0^1 \left[(1-\lambda)\widehat{v}_1(t,L) + \lambda\widehat{v}_2(t,L)\right]^{\frac{2}{n-2}}\,\mathrm{d}\lambda \tag{A.13}$$

is smooth, positive and even. It satisfies

$$0 < b \le \frac{n}{2}, \qquad b'' \ge 2\frac{(b')^2}{b} - b, \tag{A.14}$$

$$\frac{b''}{b} \le \frac{n-2}{2} - \frac{2b^2}{n} - \frac{n-4}{2}\left(\frac{b'}{b}\right)^2, \tag{A.15}$$

$$\left|\frac{b'}{b}\right|^2 \le 1 - \frac{4b^2}{n^2},$$

$$-\tanh t \le \frac{b'}{b} \le \tanh\left(\frac{n-2}{2}t\right) \qquad (t \ge 0). \tag{A.16}$$

Taking $\widehat{v}_1 = \widehat{v}_2 = \widehat{v}$, with $\widehat{v}$ even in t, gives the same estimates for the coefficient $\frac{n}{n-2}\widehat{v}(t,L)^{2/(n-2)}$ in the linearized boundary condition.

Proof. In the first case, the integrand in (A.13) is even in t for every λ. In the second case, reflection in t interchanges $\widehat{v}_1(t,L)$ and $\widehat{v}_2(t,L)$. Changing the integration variable from λ to $1-\lambda$ in (A.13) gives $b(-t) = b(t)$.

Since $\widehat{v}_i(t,L) \ge m > 0$ for $i = 1, 2$, we have

$$(1-\lambda)\widehat{v}_1(t,L) + \lambda\widehat{v}_2(t,L) \ge m \qquad (0 \le \lambda \le 1).$$

This bound and the smoothness of $\widehat{v}_i(t,L)$ allow differentiation under the integral in (A.13). The same formula gives $b(t) \ge \frac{n}{n-2}m^{\frac{2}{n-2}} > 0$. By the boundary condition in (A.1) and the fundamental theorem of calculus in λ,

$$\partial_\beta(\widehat{v}_2 - \widehat{v}_1)(t,L) = b(t)(\widehat{v}_2(t,L) - \widehat{v}_1(t,L)).$$

By (A.6), each $\widehat{v}_i(t,L)$ satisfies

$$\partial_t^2\widehat{v}_i(t,L) \ge \frac{n}{n-2}\frac{(\partial_t\widehat{v}_i(t,L))^2}{\widehat{v}_i(t,L)} - \frac{n-2}{2}\widehat{v}_i(t,L) \qquad (i = 1,2).$$

For $0 \le \lambda \le 1$, weighted Cauchy–Schwarz gives

$$\frac{\left[(1-\lambda)\partial_t\widehat{v}_1(t,L) + \lambda\partial_t\widehat{v}_2(t,L)\right]^2}{(1-\lambda)\widehat{v}_1(t,L) + \lambda\widehat{v}_2(t,L)} \le (1-\lambda)\frac{(\partial_t\widehat{v}_1(t,L))^2}{\widehat{v}_1(t,L)} + \lambda\frac{(\partial_t\widehat{v}_2(t,L))^2}{\widehat{v}_2(t,L)}.$$

Multiplying the trace inequalities by $1-\lambda$ and λ, respectively, and adding gives

$$\begin{aligned}
&\partial_t^2\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]\\
&\quad\geq \frac{n}{n-2}\frac{\Big[(1-\lambda)\partial_t\widehat{v}_1(t,L)+\lambda\partial_t\widehat{v}_2(t,L)\Big]^2}{(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)}\\
&\qquad-\frac{n-2}{2}\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big].
\end{aligned}$$

Differentiating the power $2/(n-2)$ therefore gives

$$\begin{aligned}
&\partial_t^2\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{2}{n-2}}\\
&\quad\geq 2\frac{\Big\{\partial_t\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{2}{n-2}}\Big\}^2}{\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{2}{n-2}}}\\
&\qquad-\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{2}{n-2}}.
\end{aligned}$$

Cauchy–Schwarz in the integral defining b' gives

$$\frac{n}{n-2}\int_0^1\frac{\Big\{\partial_t\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{2}{n-2}}\Big\}^2}{\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{2}{n-2}}}\,\mathrm{d}\lambda\geq\frac{(b')^2}{b}.$$

Integrating the bound for the second derivative of the integrand in (A.13) and using the Cauchy–Schwarz estimate for b' gives the differential inequality in (A.14).

For the upper estimate, use (A.5) and convexity of the power $(n+2)/(n-2)$. For every $\lambda\in[0,1]$,

$$\begin{aligned}
&\partial_t^2\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]\\
&\quad\leq\frac{(n-2)^2}{4}\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]\\
&\qquad-\frac{n}{n-2}\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{n+2}{n-2}}.
\end{aligned}$$

Differentiating the power $2/(n-2)$ gives

$$\begin{aligned}
&\partial_t^2\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{2}{n-2}}\\
&\quad\le \frac{n-2}{2}\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{2}{n-2}}\\
&\qquad-\frac{2n}{(n-2)^2}\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{6}{n-2}}\\
&\qquad-\frac{n-4}{2}\frac{\Big\{\partial_t\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{2}{n-2}}\Big\}^2}{\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{2}{n-2}}}.
\end{aligned}$$

Jensen's inequality for the cube and (A.13) give

$$\begin{aligned}
&\frac{n}{n-2}\int_0^1\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{6}{n-2}}\,\mathrm{d}\lambda\\
&\quad\ge\frac{n}{n-2}\left\{\int_0^1\Big[(1-\lambda)\widehat{v}_1(t,L)+\lambda\widehat{v}_2(t,L)\Big]^{\frac{2}{n-2}}\,\mathrm{d}\lambda\right\}^3\\
&\quad=\frac{(n-2)^2}{n^2}b^3.
\end{aligned}$$

Integrate the upper bound for the second derivative of the integrand in (A.13). Use the lower bound for the integral of the power $6/(n-2)$ and the Cauchy–Schwarz bound for b', which enters with coefficient $-(n-4)/2<0$. This gives

$$b''\le\frac{n-2}{2}b-\frac{2}{n}b^3-\frac{n-4}{2}\frac{(b')^2}{b}.$$

Dividing by $b>0$ proves (A.15).

Combining (A.14) and (A.15) gives

$$2\left(\frac{b'}{b}\right)^2-1\le\frac{b''}{b}\le\frac{n-2}{2}-\frac{2b^2}{n}-\frac{n-4}{2}\left(\frac{b'}{b}\right)^2.$$

Rearranging yields

$$\left(\frac{b'}{b}\right)^2+\frac{4b^2}{n^2}\le 1.$$

Hence $b\le n/2$ and $|b'/b|<1$. Subtracting $(b'/b)^2$ from the two bounds for b''/b gives

$$\left(\frac{b'}{b}\right)^2-1\le\left(\frac{b'}{b}\right)'\le\frac{n-2}{2}\left[1-\left(\frac{b'}{b}\right)^2\right].$$

Consequently

$$-1 \le \frac{\mathrm{d}}{\mathrm{d}t} \operatorname{arctanh}\left(\frac{b'}{b}\right) \le \frac{n-2}{2}.$$

Since b is even, $b'(0) = 0$. Integrating from 0 to $t \ge 0$ and applying the increasing function tanh proves (A.16). For $\widehat{v}_1 = \widehat{v}_2 = \widehat{v}$, (A.13) reduces to $b(t) = \frac{n}{n-2}\widehat{v}(t, L)^{2/(n-2)}$, the coefficient in the boundary condition of (2.4) in the normalization (3.3). □

A.3. **Spectral estimates.** We use β, L and ω as in Section 2.1. For cylindrically symmetric functions $f = f(\beta)$, we use the differential expression of $\mathcal{A}$ from Section 2.3 and impose the boundary condition $f'(L) = bf(L)$ for $b \ge 0$. For $n \ge 5$, we estimate the first eigenvalue and the corresponding positive eigenfunction, and give a lower bound for the gap between the first two eigenvalues. We then prove a resolvent identity for all $n \ge 3$.

For $n \ge 5$, let

$$\mathcal{Q} = \{z \in H^1(0, L) : z(0) = 0\}.$$

We use real-valued functions and write $\|\cdot\|_2$ for the norm in $L^2(0, L)$. For $z \in \mathcal{Q}$ and $b \in \mathbb{R}$, set

$$q_b[z] = \int_0^L \left(|z'|^2 + \frac{(n-2)(n-4)}{4} \csc^2\beta\, |z|^2\right) \mathrm{d}\beta - b|z(L)|^2. \tag{A.17}$$

We write $q_b[\cdot, \cdot]$ for the associated bilinear form. Hardy's inequality and $\sin\beta \ge 2\beta/\pi$ on $[0, L]$ give

$$\int_0^L \csc^2\beta\, |z|^2 \,\mathrm{d}\beta \le \frac{\pi^2}{4} \int_0^L \frac{|z|^2}{\beta^2} \,\mathrm{d}\beta \le \pi^2 \|z'\|_2^2.$$

For the trace $z(L)$, the condition $z(0) = 0$ gives, for every $\varepsilon > 0$,

$$|z(L)|^2 = 2\int_0^L zz' \,\mathrm{d}\beta \le \varepsilon\|z'\|_2^2 + \varepsilon^{-1}\|z\|_2^2. \tag{A.18}$$

For fixed b, choose $\varepsilon > 0$ with $|b|\varepsilon < 1$. Adding a sufficiently large multiple of $\|z\|_2^2$ to $q_b[z]$ gives a squared norm equivalent to $\|z\|_{H^1(0,L)}^2$ on $\mathcal{Q}$. Thus q_b is closed and bounded below. The inclusion $\mathcal{Q} \hookrightarrow L^2(0, L)$ is compact by the compact embedding $H^1(0, L) \hookrightarrow L^2(0, L)$.

Let A_b be the self-adjoint operator associated with q_b. Its differential expression on $(0, L)$ is

$$-\partial_\beta^2 + \frac{(n-2)(n-4)}{4} \csc^2\beta. \tag{A.19}$$

The domain of A_b consists of $z \in \mathcal{Q}$ for which (A.19) applied to z belongs to $L^2(0, L)$ in the sense of distributions and $z'(L) = bz(L)$. The

potential in (A.19) is bounded near L, so such a z is H^2 near L and $z'(L)$ is defined. Integration by parts in $q_b[z,\phi]$ for $\phi \in \mathcal{Q}$ vanishing near 0 gives the boundary term $(z'(L) - bz(L))\phi(L)$. This gives the condition at L. Conversely, if this term vanishes, the integration by parts identity extends to all $\phi \in \mathcal{Q}$ by density of the tests vanishing near 0, proved below.

Write $\mu_j(b)$, $j \geq 0$, for the eigenvalues of A_b in increasing order. By the min–max principle, with E ranging over linear subspaces of $\mathcal{Q}$,

$$\mu_j(b) = \min_{\substack{E \subset \mathcal{Q} \\ \dim E = j+1}} \max_{z \in E \setminus \{0\}} \frac{q_b[z]}{\|z\|_2^2}. \tag{A.20}$$

In derivatives with respect to b, we keep n and the spatial variable fixed.

For $n \geq 3$, identify the functions in $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)$ with functions of β, where the subscript sym denotes cylindrical symmetry as in Section 2.3. On such functions, the spherical L^2 inner product equals $|\mathbb{S}^{n-2}|$ times the inner product in $L^2((0,L), \omega \, \mathrm{d}\beta)$. For $b \geq 0$, let $\mathcal{A}_b$ be the self-adjoint operator in this weighted space associated with the form on $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)$ given by

$$a_b[f] = \int_0^L \omega(\beta) \left(|f'|^2 + \frac{(n-2)^2}{4} |f|^2 \right) \mathrm{d}\beta - b|f(L)|^2. \tag{A.21}$$

We write $a_b[\cdot,\cdot]$ for the associated bilinear form. For $0 < \delta < L/2$, Cauchy–Schwarz in $f(L) = f(s) + \int_s^L f'(\beta) \, \mathrm{d}\beta$ and averaging over $L - \delta \leq s \leq L$ give

$$|f(L)|^2 \leq \frac{2}{\delta} \int_{L-\delta}^L |f|^2 \, \mathrm{d}\beta + 2\delta \int_{L-\delta}^L |f'|^2 \, \mathrm{d}\beta.$$

Since ω has a positive lower bound on $[L/2, L]$, choosing δ sufficiently small gives, for each $\varepsilon > 0$,

$$|f(L)|^2 \leq \varepsilon \int_0^L \omega |f'|^2 \, \mathrm{d}\beta + C_\varepsilon \int_0^L \omega |f|^2 \, \mathrm{d}\beta. \tag{A.22}$$

For fixed b, take ε so that $b\varepsilon < 1$. Adding a sufficiently large multiple of the squared $L^2((0,L), \omega \, \mathrm{d}\beta)$ norm to $a_b[f]$ gives a squared norm equivalent to $\|f\|^2_{H^1(\mathbb{S}^{n-1}_+)}$. Thus a_b is closed and bounded below. The compact embedding $H^1(\mathbb{S}^{n-1}_+) \hookrightarrow L^2(\mathbb{S}^{n-1}_+)$ gives compactness of $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+) \hookrightarrow L^2((0,L), \omega \, \mathrm{d}\beta)$ under the identification above.

Integration by parts in (A.21) gives the interior expression

$$\mathcal{A}_b f = -\omega^{-1}(\omega f')' + \frac{(n-2)^2}{4} f \quad \text{on } (0,L). \tag{A.23}$$

Since $\omega(L) = 1$, integration by parts against tests supported near L gives $f'(L) = bf(L)$. If $\mathcal{A}_b f = \lambda f$, the identity

$$a_b[f, \phi] = \lambda \int_0^L \omega f \phi \, \mathrm{d}\beta \qquad \left(\phi \in H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)\right)$$

is the weak eigenvalue equation on $\mathbb{S}^{n-1}_+$ for $-\Delta_{\mathbb{S}^{n-1}_+} + (n-2)^2/4$ with boundary condition $\partial_\nu f = bf$. It holds in the spherical formulation for every smooth test function on $\mathbb{S}^{n-1}_+$, since averaging a test function over rotations of the first $n-1$ coordinates leaves its pairings with f unchanged. Elliptic regularity for this equation and boundary condition gives smoothness on $\mathbb{S}^{n-1}_+$.

For $n \geq 5$, multiplication by $\sin^{(n-2)/2} \beta$ is unitary from the weighted space $L^2((0, L), \omega \, \mathrm{d}\beta)$ to $L^2(0, L)$. For smooth cylindrically symmetric f, write $z = \sin^{(n-2)/2} \beta \, f$. Then $z \in \mathcal{Q}$ and $z(L) = f(L)$. By (A.21),

$$a_b[f] = \int_0^L \left(\left| z' - \frac{n-2}{2} \cot \beta \, z \right|^2 + \frac{(n-2)^2}{4} |z|^2 \right) \mathrm{d}\beta - b|z(L)|^2.$$

The boundary terms from integration by parts vanish because $\cot L = 0$ and $\cot \beta \, |z|^2 = O(\beta^{n-3})$ near 0. The resulting coefficient of $|z|^2$ is

$$\frac{(n-2)^2}{4}(1 + \cot^2 \beta) - \frac{n-2}{2} \csc^2 \beta = \frac{(n-2)(n-4)}{4} \csc^2 \beta.$$

Hence

$$a_b[f] = q_b[z], \qquad z = \sin^{\frac{n-2}{2}} \beta \, f. \tag{A.24}$$

Smooth cylindrically symmetric functions are dense in $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)$ by smoothing and averaging over rotations. Their images under multiplication by $\sin^{(n-2)/2} \beta$ include every smooth function on $[0, L]$ that vanishes near 0, since division by $\sin^{(n-2)/2} \beta$ gives a smooth cylindrically symmetric preimage. These functions are dense in $\mathcal{Q}$. Indeed, cutting off $z \in \mathcal{Q}$ on $[0, \varepsilon]$ with derivative bounded by C/ε on $[\varepsilon, 2\varepsilon]$ gives convergence in $H^1(0, L)$ by Hardy's inequality and

$$\int_0^{2\varepsilon} \frac{|z|^2}{\beta^2} \, \mathrm{d}\beta \longrightarrow 0.$$

We smooth the cutoff functions to obtain the density in $\mathcal{Q}$. Therefore (A.24) holds on the form domains and gives the unitary equivalence of $\mathcal{A}_b$ and A_b for $n \geq 5$.

Proposition A.5. *Let $n \geq 5$. For each $b \geq 0$, let e_b be the eigenfunction of A_b corresponding to $\mu_0(b)$, chosen positive on $(0, L]$ and normalized*

by $\|e_b\|_2 = 1$. *Set*

$$C(b) = e_b(L)^2, \qquad r(b) = \frac{(n-2)^2}{4} - \mu_0(b).$$

The map $b \mapsto (\mu_0(b), e_b)$ *is smooth with values in* $\mathbb{R} \times \mathcal{Q}$, *and*

$$-\frac{\mathrm{d}\mu_0}{\mathrm{d}b}(b) = C(b), \qquad b^2 \le r(b) \le b^2 + \frac{n-2}{2}, \qquad C(b) \ge 2b. \tag{A.25}$$

Moreover,

$$bC(b) \le 2r(b), \qquad \mu_1(b) - \mu_0(b) > r(b) + n - 1. \tag{A.26}$$

The form q_b *restricted to* $\{z \in \mathcal{Q} : z(L) = 0\}$ *has first eigenvalue* $n^2/4$, *with eigenfunction* $\sin^{\frac{n-2}{2}} \beta \cos \beta$. *For every* $b \ge 0$,

$$\mu_0(b) < \frac{n^2}{4} < \mu_1(b).$$

Proof. The case $j = 0$ of (A.20) allows us to choose a nonnegative first eigenfunction, since taking the absolute value preserves $q_b[z]/\|z\|_2^2$. At a zero in $(0, L)$ its derivative would vanish, and at a zero at L the same conclusion follows from $e_b'(L) = be_b(L)$. Uniqueness for $A_b e_b = \mu_0(b) e_b$ on $(0, L)$, using (A.19), gives $e_b > 0$ on $(0, L]$ and simplicity, since $e_b(L)$ fixes both data at L.

Fix $b_0 \ge 0$ and choose $M > 0$ such that $q_{b_0}[z] + M\|z\|_2^2$ is equivalent to $\|z\|_{H^1(0,L)}^2$ on $\mathcal{Q}$. By (A.17) and (A.18), the map

$$(b, \mu, e) \longmapsto \Big(q_b[e, \cdot] - \mu \langle e, \cdot \rangle_{L^2(0,L)}, \|e\|_2^2 - 1\Big)$$

is smooth from $\mathbb{R} \times \mathbb{R} \times \mathcal{Q}$ to $\mathcal{Q}^* \times \mathbb{R}$. At $(b_0, \mu_0(b_0), e_{b_0})$, testing the first component of the derivative in (μ, e) against e_{b_0} determines the variation of μ. The second component determines the projection of the variation of e onto e_{b_0}. On $\mathcal{Q} \cap e_{b_0}^{\perp}$, the eigenfunction expansion for A_{b_0} gives

$$q_{b_0}[h] - \mu_0(b_0)\|h\|_2^2 \ge \Big(\mu_1(b_0) - \mu_0(b_0)\Big)\|h\|_2^2.$$

The gap is positive by simplicity. Adding $(M + \mu_0(b_0))\|h\|_2^2$ to the left-hand side gives $q_{b_0}[h] + M\|h\|_2^2$. Thus $q_{b_0} - \mu_0(b_0)\|\cdot\|_2^2$ is coercive on $\mathcal{Q} \cap e_{b_0}^{\perp}$, and the Lax–Milgram theorem determines the remaining component of the variation of e. The derivative is therefore invertible, and the implicit function theorem gives a smooth local branch of eigenvalues and $L^2(0, L)$-normalized eigenfunctions through $(\mu_0(b_0), e_{b_0})$.

To identify this branch, use the trace bound (A.18) in the difference of the forms at b and b_0 to obtain

$$|q_b[z] - q_{b_0}[z]| = |b - b_0||z(L)|^2 \le C|b - b_0|\Big(q_{b_0}[z] + M\|z\|_2^2\Big).$$

Dividing by $\|z\|_2^2$ and applying (A.20) to the shifted forms, for $C|b-b_0| < 1$ we obtain

$$|\mu_j(b) - \mu_j(b_0)| \le C|b-b_0|(\mu_j(b_0) + M), \qquad j = 0, 1.$$

Since $\mu_0(b_0) < \mu_1(b_0)$, the eigenvalue of the smooth branch stays below $\mu_1(b)$ for b near b_0, by continuity. It therefore equals $\mu_0(b)$. Continuity of $e \mapsto e(L)$ on $\mathcal{Q}$ follows from (A.18). The normalized eigenfunction on the branch has positive value at L for b near b_0, so it equals e_b. This proves smooth dependence in $\mathbb{R} \times \mathcal{Q}$, including at $b_0 = 0$.

Differentiating $q_b[e_b, \phi] = \mu_0(b)\langle e_b, \phi\rangle_{L^2(0,L)}$ in b with $\phi \in \mathcal{Q}$ fixed and then taking $\phi = e_b$ gives

$$\frac{\mathrm{d}\mu_0}{\mathrm{d}b}(b) = -e_b(L)^2 = -C(b).$$

The terms containing $\partial_b e_b$ cancel by the same form identity with $\phi = \partial_b e_b$.

The condition $e_b(0) = 0$ and the normalization $\|e_b\|_2 = 1$ give

$$0 \le \int_0^L (\partial_\beta e_b - b e_b)^2 \, \mathrm{d}\beta = \int_0^L |\partial_\beta e_b|^2 \, \mathrm{d}\beta + b^2 - bC(b).$$

Since $\csc^2 \beta \ge 1$, (A.17) gives

$$\mu_0(b) \ge \frac{(n-2)(n-4)}{4} - b^2.$$

For $j = 0$, the min–max formula (A.20) and the identification of forms (A.24) give

$$\mu_0(b) = \min_{\substack{f \in H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+) \\ f \neq 0}} \frac{a_b[f]}{\int_0^L \omega |f|^2 \, \mathrm{d}\beta}.$$

For $b = 0$, the minimum is $(n-2)^2/4$, attained at $f = 1$ by (A.21). For $b > 0$, $f(\beta) = e^{b(\beta - L)}$ is Lipschitz in the spherical distance β and belongs to $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)$. Using this function in the quotient and setting $x = L - \beta$ in its squared $L^2((0,L), \omega \, \mathrm{d}\beta)$ norm gives

$$\begin{aligned} I(b) &= \int_0^L e^{-2bx} \cos^{n-2} x \, \mathrm{d}x < \int_0^\infty e^{-2bx} \, \mathrm{d}x = \frac{1}{2b}, \\ \mu_0(b) &\le \frac{a_b[e^{b(\beta - L)}]}{I(b)} = \frac{(n-2)^2}{4} + b^2 - \frac{b}{I(b)}. \end{aligned} \tag{A.27}$$

Together with $\mu_0(b) \ge (n-2)(n-4)/4 - b^2$, this gives $b^2 \le r(b) \le b^2 + (n-2)/2$.

By (A.24), the function $f = e_b / \sin^{(n-2)/2} \beta$ is a smooth cylindrically symmetric eigenfunction of $\mathcal{A}_b$ on $\mathbb{S}^{n-1}_+$. It is positive for $0 < \beta \le L$.

Since $\Delta_{\mathbb{S}^{n-1}_+} f - r(b) f = 0$ and $r(b) \geq 0$, the strong minimum principle gives $f(0) > 0$. Smoothness and evenness at $\beta = 0$ give

$$\begin{aligned} e_b(\beta) &= \beta^{\frac{n-2}{2}} \Big(f(0) + O(\beta^2) \Big), \\ (\log e_b)'' &= -\frac{n-2}{2} \csc^2 \beta + O(1) \qquad \text{as } \beta \downarrow 0. \end{aligned} \tag{A.28}$$

From $A_b e_b = \mu_0(b) e_b$ and (A.19),

$$(\log e_b)'' = \frac{(n-2)(n-4)}{4} \csc^2 \beta - \mu_0(b) - \Big((\log e_b)' \Big)^2. \tag{A.29}$$

At any zero of $(\log e_b)''$ in $(0, L)$, differentiation of (A.29) gives

$$(\log e_b)''' = -\frac{(n-2)(n-4)}{2} \csc^2 \beta \cot \beta < 0.$$

Thus $(\log e_b)''$, which is negative near 0 by (A.28), has no first zero in $(0, L)$. Hence $\log e_b$ is concave on $(0, L]$. The tangent bound at L, where $(\log e_b)'(L) = b$, gives $e_b(\beta)^2 \leq C(b) e^{2b(\beta - L)}$. Integrating on $(0, L)$ and using $\|e_b\|_2 = 1$ gives $C(b) \geq 2b$ for $b > 0$. For $b = 0$ this bound follows from $C(0) > 0$.

Equation (A.23) gives $(\omega f')' = r(b) \omega f$. Since $\omega(0) f'(0) = 0$ and $f'(L) = b f(L)$, integration on $(0, L)$ gives

$$b C(b) = r(b) e_b(L) \int_0^L \sin^{\frac{n-2}{2}} \beta \, e_b(\beta) \, \mathrm{d}\beta.$$

Concavity of $\log e_b$ gives $e_b(L) e_b(\beta) \leq e_b((L + \beta)/2)^2$. Consequently

$$b C(b) \leq r(b) \int_0^L e_b \left(\frac{L + \beta}{2} \right)^2 \mathrm{d}\beta = 2 r(b) \int_{L/2}^L e_b(\beta)^2 \, \mathrm{d}\beta \leq 2 r(b).$$

The identification (A.24) takes the form domain with $f(L) = 0$ onto $\{ z \in \mathcal{Q} : z(L) = 0 \}$. The Dirichlet minimum in the proof of Lemma A.6, attained at $\cos \beta$, therefore gives the claimed eigenvalue and eigenfunction for this restriction of q_b. The strict inequalities $\mu_0(b) < n^2/4 < \mu_1(b)$ follow from the same lemma, and yield

$$\mu_1(b) - \mu_0(b) > \frac{n^2}{4} - \mu_0(b) = r(b) + n - 1.$$

□

We now prove the resolvent identity for $\mathcal{A}_b$ in all dimensions $n \geq 3$.

Lemma A.6. *Let $n \geq 3$ and $b \geq 0$. Write $\lambda_0 < \lambda_1 \leq \cdots$ for the eigenvalues of $\mathcal{A}_b$, and let e_j be corresponding real orthonormal*

eigenfunctions in $L^2((0,L),\omega\,\mathrm{d}\beta)$. *Then*

$$\lambda_0 < \frac{n^2}{4} < \lambda_1.$$

Let $\mathcal{A}_{b,\perp}$ *be the restriction of* $\mathcal{A}_b$ *to* $e_0^\perp$ *with respect to the inner product* $\langle f,\phi\rangle = \int_0^L \omega f\phi\,\mathrm{d}\beta$. *Write* $\ell(f) = f(L)$ *for the trace functional on* $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)$ *and for its restriction to* $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)\cap e_0^\perp$. *For* $\lambda_0 < \xi \le n^2/4$,

$$\begin{aligned} \ell\Big((\mathcal{A}_{b,\perp} - \tfrac{n^2}{4})^{-1}\ell\Big) &= \sum_{j\ge 1}\frac{e_j(L)^2}{\lambda_j - n^2/4} = \frac{e_0(L)^2}{n^2/4-\lambda_0}, \\ \ell\Big((\mathcal{A}_{b,\perp} - \xi)^{-1}\ell\Big) &\le \frac{e_0(L)^2}{n^2/4-\lambda_0} \le \frac{e_0(L)^2}{\xi-\lambda_0}. \end{aligned} \tag{A.30}$$

For each ξ *in this range, the inverse is defined by the bilinear form* $a_b[\cdot,\cdot] - \xi\,\langle\cdot,\cdot\rangle$ *on* $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)\cap e_0^\perp$ *and maps the dual of this subspace into the subspace itself.*

Proof. The interior expression in (A.23) applied to $-\cos\beta$ gives $-(n^2/4)\cos\beta$, since $(\omega\sin\beta)' = (n-1)\omega\cos\beta$. At L, the function $-\cos\beta$ has value 0 and derivative 1. Integration by parts in (A.21) therefore gives

$$a_b[-\cos\beta,\phi] - \frac{n^2}{4}\,\langle -\cos\beta,\phi\rangle = [\omega\sin\beta\,\phi]_0^L = \phi(L) \tag{A.31}$$

for every smooth cylindrically symmetric ϕ. The term at 0 vanishes since $\omega\sin\beta = \sin^{n-1}\beta$, and at L we used $\omega(L) = \sin L = 1$. By density of smooth cylindrically symmetric functions and the trace bound (A.22), this identity holds for every $\phi\in H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)$.

For tests with $\phi(L)=0$, (A.31) is the weak eigenvalue equation for $-\cos\beta$ on the form domain $\{f\in H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+) : f(L)=0\}$. This domain is closed by (A.22) and embeds compactly into $L^2((0,L),\omega\,\mathrm{d}\beta)$. Hence the minimum of $a_b[f]/\langle f,f\rangle$ over its nonzero functions is attained. The minimizer can be chosen nonnegative because taking the absolute value preserves this quotient. Its inner product with $\cos\beta$ is positive since $\cos\beta > 0$ for $0\le\beta<L$. Orthogonality of eigenfunctions for distinct eigenvalues shows that the Dirichlet minimum is $n^2/4$.

For the unrestricted form a_b on $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)$, the first eigenvalue λ_0 is the minimum of $a_b[f]/\langle f,f\rangle$ over $f\neq 0$. Taking $f=1$ gives

$$\lambda_0 \le \frac{a_b[1]}{\int_0^L \omega\,\mathrm{d}\beta} = \frac{(n-2)^2}{4} - \frac{b}{\int_0^L\omega\,\mathrm{d}\beta} \le \frac{(n-2)^2}{4} < \frac{n^2}{4}.$$

The condition $e_j'(L) = be_j(L)$ and uniqueness for $\mathcal{A}_b e_j = \lambda_j e_j$ with (A.23) give simplicity and $e_j(L) \neq 0$. The function $e_1(L)e_0 - e_0(L)e_1$ vanishes at L and has nonzero components along both e_0 and e_1. Testing the Dirichlet Rayleigh quotient with this function gives

$$\frac{n^2}{4} \leq \frac{\lambda_0 e_1(L)^2 + \lambda_1 e_0(L)^2}{e_1(L)^2 + e_0(L)^2} < \lambda_1.$$

The quotient follows from $\langle e_i, e_j \rangle = \delta_{ij}$ and $a_b[e_i, e_j] = \lambda_i \langle e_i, e_j \rangle$. The strict inequality uses $\lambda_0 < \lambda_1$ and $e_1(L) \neq 0$.

The compact embedding of $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)$ into $L^2((0,L), \omega \, \mathrm{d}\beta)$ gives compact resolvent for $\mathcal{A}_b$. By the spectral theorem, the squared form norm is

$$a_b[f] + (1 - \lambda_0) \langle f, f \rangle = \sum_{j \geq 0} (\lambda_j - \lambda_0 + 1) |\langle f, e_j \rangle|^2.$$

This norm controls $\langle f, f \rangle^{1/2}$ because $\lambda_j - \lambda_0 + 1 \geq 1$, and is equivalent to the H^1 norm by the shifted form bound for a_b. For $j \geq 1$ and $\lambda_0 < \xi \leq n^2/4$,

$$0 < \frac{\lambda_1 - n^2/4}{\lambda_1 - \lambda_0 + 1} \leq \frac{\lambda_j - \xi}{\lambda_j - \lambda_0 + 1} \leq 1.$$

Indeed, the middle ratio is $1 - (\xi - \lambda_0 + 1)/(\lambda_j - \lambda_0 + 1)$. Thus $a_b[\cdot, \cdot] - \xi \langle \cdot, \cdot \rangle$ is bounded and coercive on $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+) \cap e_0^{\perp}$ in the form norm. The Lax–Milgram theorem gives the inverse in (A.30).

Taking $\phi = e_0$ in (A.31) gives $(\lambda_0 - n^2/4) \langle -\cos\beta, e_0 \rangle = e_0(L)$. On $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+) \cap e_0^{\perp}$ the component along e_0 contributes zero to $a_b[\cdot, \cdot] - (n^2/4) \langle \cdot, \cdot \rangle$. Restricting (A.31) to test functions in this subspace gives

$$-\cos\beta = -\frac{e_0(L)}{n^2/4 - \lambda_0} e_0 + (\mathcal{A}_{b,\perp} - \tfrac{n^2}{4})^{-1} \ell.$$

Applying $\ell(f) = f(L)$ and using $\cos L = 0$ gives

$$\ell\Big((\mathcal{A}_{b,\perp} - \tfrac{n^2}{4})^{-1} \ell\Big) = \frac{e_0(L)^2}{n^2/4 - \lambda_0}.$$

By (A.22), ℓ is continuous in the form norm. Applying ℓ to the finite sum with coefficients $e_j(L)/(\lambda_j - \lambda_0 + 1)$ in the basis (e_j) gives its squared form norm. Continuity of ℓ bounds that norm independently of the number of terms, so

$$\sum_{j \geq 0} \frac{e_j(L)^2}{\lambda_j - \lambda_0 + 1} < \infty.$$

For $u = (\mathcal{A}_{b,\perp} - \xi)^{-1}\ell$, taking $\phi = e_j$ in $a_b[u,\phi] - \xi\,\langle u,\phi\rangle = \ell(\phi)$ gives $(\lambda_j - \xi)\,\langle u, e_j\rangle = e_j(L)$ for $j \geq 1$. Hence

$$(\mathcal{A}_{b,\perp} - \xi)^{-1}\ell = \sum_{j\geq 1} \frac{e_j(L)}{\lambda_j - \xi} e_j.$$

The lower bound for $(\lambda_j - \xi)/(\lambda_j - \lambda_0 + 1)$ and the summability of $e_j(L)^2/(\lambda_j - \lambda_0 + 1)$ give convergence in the form norm. Applying ℓ term by term gives the sum of $e_j(L)^2/(\lambda_j - \xi)$. Each term increases with ξ. At $\xi = n^2/4$ the sum gives the first line of (A.30). Comparison with that value and $0 < \xi - \lambda_0 \leq n^2/4 - \lambda_0$ give the two inequalities. □

For $n \geq 5$, the unitary map in (A.24) takes $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+) \cap e_0^\perp$ onto $\mathcal{Q} \cap e_b^\perp$ and preserves the trace $\ell(f) = f(L)$, since $\sin L = 1$. By polarization of (A.24), the form $a_b[\cdot,\cdot] - \xi\,\langle\cdot,\cdot\rangle$ corresponds to $q_b[\cdot,\cdot] - \xi\langle\cdot,\cdot\rangle_{L^2(0,L)}$ on $\mathcal{Q} \cap e_b^\perp$. Thus (A.30) holds also for A_b and its restriction to $e_b^\perp$ in $L^2(0,L)$.

Appendix B. Uniqueness in dimensions five and higher

We prove the uniqueness statements used in Proposition 3.1 for $n \geq 5$. We first derive a differential inequality for the trace $Y(t,L)$ of a solution Y of (B.1). We then integrate this inequality over $[0,\infty)$ and use an expansion in the eigenfunctions for the coefficient $b(0)$ in (B.1) to prove uniqueness for odd and even solutions.

B.1. **An estimate for the trace.** Use β, L and ω as in Section 2.1. Let b be smooth and even on $\mathbb{R}$, with $0 < b \leq n/2$ and $|b'| \leq b$. Let Y be smooth and cylindrically symmetric on $\mathbb{R} \times \mathbb{S}^{n-1}_+$ and satisfy

$$\text{(B.1)} \qquad \begin{cases} \mathcal{A}Y = 0 & \text{in } \mathbb{R} \times \mathbb{S}^{n-1}_+, \\ \partial_\beta Y(t,L) = b(t) Y(t,L) & \text{for } t \in \mathbb{R}, \end{cases}$$

where $\mathcal{A}$ is defined in Section 2.3. Set

$$D_Y(t,\beta) = \sin^{\frac{n-2}{2}}\beta\, Y(t,\beta).$$

By (A.24), $t \mapsto D_Y(t,\cdot)$ is smooth with values in $\mathcal{Q} = \{z \in H^1(0,L) : z(0) = 0\}$. Also $D_Y(t,L) = Y(t,L)$ since $\sin L = 1$. Let ϕ be smooth and cylindrically symmetric on $\mathbb{S}^{n-1}_+$. By the interior equation in (B.1), integration by parts in (A.21) gives

$$a_{b(t)}[Y(t,\cdot),\phi] = \int_0^L \omega\, \partial_t^2 Y(t,\beta)\phi(\beta)\, \mathrm{d}\beta.$$

Indeed, $\omega Y_\beta \phi$ vanishes at $\beta = 0$ by smoothness and $\omega(0) = 0$. At $\beta = L$, it equals $b(t)Y(t, L)\phi(L)$ by the boundary condition in (B.1) and $\omega(L) = 1$. By density and the bilinear identity associated with (A.24),

$$\langle (D_Y)_{tt}, z\rangle = q_{b(t)}[D_Y, z] \qquad \text{for every } z \in \mathcal{Q}. \tag{B.2}$$

We write $\langle \cdot, \cdot \rangle$ and $\| \cdot \|_2$ for the inner product and norm in $L^2(0, L)$.

For each $t \in \mathbb{R}$, let $\mu = \mu_0(b(t))$ and $e = e_{b(t)}$ be the first eigenvalue of $A_{b(t)}$ and its positive normalized eigenfunction, as in Proposition A.5. In this subsection, q_b, C and r depend on t through $b(t)$. Write

$$D_Y = ue + H, \qquad (D_Y)_t = we + J,$$

where $H, J \in \mathcal{Q}$ and $H, J \perp e$ in $L^2(0, L)$. Since $\langle e_t, e\rangle = 0$ by normalization, differentiating the projections and using (B.2) gives

$$\begin{aligned} u' &= w + \langle H, e_t\rangle, \\ w' &= \mu u + \langle J, e_t\rangle, \\ \frac{\mathrm{d}}{\mathrm{d}t}\langle H, J\rangle &= q_b[H] + \|J\|_2^2 - u\langle e_t, J\rangle - w\langle e_t, H\rangle. \end{aligned} \tag{B.3}$$

The last identity follows by pairing $H_t = J + (w - u')e - ue_t$ with J and $J_t = (D_Y)_{tt} - w'e - we_t$ with H.

We use the weight $e^{-(n-1)t}$ and set

$$\begin{gathered} F = \langle H, J\rangle - \frac{n-1}{4}u^2 - \frac{w^2}{n-1}, \\ x_\pm = \frac{n-1}{2}u \pm w, \qquad P = \frac{1}{4} - \frac{\mu}{(n-1)^2}, \qquad Q = 1 - P, \\ \mathcal{E} = q_b[H] + \|J\|_2^2 - (n-1)\langle H, J\rangle. \end{gathered}$$

By (A.25) and $b \le n/2$,

$$\begin{aligned} (n-1)^2 P &= r + \frac{2n-3}{4} > 0, \\ (n-1)^2 Q &= \frac{3(n-1)^2}{4} + \mu \\ &\ge \frac{3(n-1)^2 - 6(n-1) + 2}{4} > 0. \end{aligned}$$

Since $H \perp e$, Proposition A.5 gives $q_b[H] \ge \mu_1(b)\|H\|_2^2$ with $\mu_1(b) > n^2/4$. In particular, $\mathcal{E} \ge 0$ by the identity

$$\mathcal{E} = q_b[H] - \frac{(n-1)^2}{4}\|H\|_2^2 + \left\| J - \frac{n-1}{2}H \right\|_2^2. \tag{B.4}$$

By (B.3),

$$F' - (n-1)F = Px_+^2 + Qx_-^2 + \mathcal{E} - x_+ \left\langle e_t, H + \frac{2J}{n-1} \right\rangle. \tag{B.5}$$

Fix $t \in \mathbb{R}$ and let $(e_j)_{j\geq 0}$ be an orthonormal basis of eigenfunctions of $A_{b(t)}$, with $e_0 = e$. Let $\mu_j = \mu_j(b(t))$ and $\delta = n^2/4 - \mu = r + n - 1$. The smooth dependence in Proposition A.5 allows us to differentiate the weak eigenvalue equation for e, holding e_j fixed. For $j \geq 1$, using $\partial_b q_b[z,\phi] = -z(L)\phi(L)$ from (A.17) gives

$$\langle e_t, e_j \rangle = \frac{b'\sqrt{C}\, e_j(L)}{\mu_j - \mu}. \tag{B.6}$$

Write $H + 2J/(n-1) = 2H + 2(J - (n-1)H/2)/(n-1)$. Applying Cauchy–Schwarz in the eigenfunction expansion with the two terms of (B.4) gives

$$\left\langle e_t, H + \frac{2J}{n-1} \right\rangle^2 \leq \frac{4(b')^2 C}{(n-1)^2}\, \mathcal{E} \sum_{j\geq 1} \frac{e_j(L)^2}{(\mu_j - \mu)^2} \frac{\mu_j + 3(n-1)^2/4}{\mu_j - (n-1)^2/4}.$$

Since $Q > 0$, the ratio $(\mu_j + 3(n-1)^2/4)/(\mu_j - \mu)$ is at most its value with μ_j replaced by $n^2/4$. Also $(2n-1)/(n-1)^2 \leq 9/16$. Thus

$$\frac{4}{(n-1)^2} \frac{\mu_j + 3(n-1)^2/4}{\mu_j - \mu} \leq \frac{73}{16\delta}.$$

For $z = \mu_j - n^2/4 > 0$, the arithmetic–geometric mean inequality gives

$$\mu_j - \mu \geq 2\sqrt{z\delta}, \qquad \mu_j - \frac{(n-1)^2}{4} \geq \sqrt{z(2n-1)}.$$

By (A.24), the resolvent identity (A.30) applies to $(e_j)_{j\geq 0}$. It follows that

$$\left\langle e_t, H + \frac{2J}{n-1} \right\rangle^2 \leq \frac{73(b'C)^2}{32\delta^{5/2}\sqrt{2n-1}}\, \mathcal{E}.$$

To compare the coefficient with P, use $|b'|C \leq bC \leq 2r$ by (A.26). With $s = r/(n-1)$, the ratio of the coefficient to P is at most

$$\frac{73\sqrt{2}}{16} \frac{s^2}{(1+s)^{5/2}(s + 7/16)}.$$

Here we used $2n - 1 \geq 2(n-1)$ and $(2n-3)/4 \geq 7(n-1)/16$. Since $s + 7/16 \geq (\sqrt{7}/2)\sqrt{s}$ and $s^{3/2}/(1+s)^{5/2}$ is largest at $s = 3/2$, the square of this upper bound is at most $143883/175000 < 1$. Applying this estimate in (B.5) gives

$$F' - (n-1)F \geq \frac{1}{2} Px_+^2 + Qx_-^2 + \frac{1}{2}\mathcal{E}. \tag{B.7}$$

By Cauchy–Schwarz in the eigenfunction expansion of H and (A.30),

$$\frac{\delta}{C}|H(t,L)|^2 + \frac{n^2}{4}\|H\|_2^2 \le q_b[H].$$

By (B.4), $|H(t,L)|^2 \le C\mathcal{E}/\delta$. Write $Y(t,L) = \sqrt{C}\,(x_+ + x_-)/(n-1) + H(t,L)$ and apply Cauchy–Schwarz using the three terms on the right-hand side of (B.7). The coefficient of $F' - (n-1)F$ in the bound for $b|Y(t,L)|^2/2$ is at most

$$\begin{aligned}\frac{bC}{(n-1)^2P} + \frac{bC}{2(n-1)^2Q} + \frac{bC}{\delta} \\ \le \frac{2r}{r+(2n-3)/4} + \frac{2r}{r+n-1} + \frac{r}{3(n-1)^2/4 + (n-2)^2/4 - r}.\end{aligned}$$

Here we used $bC \le 2r$ from (A.26). Since $r \le (n^2+2n-4)/4$ and $n-1 \ge 4$, we have

$$\begin{aligned}2 - \frac{2r}{r+(2n-3)/4} &\ge \frac{2(2n-3)}{(n-1)^2+6(n-1)-2} \ge \frac{7}{5(n-1)}, \\ 2 - \frac{2r}{r+n-1} &\ge \frac{8(n-1)}{(n-1)^2+8(n-1)-1} \ge \frac{8}{3(n-1)}.\end{aligned}$$

The third fraction is at most

$$\frac{(n-1)^2+4(n-1)-1}{3(n-1)^2-6(n-1)+2} \le \frac{1}{3} + \frac{4}{n-1},$$

since $3(n-1)^2-6(n-1)+2 \ge 3(n-1)^2/2$. The three fractions therefore sum to less than $13/3$. This proves

$$b|Y(t,L)|^2 \le \frac{26}{3}\Big(F' - (n-1)F\Big). \tag{B.8}$$

B.2. **Uniqueness for odd and even solutions.**

Lemma B.1. *Let $n \ge 5$ and let b be smooth and even on $\mathbb{R}$, with $0 < b \le n/2$ and $|b'| \le b$. Let Y be a smooth cylindrically symmetric solution of (B.1) on $\mathbb{R}\times\mathbb{S}^{n-1}_+$. Suppose that Y and $\partial_t Y$ have polynomial growth in the sense of (2.5). Then $Y \equiv 0$ in either of the following cases.*

- (i) *Y is odd in t, $\partial_t Y(0,L) = 0$, and $|b'(t)|/b(t) \le (n-2)t/2$ for $t \ge 0$.*
- (ii) *Y is even in t and $Y(0,L) = 0$.*

Proof. The growth assumption gives $e^{-(n-1)t}F(t) \to 0$ as $t \to \infty$. If Y is odd, then $D_Y(0,\cdot) = 0$ and $F(0) = -w(0)^2/(n-1)$. If Y is even,

then $(D_Y)_t(0,\cdot) = 0$ and $F(0) = -(n-1)u(0)^2/4$. Integrating (B.8) with weight $e^{-(n-1)t}$ gives

(B.9)
$$I := \int_0^\infty e^{-(n-1)t} b(t) |Y(t,L)|^2 \, \mathrm{d}t \le \begin{cases} \dfrac{26}{3(n-1)} w(0)^2 & \text{if } Y \text{ is odd,} \\ \dfrac{13(n-1)}{6} u(0)^2 & \text{if } Y \text{ is even.} \end{cases}$$

For the rest of the proof, let $\mu_j = \mu_j(b(0))$ and fix an orthonormal basis $(e_j)_{j\ge 0}$ of eigenfunctions of $A_{b(0)}$, with $e_0 = e_{b(0)}$. We evaluate C, r and $\delta = n^2/4 - \mu_0$ at $b(0)$. Let $V = (D_Y)_t(0,\cdot)$ if Y is odd and $V = D_Y(0,\cdot)$ if Y is even. Set $V_j = \langle V, e_j \rangle$. Thus $V_0 = w(0)$ in the odd case and $V_0 = u(0)$ in the even case.

In both cases, $V(L) = 0$ by the hypothesis at $(0,L)$. The variational characterization of $n^2/4$ on $\{z \in \mathcal{Q} : z(L) = 0\}$ in Proposition A.5 gives $q_{b(0)}[V] \ge (n^2/4)\|V\|_2^2$. Expanding in the fixed basis gives

$$\delta |V_0|^2 \le \sum_{j\ge 1} (\mu_j - n^2/4) |V_j|^2. \tag{B.10}$$

For $j \ge 1$, set $\kappa_j = \sqrt{\mu_j} > n/2$ and $Y_j(t) = \langle D_Y(t,\cdot), e_j \rangle$. By (B.2) with $z = e_j$,

$$-Y_j'' + \kappa_j^2 Y_j = e_j(L)\big(b(t) - b(0)\big) Y(t,L). \tag{B.11}$$

The right-hand side is $q_{b(0)}[D_Y, e_j] - q_{b(t)}[D_Y, e_j]$ by (A.17). By parity, $Y_j(0) = 0$ if Y is odd and $Y_j'(0) = 0$ if Y is even. Multiply (B.11) by $e^{-\kappa_j t}$ and integrate by parts on $[0,\infty)$. The polynomial growth of Y and $\partial_t Y$ gives

$$Y_j'(0) + \kappa_j Y_j(0) = e_j(L) \int_0^\infty e^{-\kappa_j t} \big(b(t) - b(0)\big) Y(t,L) \, \mathrm{d}t. \tag{B.12}$$

The left-hand side is V_j if Y is odd and $\kappa_j V_j$ if Y is even.

Integrating $|b'|/b \le 1$ on $[0,t]$ gives $|\log(b(t)/b(0))| \le t$. Since $\sinh^2$ is even and increasing on $[0,\infty)$, we have

(B.13)
$$\frac{|b(t) - b(0)|^2}{b(t)} = 4b(0) \sinh^2 \frac{\log b(t) - \log b(0)}{2} \le b(0) e^t (1 - e^{-t})^2.$$

In case (i), integrating $|b'(t)|/b(t) \le (n-2)t/2$ gives $|\log(b(t)/b(0))| \le (n-2)t^2/4$. By convexity, $\sinh(\lambda t/2) \le \lambda \sinh(t/2)$ for $0 \le \lambda \le 1$. Thus in the odd case the right-hand side of (B.13) can be multiplied by $\min\{1, (n-2)^2 t^2/16\}$.

Applying Cauchy–Schwarz in (B.12) with the integral I in (B.9) gives

$$|Y_j'(0)+\kappa_j Y_j(0)|^2 \le e_j(L)^2 I \int_0^\infty e^{-(2\kappa_j-n+1)t}\frac{|b(t)-b(0)|^2}{b(t)}\,\mathrm{d}t.$$

Put $x=\kappa_j-n/2>0$, so that $\mu_j-n^2/4=x(x+n)$. Using (B.13), with the additional factor $\min\{1,(n-2)^2t^2/16\}$ in the odd case, reduces the estimates for odd and even Y to the first and second inequalities below, respectively.

(B.14)
$$x^2(x+n)^2\int_0^\infty e^{-2xt}(1-e^{-t})^2\min\{1,(n-2)^2t^2/16\}\,\mathrm{d}t \le \frac{15n^2}{112},$$
$$\frac{x^2(x+n)^2}{(x+n/2)^2}\int_0^\infty e^{-2xt}(1-e^{-t})^2\,\mathrm{d}t \le \frac{3}{8}.$$

Expanding the square in the integral without the minimum gives

$$\int_0^\infty e^{-2xt}(1-e^{-t})^2\,\mathrm{d}t = \frac{1}{2x(x+1)(2x+1)}.$$

For $0<x\le 5/2$, we omit the minimum, divide the bound for the first line of (B.14) by n^2, and use $n\ge 5$. This gives at most $\frac{x(x+5)^2}{50(x+1)(2x+1)}$, which is at most $15/112$ since

$$750(x+1)(2x+1)-112x(x+5)^2=(5-2x)(56x^2-50x+150)\ge 0.$$

For $5/2\le x\le n/2$, use $1+x/n\le 3/2$ and the decrease of $\frac{x}{(x+1)(2x+1)}$ on $[5/2,\infty)$ to obtain the same bound. For $x\ge n/2$, use $1-e^{-t}\le t$ and integrate t^4e^{-2xt}. The left-hand side of the first line, divided by n^2, is then at most

$$\frac{3(n-2)^2(x+n)^2}{64n^2x^3}\le\frac{27(n-2)^2}{32n^3}\le\frac{1}{16}.$$

The first inequality uses $x+n\le 3x$ and $x\ge n/2$. The derivative of $(n-2)^2/n^3$ is $(n-2)(6-n)/n^4$, so its maximum on $[5,\infty)$ occurs at $n=6$. For the second line of (B.14), we use $(x+n)/(x+n/2)\le 2$ and $\frac{2x}{(x+1)(2x+1)}\le 3/8$, which follows from $6x^2-7x+3>0$.

The inequalities (B.14) bound the sum in (B.10) by

$$\begin{cases}\dfrac{15n^2}{112}\dfrac{b(0)C}{\delta}I & \text{if } Y \text{ is odd},\\[2ex] \dfrac{3}{8}\dfrac{b(0)C}{\delta}I & \text{if } Y \text{ is even}.\end{cases}$$

Indeed, in both cases the spectral sum is $\sum_{j\geq 1} e_j(L)^2/(\mu_j - n^2/4) = C/\delta$ by (A.30). Using $b(0)C \leq 2r$ from (A.26) gives

$$\frac{b(0)C}{\delta^2} \leq \frac{2r}{(r+n-1)^2} \leq \frac{1}{2(n-1)}.$$

Consequently, by (B.9),

$$\sum_{j\geq 1}(\mu_j - n^2/4)|V_j|^2 \leq \begin{cases} \dfrac{1625}{1792}\delta|V_0|^2 & \text{if } Y \text{ is odd,} \\ \dfrac{13}{32}\delta|V_0|^2 & \text{if } Y \text{ is even.} \end{cases}$$

In the odd case we used $n/(n-1) \leq 5/4$. Both constants are less than 1, so (B.10) gives $V_0 = 0$ and hence $F(0) = 0$. Integrating (B.7) with weight $e^{-(n-1)t}$ gives $x_+ = x_- = 0$ and $\mathcal{E} = 0$ on $[0,\infty)$. Thus $u = w = 0$, and (B.4) gives $H = J = 0$ since $\mu_1(b(t)) > n^2/4$. By parity, $Y \equiv 0$ on $\mathbb{R}\times\mathbb{S}^{n-1}_+$. □

The growth assumption in Lemma B.1 can be replaced by

$$\liminf_{t\to\infty} e^{-(n-1)t/2}\Big(\|D_Y(t,\cdot)\|_2 + \|(D_Y)_t(t,\cdot)\|_2\Big) = 0.$$

Choose $t_i \to \infty$ along which the expression in the limit tends to zero. Then $e^{-(n-1)t_i}F(t_i) \to 0$, and the nonnegativity in (B.7) gives (B.9). Since $\kappa_j > (n-1)/2$, the boundary term in the integration leading to (B.12) vanishes along the same sequence. The Cauchy–Schwarz estimate using (B.9) and (B.14) gives absolute convergence of the integral in (B.12).

Appendix C. Uniqueness in dimensions three and four

We prove the uniqueness statements used in Proposition 3.1 for $n = 3, 4$. We expand $Y(t,\cdot)$ in the eigenfunctions of $\mathcal{A}_{n/4}$, defined by (A.21). For each parity, the condition at $(0, L)$ determines an integral equation for $Y(t, L)$. We estimate its kernel separately in dimensions three and four.

Theorem C.1. *Let $n \in \{3, 4\}$ and let b be smooth and even on $\mathbb{R}$, with $0 \leq b \leq n/2$. Let Y be smooth and cylindrically symmetric on $\mathbb{R}\times\mathbb{S}^{n-1}_+$, with polynomial growth in the sense of (2.5). Suppose that*

$$\text{(C.1)} \qquad \begin{cases} \mathcal{A}Y = 0 & \text{in } \mathbb{R}\times\mathbb{S}^{n-1}_+, \\ \partial_\beta Y(t,L) = b(t)Y(t,L) & \text{for } t \in \mathbb{R}. \end{cases}$$

Here $\mathcal{A}$ is as in Section 2.3. Then $Y \equiv 0$ on $\mathbb{R}\times\mathbb{S}^{n-1}_+$ in either of the following cases.

(i) *Y is odd in t and $\partial_t Y(0,L) = 0$.*

(ii) *Y is even in t and $Y(0, L) = 0$.*

Proof. Use β, L and ω as in Section 2.1. Set $b_* = n/4$. Let $\mathcal{A}_{b_*}$ be the operator associated with a_{b_*} in (A.21), with form domain $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)$. The inner product $\langle \cdot, \cdot \rangle$ is taken in $L^2((0, L), \omega \, \mathrm{d}\beta)$.

Testing (A.21) with the constant function 1 gives

$$a_{b_*}[1] = \begin{cases} \dfrac{1}{4}\displaystyle\int_0^L \sin\beta \, \mathrm{d}\beta - \frac{3}{4} = -\frac{1}{2} & (n = 3), \\ \displaystyle\int_0^L \sin^2\beta \, \mathrm{d}\beta - 1 = \frac{L}{2} - 1 < 0 & (n = 4). \end{cases}$$

Let λ_j and e_j, $j \geq 0$, be the eigenvalues and real orthonormal eigenfunctions of $\mathcal{A}_{b_*}$. By the Rayleigh quotient and Lemma A.6, $\lambda_0 < 0$ and $\lambda_1 > n^2/4$. We choose e_0 positive on $(0, L]$ and the signs of the other eigenfunctions so that $e_j(L) > 0$. The absolute value of a Rayleigh minimizer is also a minimizer. If this nonnegative eigenfunction vanishes at some $\beta \in (0, L)$, then its derivative vanishes there and uniqueness for the eigenfunction equation gives $e_0 = 0$. At $\beta = L$, the condition $e_j'(L) = b_* e_j(L)$ and the same uniqueness argument give $e_j(L) \neq 0$ for every $j \geq 0$. Write $\kappa_0 = \sqrt{-\lambda_0}$, $k_j = \sqrt{\lambda_j}$ for $j \geq 1$, and $C_j = e_j(L)^2$. By (A.30),

$$A_* := \sum_{j \geq 1} \frac{C_j}{k_j^2 - n^2/4} = \frac{C_0}{n^2/4 + \kappa_0^2}. \tag{C.2}$$

Let $F(t) = (b(t) - b_*)Y(t, L)$ and $Y_j(t) = \langle Y(t, \cdot), e_j \rangle$. Testing (C.1) against e_j and integrating by parts in β gives

$$-Y_j'' + \lambda_j Y_j = e_j(L) F. \tag{C.3}$$

At $\beta = 0$ the boundary term vanishes by smoothness and $\omega(0) = 0$. At $\beta = L$ the difference of the boundary terms is $F(t) e_j(L)$ by (C.1) and $e_j'(L) = b_* e_j(L)$. For $j \geq 1$, the conditions at $t = 0$ are $Y_j(0) = 0$ in the odd case and $Y_j'(0) = 0$ in the even case. Thus

$$\begin{aligned} Y_j(t) &= e_j(L) \int_0^\infty G_{k_j}^{D/N}(t, \tau) F(\tau) \, \mathrm{d}\tau, \\ G_{k_j}^{D/N}(t, \tau) &= \frac{e^{-k_j|t-\tau|} \mp e^{-k_j(t+\tau)}}{2k_j}. \end{aligned} \tag{C.4}$$

The minus sign gives the Dirichlet condition and the plus sign gives the Neumann condition at $t = 0$. The jump of $\partial_t G_{k_j}^{D/N}$ at $t = \tau$ is -1, so $(-\partial_t^2 + k_j^2) G_{k_j}^{D/N}(t, \tau) = \delta_\tau$ in distributions on $(0, \infty)$, where δ_τ is the Dirac measure at τ. Both kernels are nonnegative since $|t - \tau| \leq t + \tau$

for $t, \tau \geq 0$. The integral in (C.4) has polynomial growth, since F has polynomial growth and the kernel decays exponentially in $|t-\tau|$. Applying $-\partial_t^2 + k_j^2$ to the difference between Y_j and the first line of (C.4) gives zero. Polynomial growth excludes the term in $e^{k_j t}$, and the condition at 0 excludes the term in $e^{-k_j t}$.

For $\lambda > n/2$, use G_λ^D and G_λ^N from (C.4) with k_j replaced by λ. Then

$$\begin{aligned} \int_0^\infty G_\lambda^D(t,\tau)\sinh(n\tau/2)\ \mathrm{d}\tau &= \frac{\sinh(nt/2)}{\lambda^2 - n^2/4}, \\ \int_0^\infty G_\lambda^N(t,\tau)\cosh(n\tau/2)\ \mathrm{d}\tau &= \frac{\cosh(nt/2)}{\lambda^2 - n^2/4}. \end{aligned} \tag{C.5}$$

Applying $-\partial_t^2 + \lambda^2$ to both sides gives $\sinh(nt/2)$ in the first line and $\cosh(nt/2)$ in the second. Both sides of the first line vanish at $t = 0$, and both sides of the second have zero t-derivative there. The growth bound $O(e^{nt/2})$ and $\lambda > n/2$ give uniqueness. Let $h(t) = \sinh(nt/2)$ if Y is odd and $h(t) = \cosh(nt/2)$ if Y is even. Smoothness, parity and polynomial growth give $|F| \leq Mh$ on $[0,\infty)$ for some $M > 0$. By (C.5), the jth term $e_j(L)Y_j(t)$ for $j \geq 1$ is bounded in absolute value by $Mh(t)C_j/(k_j^2 - n^2/4)$. The series converges locally uniformly in t by (C.2). Its sum, including the term $j = 0$, is $Y(t, L)$ because the expansion of $Y(t,\cdot)$ converges in the shifted form norm from the proof of Lemma A.6. The map $f \mapsto f(L)$ is continuous in that norm by (A.22).

For $j \geq 1$, multiply (C.3) by $e^{-k_j t}$ and integrate by parts on $[0,\infty)$ to obtain

$$Y_j'(0) + k_j Y_j(0) = e_j(L)\int_0^\infty e^{-k_j\tau}F(\tau)\ \mathrm{d}\tau.$$

The term at infinity vanishes since Y_j and F have polynomial growth and (C.3) gives polynomial growth of Y_j'' and hence of Y_j'. Expand $\partial_t Y(0,\cdot)$ in the odd case and $Y(0,\cdot)$ in the even case in the same shifted form norm. Taking the trace at $\beta = L$ and using parity gives

$$\begin{aligned} \partial_t Y(0,L) &= e_0(L)Y_0'(0) + \sum_{j\geq 1} C_j \int_0^\infty e^{-k_j\tau}F(\tau)\ \mathrm{d}\tau \quad \text{if } Y \text{ is odd}, \\ Y(0,L) &= e_0(L)Y_0(0) + \sum_{j\geq 1} \frac{C_j}{k_j}\int_0^\infty e^{-k_j\tau}F(\tau)\ \mathrm{d}\tau \quad \text{if } Y \text{ is even}. \end{aligned} \tag{C.6}$$

Let $S(t) = \sin(\kappa_0 t)/\kappa_0$ and $P = \pi/\kappa_0$. By (C.3), $Y_0'' + \kappa_0^2 Y_0 = -e_0(L)F$. Its solution has homogeneous term $Y_0'(0)S(t)$ in the odd case and $Y_0(0)\cos(\kappa_0 t)$ in the even case. In either case the term with forcing is $-e_0(L)\int_0^t S(t-\tau)F(\tau)\ \mathrm{d}\tau$. The hypotheses at $(0, L)$ and (C.6) determine the homogeneous term. Using it in the expansion of $Y(t, L)$

gives, for $t \geq 0$,

$$\begin{aligned} Y(t,L) &= \int_0^\infty K^D(t,\tau)F(\tau)\,\mathrm{d}\tau \quad \text{if } Y \text{ is odd,} \\ Y(t,L) &= \int_0^\infty K^N(t,\tau)F(\tau)\,\mathrm{d}\tau \quad \text{if } Y \text{ is even,} \end{aligned} \tag{C.7}$$

where

$$\begin{aligned} K^D(t,\tau) &= -C_0 S(t-\tau)\mathbf{1}_{\{\tau<t\}} \\ &\quad + \sum_{j\geq 1} C_j\Big(G^D_{k_j}(t,\tau) - S(t)e^{-k_j\tau}\Big), \end{aligned} \tag{C.8}$$

$$\begin{aligned} K^N(t,\tau) &= -C_0 S(t-\tau)\mathbf{1}_{\{\tau<t\}} \\ &\quad + \sum_{j\geq 1} C_j\left(G^N_{k_j}(t,\tau) - \frac{\cos(\kappa_0 t)}{k_j}e^{-k_j\tau}\right). \end{aligned} \tag{C.9}$$

For each parity, the corresponding series converges in $L^1((0,\infty), h(\tau)\,\mathrm{d}\tau)$ by (C.2), (C.5) and

$$\begin{aligned} \int_0^\infty e^{-k_j\tau}\sinh(n\tau/2)\,\mathrm{d}\tau &= \frac{n/2}{k_j^2 - n^2/4}, \\ \int_0^\infty e^{-k_j\tau}\cosh(n\tau/2)\,\mathrm{d}\tau &= \frac{k_j}{k_j^2 - n^2/4}. \end{aligned}$$

Since $|F| \leq Mh$, these estimates also justify integration of the series in (C.7).

Bounds for the kernels. Define

$$\begin{aligned} Q^D(t) &= \frac{\displaystyle\int_0^\infty |K^D(t,\tau)|\sinh(n\tau/2)\,\mathrm{d}\tau}{\sinh(nt/2)} \quad (t>0), \\ Q^N(t) &= \frac{\displaystyle\int_0^\infty |K^N(t,\tau)|\cosh(n\tau/2)\,\mathrm{d}\tau}{\cosh(nt/2)} \quad (t\geq 0). \end{aligned}$$

We bound b_*Q^D on $(0,\infty)$ and b_*Q^N on $[0,\infty)$. By (C.2) and (C.5), the series in (C.8) and (C.9) contribute at most $A_*\sinh(nt/2) + (n/2)A_*|S(t)|$ and $A_*\cosh(nt/2) + A_*|\cos(\kappa_0 t)|$ to the respective integrals. For the remaining term, set

$$\begin{aligned} I^D(t) &= \int_0^t |S(t-\tau)|\sinh(n\tau/2)\,\mathrm{d}\tau, \\ I^N(t) &= \int_0^t |S(t-\tau)|\cosh(n\tau/2)\,\mathrm{d}\tau. \end{aligned}$$

Since $S \geq 0$ on $[0, P]$, applying $\partial_t^2 + \kappa_0^2$ to I^D and I^N gives $\sinh(nt/2)$ and $\cosh(nt/2)$, respectively, on this interval. Both integrals and their first derivatives vanish at 0. Hence on $[0, P]$,

$$\begin{aligned}(n^2/4 + \kappa_0^2) I^D(t) &= \sinh(nt/2) - \frac{n}{2} S(t), \\ (n^2/4 + \kappa_0^2) I^N(t) &= \cosh(nt/2) - \cos(\kappa_0 t).\end{aligned} \tag{C.10}$$

For $t = mP + \delta$, where $m \geq 0$ is an integer and $0 \leq \delta < P$, the identity

$$|S(r)| = S(r) + 2 \sum_{1 \leq j \leq r/P} S(r - jP), \qquad r \geq 0,$$

follows from $S(r - jP) = (-1)^j S(r)$. The convolutions with S in place of $|S|$ satisfy (C.10) on $[0, \infty)$. Applying the identity for $|S|$ then gives

$$\begin{aligned}(n^2/4 + \kappa_0^2) I^D(t) &= \sinh(nt/2) - \frac{n}{2} |S(t)| + 2 \sum_{j=1}^{m} \sinh(n(t - jP)/2), \\ (n^2/4 + \kappa_0^2) I^N(t) &= \cosh(nt/2) - \cos(\kappa_0 \delta) + 2 \sum_{j=1}^{m} \cosh(n(t - jP)/2).\end{aligned}$$

For $1 \leq j \leq m$,

$$\begin{aligned}\frac{\sinh(n(t - jP)/2)}{\sinh(nt/2)} &\leq e^{-njP/2}, \\ \frac{\cosh(n(t - jP)/2)}{\cosh(nt/2)} &\leq 2e^{-njP/2}.\end{aligned}$$

Summing the geometric series gives, for $t \geq P$,

$$\begin{aligned}(n^2/4 + \kappa_0^2) I^D(t) &\leq \sinh(nt/2) + \frac{2 \sinh(nt/2)}{e^{nP/2} - 1} - \frac{n}{2} |S(t)|, \\ (n^2/4 + \kappa_0^2) I^N(t) &\leq \cosh(nt/2) + \frac{4 \cosh(nt/2)}{e^{nP/2} - 1} + 1.\end{aligned} \tag{C.11}$$

Combining these bounds with (C.10) and the estimates of the two series gives

$$\begin{aligned}\frac{Q^D(t)}{2A_*} &\leq 1 && (0 < t \leq P), \\ \frac{Q^D(t)}{2A_*} &\leq 1 + \frac{1}{e^{nP/2} - 1} && (t \geq P),\end{aligned}$$

$$\begin{aligned}
\frac{Q^N(t)}{2A_*} &\le 1 && (0 \le t \le P/2),\\
\frac{Q^N(t)}{2A_*} &\le 1 + \frac{4\kappa_0}{n} e^{-1-nP/4} && (P/2 \le t \le P),\\
\frac{Q^N(t)}{2A_*} &\le 1 + \frac{2}{e^{nP/2}-1} + \frac{1}{\cosh(nP/2)} && (t \ge P).
\end{aligned} \tag{C.12}$$

In the fourth line, use $|\cos(\kappa_0 t)| \le \kappa_0(t - P/2)$, $1/\cosh(nt/2) \le 2e^{-nt/2}$, and $\sup_{s\ge 0} s e^{-ns/2} = 2/(ne)$.

Dimension three. Here $b_* = 3/4$. In the variable $x = \cos\beta$ the expression (A.23) is $T = -\partial_x((1-x^2)\partial_x) + 1/4$ on $(0,1)$. Consider $f(x) = e^{-9x/32}(1+x)^{-15/32}$. Since $f'(0) = -3f(0)/4$, the function $f(\cos\beta)$ belongs to $\mathrm{Dom}(\mathcal{A}_{b_*})$. Differentiation gives

$$\frac{(T + 25/32)f}{f} = \frac{x(1-x)}{1+x}\left(\frac{9}{64} - \frac{81x}{1024}\right) > 0 \qquad (0 < x < 1).$$

Pairing with e_0 and using self-adjointness gives

$$0 < \langle (\mathcal{A}_{b_*} + 25/32) f(\cos\beta), e_0\rangle = (25/32 - \kappa_0^2)\,\langle f(\cos\beta), e_0\rangle\,.$$

Thus $\kappa_0^2 < 25/32$. The eigenfunction equation integrated on $(0, L)$ gives

$$\frac{3}{4} e_0(L) = (\kappa_0^2 + 1/4)\int_0^L \sin\beta\, e_0 \,\mathrm{d}\beta.$$

Since $\|e_0\|_{L^2((0,L),\omega\,\mathrm{d}\beta)} = 1$ and $\int_0^L \sin\beta\,\mathrm{d}\beta = 1$, the integral is at most 1. By (C.2),

$$2b_*A_* \le \frac{8(\kappa_0^2 + 1/4)^2}{3(\kappa_0^2 + 9/4)} < \frac{363}{388} < \frac{15}{16}.$$

The second inequality uses $\kappa_0^2 < 25/32$ and the increase of $(x+1/4)^2/(x+9/4)$ for $x \ge 0$.

Also $\kappa_0 < 9/10$ gives $3P/2 > 5$. The fourth and fifth bounds in (C.12) are at most $1 + (6/5)e^{-7/2} < 1 + 1/20$ and $1 + 2/99 + 1/50 < 1 + 1/20$, respectively. Here $e^5 > 100$ and $e^{7/2} > 24$ follow from $e > 8/3$. These estimates give $b_*Q^D(t) < 63/64$ for $t > 0$ and $b_*Q^N(t) < 63/64$ for $t \ge 0$.

Dimension four. Here $b_* = 1$. By (A.23), $\psi_j = \sin\beta\, e_j$ satisfies

$$-\psi_j'' = \lambda_j\psi_j \quad \text{on } (0,L), \qquad \psi_j(0) = 0, \qquad \psi_j'(L) = \psi_j(L).$$

Multiplication by $\sin\beta$ preserves the norm from $L^2((0,L),\omega\,\mathrm{d}\beta)$ to $L^2(0,L)$. Conversely, the sine and hyperbolic sine solutions with $\psi_j(0) =$

0 give smooth cylindrically symmetric functions after division by $\sin\beta$, by their expansions at $\beta = 0$. Thus

(C.13)
$$\kappa_0 \coth(L\kappa_0) = 1, \qquad k_j \cot(Lk_j) = 1, \qquad 2j < k_j < 2j+1 \quad (j \geq 1).$$

Indeed, the derivative of $x\cot(Lx)$ is $\big(\sin(2Lx)/2 - Lx\big)/\sin^2(Lx) < 0$ where it is defined. The function decreases from $+\infty$ to 0 on $(2j, 2j+1)$ for $j \geq 1$. It is at most $1/L < 1$ on $(0,1)$ and nonpositive wherever the cotangent is nonpositive. Its values at $5/2$ and $17/6$ are $5/2 > 1$ and $17(2-\sqrt{3})/6 < 1$, respectively. Hence $5/2 < k_1 < 17/6$.

The eigenfunctions ψ_j are multiples of $\sinh(\kappa_0\beta)$ for $j = 0$ and $\sin(k_j\beta)$ for $j \geq 1$. Integrating their squares and using (C.13) gives

$$\int_0^L \sinh^2(\kappa_0\beta)\, \mathrm{d}\beta = \frac{\sinh^2(L\kappa_0)}{2\kappa_0^2} - \frac{L}{2},$$
$$\int_0^L \sin^2(k_j\beta)\, \mathrm{d}\beta = \frac{L}{2} - \frac{\sin^2(Lk_j)}{2k_j^2}.$$

Since $\sin L = 1$, division of the squared values at L by these integrals gives

$$C_0 = \frac{2\kappa_0^2}{1 - L + L\kappa_0^2}, \qquad C_j = \frac{2}{L + (L-1)/k_j^2} \quad (j \geq 1). \tag{C.14}$$

We claim that

$$\frac{441}{500} < \kappa_0 < \frac{89}{100}, \qquad P > \frac{7}{2},$$
$$C_0 > 2, \qquad C_1 > 1, \qquad 2A_* < 1. \tag{C.15}$$

By (C.13), $\tanh(L\kappa_0) = \kappa_0$. The strict concavity of tanh on $(0,\infty)$ implies that $\tanh(Ls)/s$ decreases strictly for $s > 0$.

The exponential series gives $1957/720 < e < 11/4$ and, for $s = 23/100$,

$$e^s \leq 1 + s + \frac{s^2}{2} + \frac{s^3}{6} + \frac{s^4}{24(1-s/5)} < \frac{1259}{1000}.$$

Indeed, the terms of degree at least four have successive ratios at most $s/5$. Consequently

$$e^{277/100} > \left(\frac{1957}{720}\right)^3 \frac{1000}{1259} > \frac{941}{59}.$$

Since $\pi > 223/71$, we have $2L(441/500) > 277/100$. The ratio $(1 + 441/500)/(1 - 441/500)$ equals $941/59$. Hence $\tanh(L(441/500)) >$

$441/500$ and $\kappa_0 > 441/500$. For the upper bound, $L < 11/7$ gives $2L(89/100) < 3 - 1/5$. Since

$$e^{3-1/5} < \frac{(11/4)^3}{1 + 1/5 + 1/50} < \frac{189}{11} = \frac{1 + 89/100}{1 - 89/100},$$

we have $\tanh(L(89/100)) < 89/100$. Thus $\kappa_0 < 89/100$ and $P > 7/2$.

The first denominator in (C.14) is positive by the integral formula for C_0. It is less than κ_0^2 since $L > 1$ and $\kappa_0 < 1$, so $C_0 > 2$. Also $k_1 > 5/2$ and $L < 11/7$ give $C_1 > 350/291 > 1$.

To prove $2A_* < 1$, write $x = \kappa_0^2$. Equations (C.2) and (C.14) give

$$\frac{4}{2A_*} = (4 + x)\frac{1 - L + Lx}{x}.$$

By (C.13), $L = \operatorname{arctanh}(\sqrt{x})/\sqrt{x}$. Thus the right-hand side is the value at $x = \kappa_0^2$ of the strictly increasing function on $(0, 1)$ given by

$$(4 + x)\sum_{j=0}^{\infty} \frac{2x^j}{(2j + 1)(2j + 3)}.$$

It suffices to prove that this function is greater than 4 at $x = (441/500)^2$. For $s = 441/500$, the inequality $e^{277/100} > 941/59$ gives $2\operatorname{arctanh}(s) < 277/100$. Since

$$\frac{277}{100} < \frac{2s(4 - 3s^2)}{(4 + s^2)(1 - s^2)},$$

we have

$$(4 + s^2)\frac{1 - (1 - s^2)\operatorname{arctanh}(s)/s}{s^2} > 4.$$

This proves (C.15).

Raising the following inequalities to the fifth power reduces them to comparisons with $(11/4)^2$ and $(11/4)^3$, respectively. Thus $e < 11/4$ gives

$$e^{2/5} < \frac{3}{2}, \qquad e^{3/5} < 2. \tag{C.16}$$

Since $P > 7/2$ and $(1957/720)^7 > 1000$, we have $e^{2P} > 1000$. In the fourth line of (C.12), $\kappa_0 e^{-1-P} < 1/80$ since $\kappa_0 < 1$ and $e^{9/2} > 80$. The last line is at most $1 + 2/999 + 1/500 < 81/80$. Together with $2A_* < 1$ these estimates give

$$Q^D(t) < \frac{1000}{999} \quad (t > 0), \qquad Q^N(t) < \frac{81}{80} \quad (t \geq 0). \tag{C.17}$$

For $0 \leq t \leq 3/2$, with $t > 0$ in the odd case, the first and third lines of (C.12) give the stronger bound $2A_* < 1$ for both quantities since $3/2 < P/2$.

Let $t \geq 3/2$. We use the terms with $j = 0$ and $j = 1$ in (C.8) and (C.9) on the interval $1/20 \leq t-\tau \leq 1/5$. On this interval the $j = 0$ term is negative. The inequality $S(t-\tau) \geq (t-\tau)(1-\kappa_0^2(t-\tau)^2/6)$ and $C_0 > 2$ give $C_0 S(t-\tau) > 149/1500$. Also $\tau \geq 13/10$ and $k_1 > 5/2$, so $k_1\tau > 3$ and $e^{-k_1\tau} < 1/20$. Since $k_1 < 17/6$, we have $k_1(t-\tau) < 3/5$ and by (C.16),

$$\frac{e^{-k_1(t-\tau)}}{2k_1} > \frac{3}{34}.$$

Equation (C.4) now gives $G^D_{k_1}(t,\tau) > (399/400)(3/34)$ and $G^N_{k_1}(t,\tau) > 3/34$. Since $|S(t)| < 8/7$, $k_1^{-1} < 2/5$ and $C_1 > 1$, the $j = 1$ terms of (C.8) and (C.9) are bounded below by

$$\frac{399}{400}\frac{3}{34} - \frac{2}{35} > \frac{3}{100}, \qquad \frac{3}{34} - \frac{1}{50} > \frac{1}{15},$$

respectively. On the same interval,

$$\frac{\sinh(2\tau)}{\sinh(2t)} \geq e^{-2(t-\tau)}(1-e^{-4\tau}) > \frac{13}{20},$$
$$\frac{\cosh(2\tau)}{\cosh(2t)} \geq e^{-2(t-\tau)} > \frac{2}{3},$$

by (C.16) and $e^{4\tau} > 40$.

For real numbers u, w of opposite sign, $|u| + |w| - |u+w| = 2\min(|u|,|w|)$. The terms with $j = 0$ and $j = 1$ have opposite signs throughout the interval $1/20 \leq t-\tau \leq 1/5$. Its length is $3/20$, so combining these terms before taking the absolute value decreases the bound for $Q^D(t)$ by at least $2(3/100)(13/20)(3/20) = 117/20000$. It decreases the bound for $Q^N(t)$ by at least $2(1/15)(2/3)(3/20) = 1/75$. In particular, for $t \geq 3/2$,

$$Q^D(t) < \frac{1000}{999} - \frac{117}{20000} < 1,$$
$$Q^N(t) < \frac{81}{80} - \frac{1}{75} = \frac{1199}{1200} < 1.$$

For the weight h chosen above, set

$$\|Y(\cdot,L)\|_h = \sup_{\substack{t\geq 0 \\ h(t)>0}} \frac{|Y(t,L)|}{h(t)}.$$

This is finite by smoothness, parity and polynomial growth. Since $0 \leq b \leq 2b_*$, we have $|F(t)| \leq b_*|Y(t,L)|$. By (C.7), the ratio $|Y(t,L)|/h(t)$ is bounded by $b_* Q^D(t)\|Y(\cdot,L)\|_h$ in the odd case and

by $b_* Q^N(t)\|Y(\cdot, L)\|_h$ in the even case. Taking the supremum in t and using the bounds on $b_* Q^D$ and $b_* Q^N$ gives

$$\begin{aligned} \|Y(\cdot, L)\|_h &\leq \frac{63}{64}\|Y(\cdot, L)\|_h \quad (n = 3), \\ \|Y(\cdot, L)\|_h &\leq q\|Y(\cdot, L)\|_h \quad (n = 4), \end{aligned} \tag{C.18}$$

where

$$q = \max\left\{2A_*, \frac{1000}{999} - \frac{117}{20000}, \frac{1199}{1200}\right\} < 1.$$

Hence $Y(t, L) = 0$ and $F(t) = 0$ on $[0, \infty)$. By (C.4), $Y_j = 0$ for $j \geq 1$. The expansion at L then gives $e_0(L)Y_0(t) = Y(t, L) = 0$. Since $e_0(L) > 0$, completeness and parity give $Y \equiv 0$ on $\mathbb{R} \times \mathbb{S}^{n-1}_+$. □

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Instituto de Matemática e Estatística, Universidade Federal Fluminense, Niterói, RJ, Brasil, 24210-201.
Email address: felipebogo@id.uff.br